

Contents

Volume 2 - Applications Guide: Appendix A - Project Analysis Case Studies

A1 – Project Analysis Case Study 1:

1	Upgrading a Gravel Road to a Bituminous Pavement	1
2	Locate the case study data	2
3	Review the case study input data	3
3.1	General	4
3.2	Select Sections	4
3.3	Select Vehicles	9
3.4	Define Normal Traffic	12
3.5	Alternatives	14
3.6	Overview of project alternatives	19
3.7	Project Alternative: Without Project	20
3.8	Project Alternative: With Project	33
4	Run HDM-4 and examine the results	61
4.1	Setup Run	61
4.2	Run Analysis	61
4.3	Deterioration and works effects	62
4.4	Cost streams	64

A2 – Project Analysis Case Study 2:

1	Optimum Rehabilitation Standards for a Paved Road	1
1.1	Locate the case study data	1
1.2	Review the case study input data	1
1.3	Run HDM-4 and examine the results	38
1.4	Generate reports	39
1.5	Selected reports	42

A3 – Project Analysis Case Study 3:

1	Traffic Volume Capacity Improvements	1
1.1	Locate the case study data	1
1.2	Review the case study input data	1
1.3	Run HDM-4 and examine the results	21
1.4	Generate reports	21
 A4 – Project Analysis Case Study 4:		
1	New Bypass Construction (traffic diversion effects)	1
1.1	Locate the case study data	3
1.2	Review the case study input data	3
1.3	Run HDM-4 and examine the results	15

Appendix A1 - Project Analysis Case Study 1

1 Upgrading a Gravel Road to a Bituminous Pavement

This case study presents the economic analysis of a project to upgrade an existing gravel road to a paved standard. The existing road is 50 km long and passes through varying topography. For analysis purposes, three sections, based on geometry, pavement condition, and traffic volume can represent the road. Traffic and condition data are available from surveys undertaken in 1998. The gravel thickness in 1998 was 150 mm.

The objective of the case study is to demonstrate the structure and capabilities of HDM-4, and the steps needed to undertake the project appraisal.

The purpose of the appraisal is to assess the economic benefits resulting from the proposed investment. (This differs from a financial appraisal that is concerned with the means of financing a project and the financial profitability of the project). The economic feasibility of the project is assessed by comparison against a base-line project alternative (that is, a without project alternative). The project alternatives are:

- **Without Project** (see Section 3.7)

Maintain existing gravel road.

- **With Project** (see Section 3.8)

Maintain existing gravel road before upgrading to a bituminous pavement, followed by maintenance of the bituminous pavement.

To demonstrate this case study the following steps are followed:

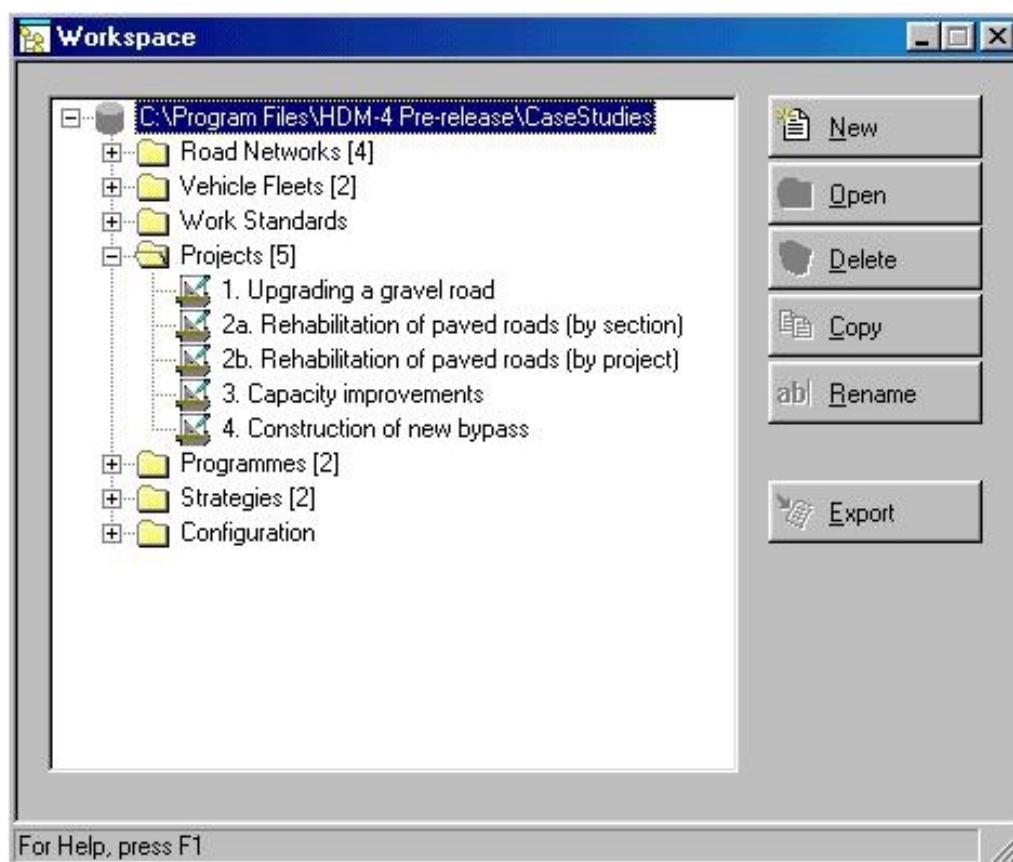
- **Locate the case study data** (see Section 2)
- **Review the case study input data** (see Section 3)
- **Run HDM-4 and examine the results** (see Section 4)

2 Locate the case study data

The case study data are included in the default daTabase installed with the HDM-4 software. The data for this case study are located in the **Projects folder** in the **Case Studies Workspace**. The name of the case study is **Case Study 1: Upgrading a gravel road**.

To locate the data for this case study:

- Go to the **Case Studies Workspace**
- Open the **Projects folder**
- Double-click on the case study named **Case Study 1: Upgrading a gravel road**



3 Review the case study input data

The case study input can be reviewed under the **HDM-4 Projects** work flow buttons and associated Tabs as follows:

- **Define Project Details**

The following Tab pages may be displayed:

- ☐ General (see Section 3.1)
- ☐ Select Sections (see Section 3.2)
- ☐ Select Vehicles (see Section 3.3)
- ☐ Define Normal Traffic (see Section 3.4)

- **Specify Alternatives**

The following Tab page may be displayed:

- ☐ Alternatives (see Section 3.5)

- **Analyse Projects**

The following Tab pages may be displayed:

- ☐ Setup Run (see Section 4.1)
- ☐ Run Analysis (see Section 4.2)

- **Generate Reports**

- ☐ Select Reports

It is recommended that the user keep a record of all input data in hard copy format, noting the sources of all information. This will assist when input data is checked (for example, on HDM-4 reports) and outputs reviewed, possibly by different personnel.

3.1 General

The screenshot shows a software window titled "Project: 1. Upgrading a gravel road". On the left is a vertical toolbar with icons and labels: "Define Project Details" (green arrow), "Specify Alternatives" (green arrow), "Analyse Projects" (green arrow), "Generate Reports" (document icon), "Save" (floppy disk icon), and "Close" (red X icon). The main area has four tabs: "General" (selected), "Select Sections", "Select Vehicles", and "Define Normal Traffic".

Under the "General" tab, the following fields are visible:

- Description:** "Project Case Study 1 : Upgrading a gravel road."
- Analyse by:** Radio buttons for "Section" and "Project". "Project" is selected.
- Start year:** "2000" | **Duration:** "20" years
- Road Network:** Dropdown menu showing "Northern Province Project Road Sections".
- Vehicle Fleet:** Dropdown menu showing "Northern Province Vehicles".
- Currencies:**
 - Fleet:** "US Dollar" × "1" = output currency
 - Works:** "US Dollar" × "1" = output currency
 - Output:** "US Dollar" (dropdown menu)

At the bottom, a status bar reads: "Start of year of analysis period (otherwise referred to as the Project Start Year)".

This Tab page confirms the project title and type of analysis required - project analysis in this case, as the task is to compare, for all three road sections, the upgrade alternative against the do-minimum alternative.

The project start year has been defined as 2000 with the analysis period (duration) specified as 20 years. The road network to be used for the project has been pre-defined under the name **Northern Province** in the **Road Network** folder. Similarly the vehicle fleet has been pre-defined in the **Vehicle Fleet** folder under the same name. The road sections (within the Northern Province road network) and vehicles types (from the Northern Province vehicle fleet) to be used for the analysis are selected under the **Select Sections** and **Select Vehicles** Tabs respectively, as discussed below:

Finally the **General** Tab page confirms that the input and output currencies are both US dollars.

3.2 Select Sections

The road section is the basic entity for all calculations of pavement deterioration, construction and maintenance costs, and economic analysis. Each section is considered homogeneous in terms of its physical attributes (for example, road class, climate, carriageway width, geometry, pavement condition, traffic flow, and axle loading).

For this case study, the road has been divided into three sections based on those physical attributes that vary along its length. Table A1.1 contains a summary of the key physical attributes for each section.

Note that in this case study, no shoulders are specified, and the full width of gravel surfacing is assigned as carriageway width. In practice it is often difficult to identify shoulders on a gravel road.

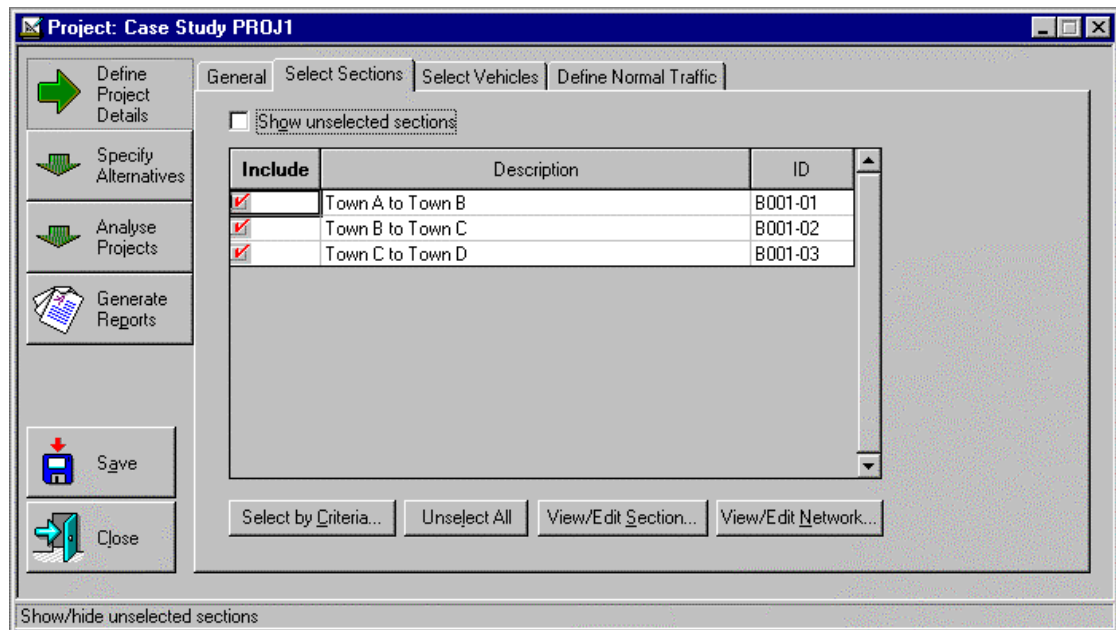
Table A1.1
Case study 1 - Summary of section attributes

Section ID	B001-01	B001-02	B001-03
General			
Section Name	Town A to Town B	Town B to Town C	Town C to Town D
Length (km)	20	10	20
Carriageway width (m)	7	6	8
Shoulder width (m)	0	0	0
Number of lanes	2	2	2
Number of shoulders	0	0	0
AADT (1998)	400	200	600
Flow direction	2 - way	2 - way	2-way
Geometry (before upgrading)			
Rise and Fall (m/km)	10	15	30
*No rises and falls (n°/km)	0.1	0.4	1.0
Av horiz curv (deg/km)	50	300	15
*Superelevation %	3.0	3.0	3.0
*σ _{dral}	0.1	0.1	0.1
Speed limit (kph)	100	80	100
*Speed limit enforcement	1.1	1.1	1.1
Altitude (m)	120	295	370
Condition 1998			
Gravel Thickness	150	150	150
Roughness (IRI, m/km)	8	8	8

Notes:

- 1 Geometry Data marked with an * (asterisk) is stored under **Section/Details/Alignment**. Other geometric data are held under **Section/Geometry**.

Traffic flow (AADT) used in project analysis is defined by section at project level under **Define Project Details/Define Normal Traffic**, and will often be an update of base traffic data retained at section level under **Network/Section/Definition**.



The **Select Sections** Tab page displays those sections of the **Northern Province** road network that will be included in the analysis. For this case study, the three sections listed should have a **tick** in the **Include** column to confirm their selection.

Each section has been assigned a unique Description and ID, either or both of which may be used to identify the section on HDM-4 reports. In this case study, the section description identifies towns at the ends of each section. It is helpful if the Section ID includes reference to the road class and road number (denoted by B001 in this case study).

By double-clicking on one section in the **Select Section** Tab page, the characteristics of that section can be reviewed under four Tabs (**Definition**, **Geometry**, **Pavement**, and **Condition**). The corresponding Tab pages are reviewed below for one of the sections from this case study (**Section Description: Town A to Town B**).

■ Section/Definition

Lists basic characteristics such as speed-flow type and traffic-flow pattern, also carriageway length and width. The pavement type is defined as Gravel (HDM-4 pavement type GRUP, Granular Unsealed Pavement).

Section pavement details are stored under the **Pavement** Tab.

Section: Town A to Town B

Definition | Geometry | Pavement | Condition

Description: Length: km

ID: Carriageway width: m

Speed flow type: Shoulder width: m

Traffic flow pattern: Number of Lanes:

Road class:

Climate zone:

Surface class:

Pavement Type:

Traffic

ADT: Year:

Flow direction:

Details... OK Cancel Help

Name of section

■ **Section/Geometry**

Summarises existing geometric parameters (before upgrading) including those for horizontal and vertical alignment. Note that certain geometric parameters (marked with * in Table A1.1) are held under **Section/Details/Alignment**.

□ **Section/Geometry screen**

Section: Town A to Town B

Definition | Geometry | Pavement | Condition

Rise + Fall: m/km

Avg horiz curvature: deg/km

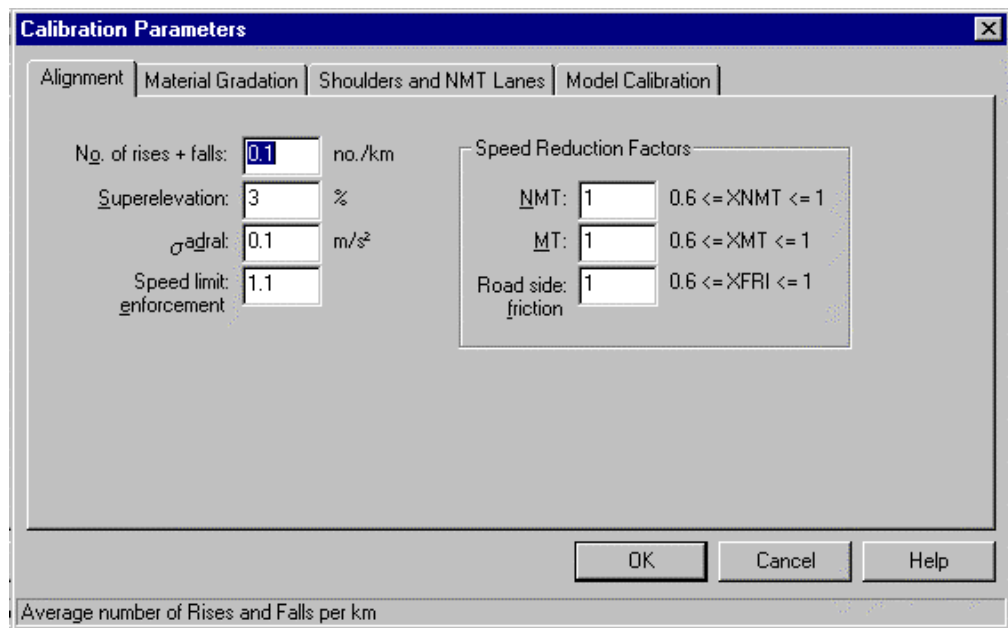
Speed limit: km/h

Altitude: m

Details... OK Cancel Help

Average road Rise plus Fall

- Section/Geometry/Alignment screen



Calibration Parameters

Alignment | Material Gradation | Shoulders and NMT Lanes | Model Calibration

Ng. of rises + falls: no./km
 Superelevation: %
 Gradat: m/s²
 Speed limit enforcement:

Speed Reduction Factors

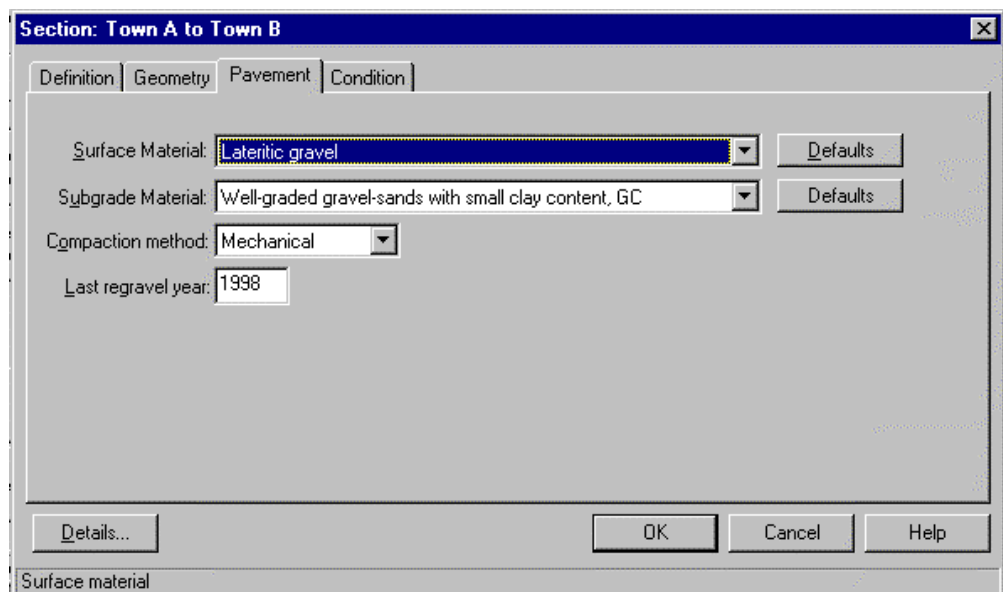
NMT: 0.6 <= XNMT <= 1
 MT: 0.6 <= XMT <= 1
 Road side friction: 0.6 <= XFRI <= 1

OK Cancel Help

Average number of Rises and Falls per km

■ Pavement

- Section/Pavement screen



Section: Town A to Town B

Definition | Geometry | Pavement | Condition

Surface Material: Defaults
 Subgrade Material: Defaults
 Compaction method:
 Last regravelling year:

Details... OK Cancel Help

Surface material

Gives pavement layer descriptions and year of last regravelling. The material properties can be reviewed under **Section/Details/Material Gradation**.

- ❑ Section/Details/Material Gradation screen

Calibration Parameters

Alignment | **Material Gradation** | Shoulders and NMT Lanes | Model Calibration

Surface

Surface Material: Lateritic gravel [Defaults]

Max particle size: 21.9 mm Plasticity index: 10.1 %

% passing 2.00mm sieve: 51.1 % % passing 0.425mm sieve: 41.6 % % passing 0.075mm sieve: 25.5 %

Subgrade

Subgrade Material: Well-graded gravel-sands with small clay content, GC [Defaults]

Max particle size: 13 mm Plasticity index: 15 %

% passing 2.00mm sieve: 60 % % passing 0.425mm sieve: 40 % % passing 0.075mm sieve: 18 %

[OK] [Cancel] [Help]

Details of pavement construction

- **Condition**

Confirms gravel thickness 150 mm and roughness 8 m/km (IRI) in 1998. Note that, to enable road condition to be modelled through the analysis period, condition data must be specified for a date prior to the start of the analysis period. In this case study the condition data refer to the end of 1998, which is prior to the analysis start year 2000.

- ❑ Section/Condition screen

Section: Town A to Town B

Definition | Geometry | Pavement | **Condition**

Condition at end of year	1998
Gravel Thickness (mm)	150.00
Roughness (IRI - m/km)	8.00

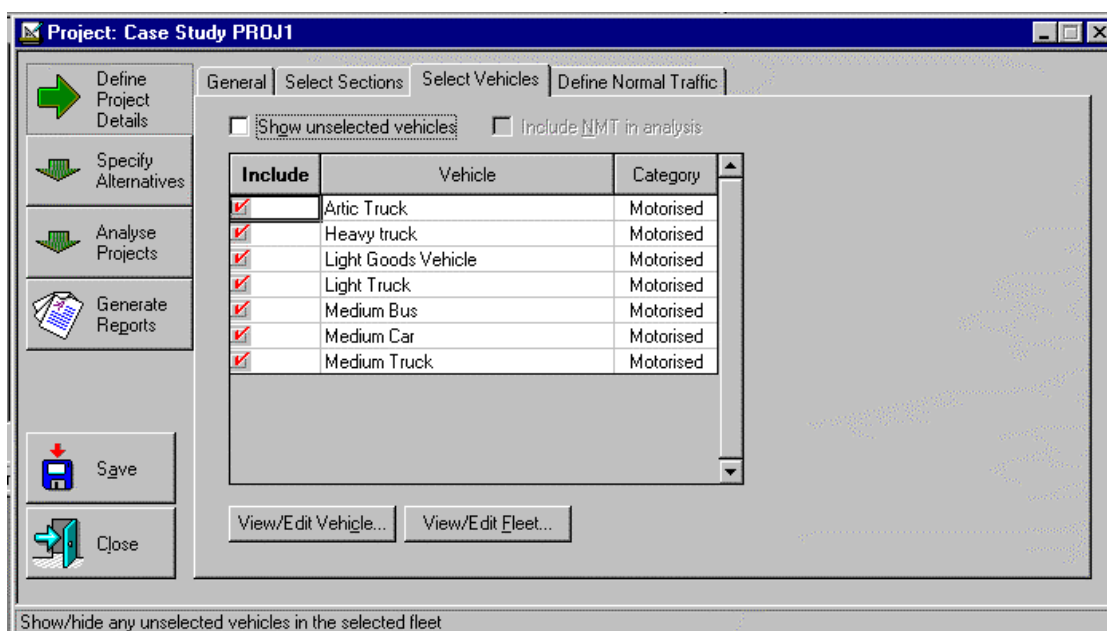
[Add New Year] [Delete Year] [Sort Years]

[Details...] [OK] [Cancel] [Help]

Add a column for a new year of condition data

3.3 Select Vehicles

This Tab page displays those vehicles that were selected (from the pre-defined **Northern Province Vehicle Fleet**) for this case study analysis.

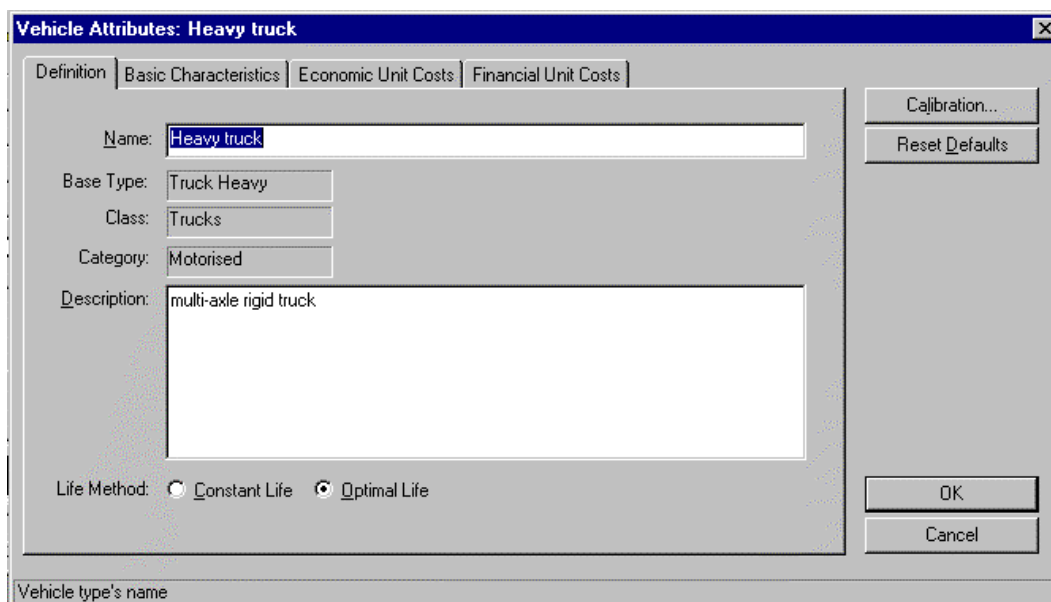


The **tick** in the **Include** column confirms the selection. This list defines the range of vehicle types that can be assigned to each road section, although the full range need not be used. The assignment of traffic by section is effected under the **Define Normal Traffic** Tab, discussed later.

Vehicle attributes for a particular vehicle type can be reviewed by double-clicking on the appropriate vehicle type descriptions. The vehicle attributes are held under four Tabs: **Definition**, **Basic Characteristics**, **Economic Unit Costs**, and **Financial Unit Costs**.

An example of each Tab page for one vehicle type is shown below:

- **Vehicle Attributes/Definition screen**



■ Vehicle Attributes/Basic Characteristics screen

Vehicle Attributes: Heavy truck

Definition | Basic Characteristics | Economic Unit Costs | Financial Unit Costs

Physical

Passenger Car Space Equiv: 1.6

No. of wheels: 10

No. of axles: 3

Tyres

Tyre type: Bias-ply

Base no. of recaps: 1.3

Retread cost: 15 %

Utilisation

Annual km: 86000 km

Working hours: 2050 hrs

Average life: 14 years

Calculate...

Private use: 0 %

Passengers: 0 persons

Work related passenger-trips: 0 %

Loading

ESAL: 2.28

Calculate...

Operating weight: 13 tonnes

OK

Cancel

Passenger Car Space Equivalents factor (PCSE)

■ Vehicle Attributes/Economic Unit Costs screen

Vehicle Attributes: Heavy truck

Definition | Basic Characteristics | Economic Unit Costs | Financial Unit Costs

Vehicle resources

New vehicle: 80000

Replacement tyre: 375

Fuel: 0.5 per litre

Lubricating oil: 2 per litre

Maintenance labour: 3 per hour

Crew wages: 4 per hour

Annual overhead: 5000

Annual interest: 6 %

Time Value

Passenger: 0 per hour

working time

Passenger: 0 per hour

non-working time

Cargo: 0.3 per hour

(all costs should be in the currency specified in the Fleet Info dialog)

OK

Cancel

The average purchase cost of a new vehicle of this type

■ Vehicle Attributes/Financial Unit Costs screen

Vehicle Attributes: Heavy truck

Definition | Basic Characteristics | Economic Unit Costs | **Financial Unit Costs**

Vehicle resources:

New vehicle:	94118	Maintenance labour:	3.53	per hour
Replacement tyre:	441	Crew wages:	4.7	per hour
Fuel:	0.59	per litre	Annual overhead:	5882
Lubricating oil:	2.35	per litre	Annual interest:	6 %

(all costs should be in the currency specified in the Fleet Info dialog)

Calibration...
Reset Defaults

OK
Cancel

The average purchase cost of a new vehicle of this type

Note that the vehicle attributes are defined for a representative vehicle within each vehicle class. For example, under **Basic Characteristics**, the loading details (No of equivalent standard axles and operating weight) are based on the average weight of vehicles in the vehicle class.

3.4 Define Normal Traffic

This Tab page gives the traffic volume (AADT) on each road section in the specified year. When setting up a specific case study, the base AADT held by section is automatically assigned to the **Define Normal Traffic** Tab page. However the user may update the AADT/year information for the purposes of the case study. In this event the AADT data held at section level (under **Network/Section/Definition**) remains unchanged.

Project: Case Study PROJ1

General | Select Sections | Select Vehicles | **Define Normal Traffic**

Section	ID	Motorised AADT	NMT AADT	Year
Town A to Town B	B001-01	400.00	0.00	1998
Town B to Town C	B001-02	200.00	0.00	1998
Town C to Town D	B001-03	600.00	0.00	1998

Define Project Details
Specify Alternatives
Analyse Projects
Generate Reports
Save
Close

Growth Periods and Compositions
Copy Paste Edit Section Traffic Details...

Normal Traffic for the Project

By double-clicking on the row representing a specific section, the traffic composition in the specified year (by % of each vehicle type) can be reviewed, together with vehicle growth rates. Thus for section A-B, the normal traffic details are as given below:

■ **Define Normal Traffic/Motorised screen**

Normal Traffic Details

Motorised

Details for section: Town A to Town B

Total AADT: Year:

Growth Periods

Vehicle	Initial Composition (%)	Annual % increase from 2000
Medium Bus	5.00	2.00
Artic Truck	2.50	2.00
Heavy truck	2.50	2.00
Medium Truck	10.00	2.00
Light Truck	10.00	2.00
Light Goods Vehicle	20.00	4.00
Medium Car	50.00	4.00

Buttons: Add New Period, Delete Period, Edit Period...

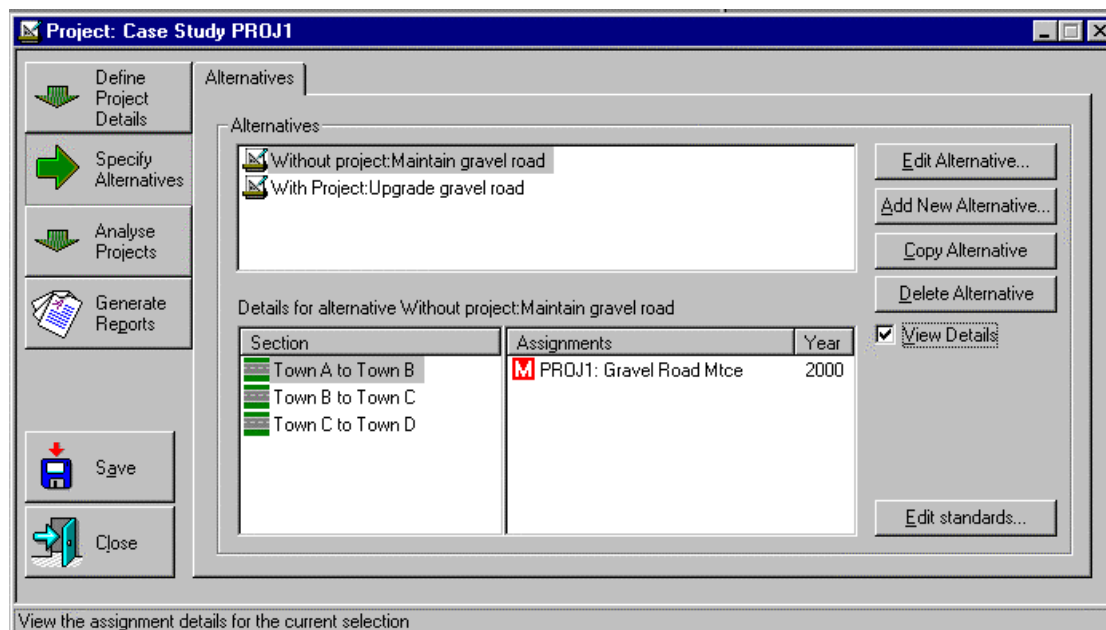
Buttons: OK, Cancel, Help

The annual average daily traffic on the road section

For this case study, only one growth period is indicated over the 20-year analysis period. Several growth periods may be defined within the analysis period. The user is encouraged to define future growth rates (by vehicle type) as accurately as possible. Note that negative growth rates are permitted.

3.5 Alternatives

The **Alternatives** Tab is displayed after selecting the **Specify Alternatives** button.



The two project alternatives considered in this case study are defined below. The first alternative, **Without Project**, represents a continuation of current maintenance practice. The second alternative, **With Project**, represents the implementation of the project to upgrade the existing gravel road to paved standard.

- **Without Project** (see Section 3.7)

This project alternative comprises grading every six months; spot regravelling to replace 30% of material lost each year (if the gravel thickness falls below 100 mm, and gravel resurfacing (whenever the gravel thickness falls below 50 mm).

Note that in a given year, if gravel resurfacing is triggered, this operation supersedes grading and spot regravelling.

- **With Project** (see Section 3.8)

This project alternative includes upgrading the road in the year 2004 (duration of works is two years for road sections B001-01 and B001-03, and one year for section B001-02). Before upgrading, the existing gravel road will be maintained, by grading, every six months and spot regravelling to replace 30% of material lost each year (if the gravel thickness falls below 100 mm). The latter works will maintain some protection of the subgrade prior to upgrading.

After upgrading, the road will receive routine maintenance in the form of crack sealing (if wide structural cracking reaches 5%) and patching (if the severely damaged area reaches 5%).

The analysis period is 20 years, starting in the year 2000, and analysis is by project as specified under the **General** Tab.

The **Alternatives** Tab page is split into two boxes. The upper box shows the names of the two project alternatives that have been set up for this case study.

The bottom box shows details, by section, of the **Road Works Standards** (assignments) associated with each project alternative. Note that the assignments box shows the Road Works Standards corresponding to the selected project alternative and section. For the selected alternative, the Road Works Standards assigned to a different section can be reviewed by selecting that section.

A schematic overview of the Project Alternatives and their constituent Road Works Standards is shown in Figure A1.1. Note that Road Works Standards are sub-divided into Maintenance Standards denoted by **M**, and Improvement Standards denoted by **I**.

The Road Works Standards and associated works assigned to each project alternative are summarised in Table A1.2. A summary of the intervention limits for maintenance works is given in Table A1.3. These Tables allow the user to check consistency of data across alternatives.

Each Road Works Standard has an associated date, assigned at project level, which represents the year from which the **standard** takes effect. For the **Without Project** alternative, each of the three road sections has been assigned a Maintenance Standard named **Gravel Road Maintenance**. This assignment commences in the year 2000, which is the first year of the 20-year analysis period (2000-2019).

For the **With Project** alternative, each section has been assigned a set of Road Works Standards as follows:

- **M** Maintenance Standard for gravel road prior to upgrading
- **I** Improvement Standard representing the upgrading works
- **M** Maintenance Standard for paved road after upgrading

Note that (for a given feature) only one Maintenance Standard or Improvement Standard will be effective in any analysis year. However a Maintenance Standard may include more than one **works item** which could be implemented in a given year, for example, grading and spot regravelling on unsealed roads, crack sealing and patching on bituminous roads. Details of the logical rules associated with implementation of works are given in Chapter D1 of the [Analytical Framework and Model Descriptions](#).

Calendar year	Without project	With project		
	All sections	Section B001-01	Section B001-02	Section B001-03
1998	Traffic and Condition data			
1999				
2000	(1)	(2)	(2)	(2)
2001	(1)	(2)	(2)	(2)
2002	(1)	(2)	(2)	(2)
2003	(1)	(2)	(2)	(2)
2004	(1)	(3)	(4)	(5)
2005	(1)	(3)	(6)	(5)
2006	(1)	(6)	(6)	(6)
2007	(1)	(6)	(6)	(6)
2008	(1)	(6)	(6)	(6)
2009	(1)	(6)	(6)	(6)
2010	(1)	(6)	(6)	(6)
2011	(1)	(6)	(6)	(6)
2012	(1)	(6)	(6)	(6)
2013	(1)	(6)	(6)	(6)
2014	(1)	(6)	(6)	(6)
2015	(1)	(6)	(6)	(6)
2016	(1)	(6)	(6)	(6)
2017	(1)	(6)	(6)	(6)
2018	(1)	(6)	(6)	(6)
2019	(1)	(6)	(6)	(6)

(1)		Maintenance Standard: Gravel road maintenance
(2)		Maintenance Standard: Maintenance before upgrading
(3)		Improvement Standard: Pave section B001-01
(4)		Improvement Standard: Pave section B001-02
(5)		Improvement Standard: Pave section B001-03
(6)		Maintenance Standard: Crack sealing and patching paved road

Figure A1.1 Case Study 1: Overview of project alternatives and road work standards

Table A1.2
Case Study 1: Details of road works standards for each project alternative

Project alternative	Section ID		Road Works Standards	Effective from year	Maintenance works/ Improvement type
Without Project Maintain Gravel Road	B001-01	M	Gravel Road Maintenance (GRAVEL)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
					Gravel resurfacing (RESURF)
	B001-02	M	Gravel Road Maintenance (GRAVEL)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
					Gravel resurfacing (RESURF)
	B001-03	M	Gravel Road Maintenance (GRAVEL)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
					Gravel resurfacing (RESURF)
With Project Upgrade Gravel Road	B001-01	M	Maintenance before upgrading (BEFORE)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
		I	Pave Section B001-01 in 2004 (PAVE01)	2004	Upgrading
		M	Crack sealing and patching paved road (SEAPAT)	2006	Crack sealing (SEAL)
					Patching (PATCH)
	B001-02	M	Maintenance before upgrading (BEFORE)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
		I	Pave Section B001-02 in 2004 (PAVE02)	2004	Upgrading
		M	Crack sealing and patching paved road (SEAPAT)	2005	Crack sealing (SEAL)
					Patching (PATCH)
	B001-03	M	Maintenance before upgrading (BEFORE)	2000	Grading (GRADE6)
					Spot regravelling (SPG100)
		I	Pave Section B001-03 in 2004 (PAVE03)	2004	Upgrading
		M	Crack sealing and patching paved road (SEAPAT)	2006	Crack sealing (SEAL)
					Patching (PATCH)

Table A1.3
Case study 1: Intervention limits for maintenance works

	Road works standard	Effective from year	Maintenance works	* S/R	Last Year	Max IRI	Max Quantity	Interval		AADT	
								Min	Max	Min	Max
M	Gravel Road Maintenance	2000	Grading	S	2019	30	n/a	180 days	750 days	0	100,000
			Spot regravelling	R	2019	30	100 m ³ /km/yr	n/a	n/a	0	100,000
			Gravel resurfacing	R	2017	30	n/a	3 yrs	99 yrs	0	100,000
M	Maintenance before upgrading	2000	Grading	S	2003	30	n/a	180 days	10000 days	0	100,000
			Spot regravelling	R	2003	30	100 m ³ /km/yr	n/a	n/a	0	100,000
M	Crack sealing and patching paved road	2006	Crack sealing	R	2019	12.5	1500 m ² /km/yr	n/a	n/a	0	100,000
			Patching	R	2019	12.5	1500 m ² /km/yr	n/a	n/a	0	100,000

Notes:

n/a not applicable

* S = Scheduled intervention
 R = Responsive intervention

3.6 Overview of project alternatives

The use of the HDM-4 software to set up the Road Works Standards for this case study is described in more detail below. Each Project Alternative includes one or more Maintenance Standards and/or Improvement Standards that are assigned to be effective from a specified date. Each Maintenance Standard may define several Works Items, and each Improvement Standard defines a specific improvement type.

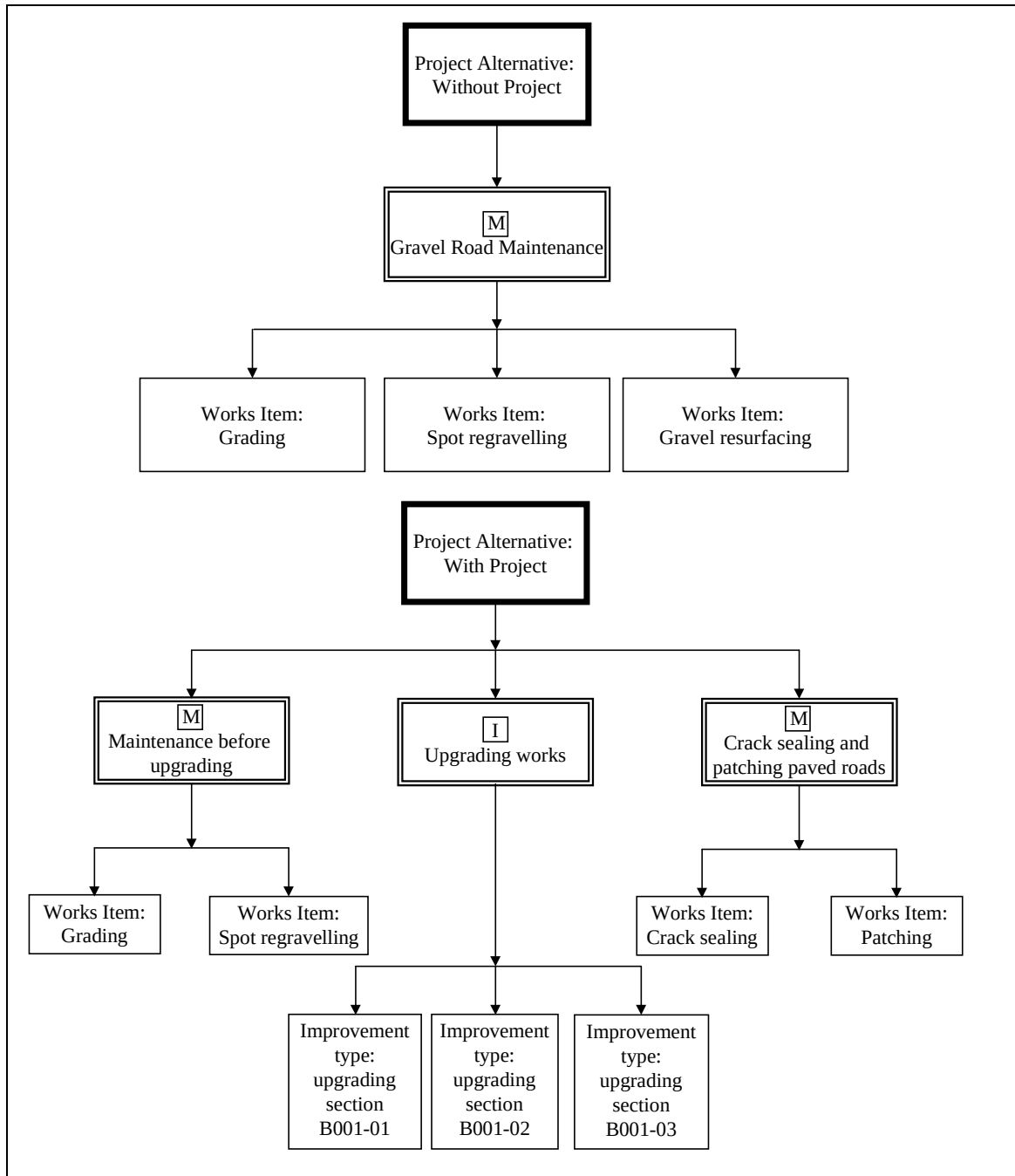


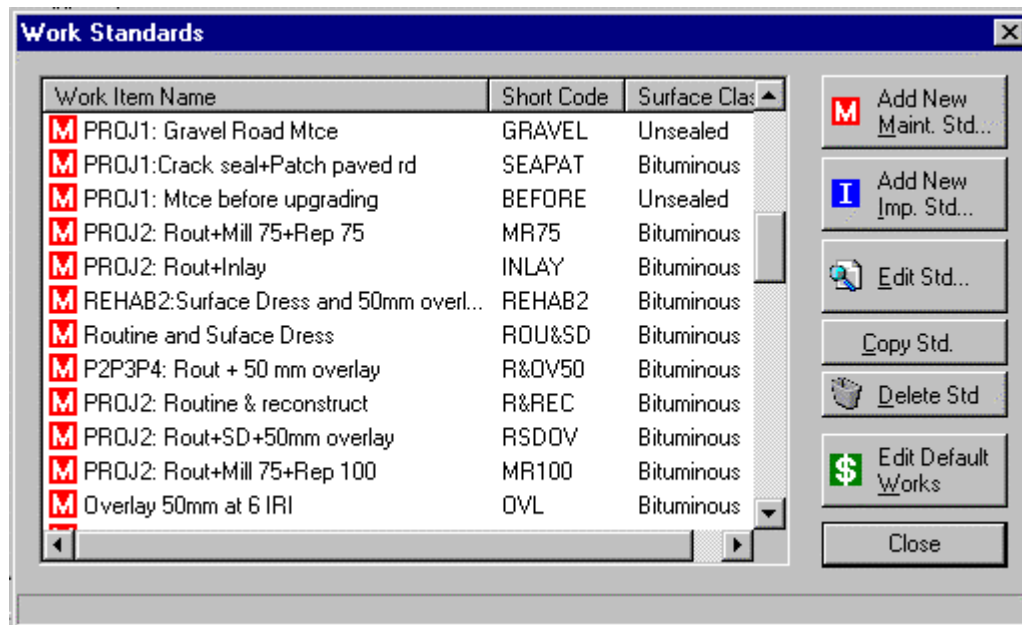
Figure A1.2 Schematic diagram showing details of Project Alternatives for Case Study 1

3.7 Project Alternative: Without Project

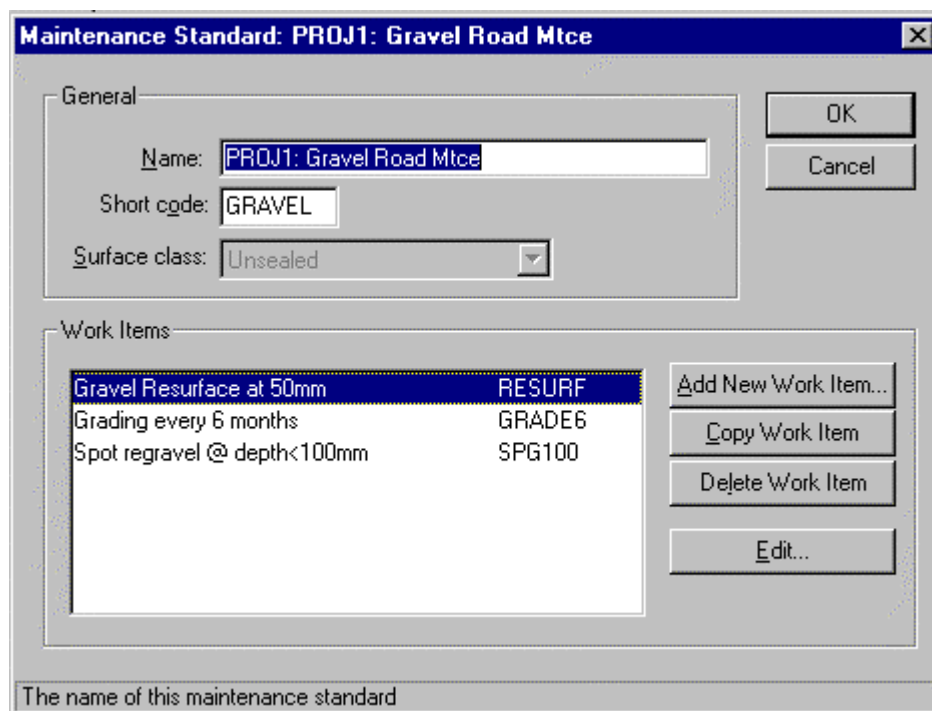
This Project Alternative includes one Maintenance Standard associated with the existing gravel road.

3.7.1 Maintenance Standard: Gravel Road Maintenance

The specification for this Maintenance Standard can be reviewed via the **Edit Standards** button (or alternatively via **HDM-4 Workspace**). This displays a list of the names of all Maintenance Standards that have been pre-defined within the **Road Works Standards** folder.



By double-clicking on **Gravel Road Maintenance**, the window for the specification of this maintenance standard is displayed.



This window shows the names and codes assigned to the **Gravel Road Maintenance** Maintenance Standard and constituent Works Items. These names and codes appear on certain HDM-4 reports so it is essential that they are unique within a Maintenance Standard. The software checks for unique names and codes when Maintenance Standards are set up.

The window lists the three works items that are associated with the **Gravel Road Maintenance** Maintenance Standard. As previously described, the **Without Project** alternative applies grading every six months, spot regravelling to replace 30% material lost each year (if the gravel thickness falls below 100 mm), and gravel resurfacing (if the gravel thickness falls below 50 mm). The original gravel thickness was 150 mm (1998).

Note that in a given year, spot regravelling and grading are both permitted as these are complimentary works. However if gravel resurfacing is triggered, this will **supersede** the other works. Hence in a given year, the possible works under the **Gravel Road Maintenance** Maintenance Standard are:

- **Grading every six months, or**
- **Spot regravelling AND Grading every six months, or**
- **Gravel resurfacing**

The **Works** Items and **Operations** associated with the **Maintenance Standard** are shown in Figure A1.3. Full details of the logical rules associated with the implementation of works are given in Chapter D1 of the [Analytical Framework and Model Descriptions](#).

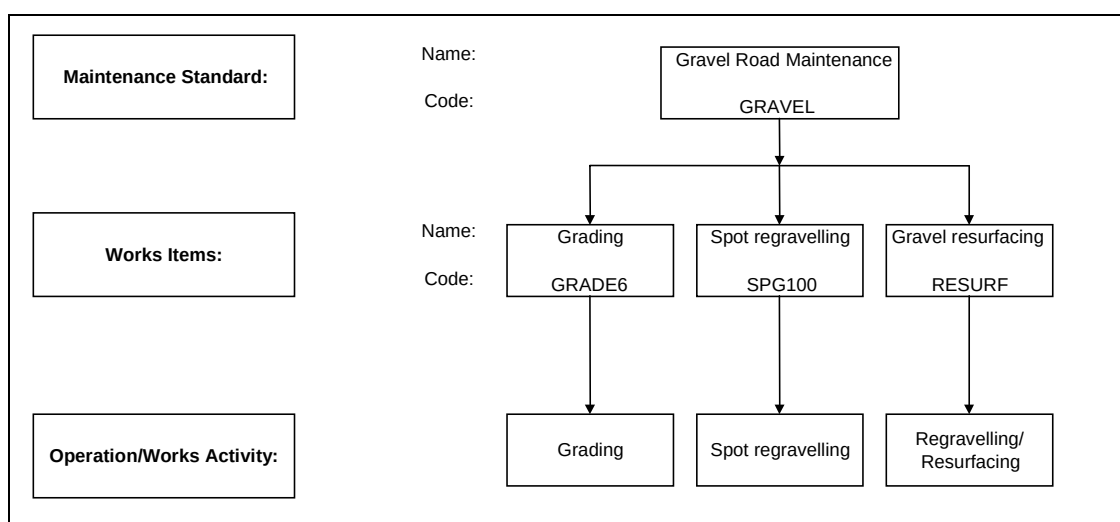


Figure A1.3 Hierarchy within a Maintenance Standard

Note that, when setting up a Maintenance Standard, the user defines the name and code for the Maintenance Standard and Maintenance Works. The name of the Operation/Works Activity is selected from a pre-defined list. It is useful if the name/code assigned to the Works Item also includes information on intervention (for example, grading every six months). It may also be useful if the name given to the Maintenance Standard refers to the surface type (for example, Gravel Road Maintenance).

Works Item: Grading

By double-clicking on a particular works item, the Maintenance Works Item specification window is displayed. The specification for **Grading** is contained under four Tabs, **General**, **Design**, **Intervention**, and **Costs**:

■ Grading/General

The **General** Tab page for Works Item **Grading** is shown below:

The screenshot shows a dialog box titled "Maintenance Works Item: Grading every 6 months". It has four tabs: "General", "Design", "Intervention", and "Costs". The "General" tab is active. The fields are as follows:

- Name: Grading every 6 months
- Short code: GRADE6
- Surface: Unsealed
- Feature type: Carriageway (dropdown menu)
- Operation: Grading (dropdown menu)
- Intervention type: ☒ Scheduled, ☐ Responsive

At the bottom are buttons for "OK", "Cancel", "Apply", and "Help". A status bar at the very bottom says "The name of this works item".

This confirms the Name and Short Code of the Works Item. The operation type/works activity (specified from a pre-defined list) is confirmed as **grading**, and intervention type is **scheduled**.

■ Grading/Design

The **Design** Tab page for grading specifies the compaction method as mechanical.

■ Grading/Intervention

The **Intervention** Tab page confirms the frequency at which the grading works will be undertaken (that is, 180 days).

The screenshot shows a dialog box titled "Maintenance Works Item: Grading every 6 months" with a close button (X). It has four tabs: General, Design, Intervention (selected), and Costs. The "Intervention criteria" section has two radio buttons: "Time interval:" (selected) with a value of 180 Days, and "Traffic interval:" with a value of 200 vehicles. The "Limits" section contains several input fields: "Last year:" with 2019 year, "Max. roughness:" with 30 IRI (m/km), "Max. quantity:" with 5000 m²/km/year, "Interval:" with 180 days, and "ADT:" with 0. There are also "Minimum" and "Maximum" columns for "Interval:" (180 and 750 days) and "ADT:" (0 and 100000). At the bottom are buttons for OK, Cancel, Apply, and Help. A status bar at the very bottom reads "General maintenance works parameters".

In general, scheduled grading works are timed from the **last regravelling year**, specified under **Section/Pavement** (1998 in this case study). However the works can only be implemented after the date when the Maintenance Standard becomes effective (specified at project level, 2000 for this case study).

Note that for scheduled works, the frequency of works can also be specified in terms of traffic interval (that is, number of vehicles between successive gradings).

The **Intervention** screen also shows limits outside which the grading works would **not** be triggered:

❑ Last year

The last year in which grading should be considered. For this Maintenance Standard the last year for grading is set at 2019 (the last year in the analysis period).

This parameter could be used to ensure that a given works is not triggered too close to the end of the analysis period.

❑ Maximum roughness

Sets the maximum roughness at which grading would be undertaken (30 m/km IRI).

❑ Interval

The minimum interval between successive grading operations has been set at 180 days (six months).

The maximum interval between successive grading operations, above which grading will not be implemented, has been specified as 750 days (just over two years) to allow for the fact that grading will not be triggered in a year when gravel resurfacing

is implemented. If the specified interval is exceeded, grading will not be triggered, but other Works Items under this Maintenance Standard would be considered.

❑ **AADT**

Allows the user to specify the AADT range over which the works are considered appropriate. For this case study, the range is specified as 0-100,000 to avoid elimination of grading based on this parameter.

■ **Grading/Costs**

The unit costs for grading works (expressed in US dollars per km) can be reviewed under the **Costs** Tab. Note that default costs for all works operations are specified under the **HDM Workspace** option **Road Works Standards/Default Works Costs & Energy Consumption**.

Maintenance Works Item: Grading every 6 months

General | Design | Intervention | **Costs**

Economic Financial
 Unit cost: 120 144 per km

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 1 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

By clicking the **OK** button (on the Costs screen), the user is returned to the entry screen for the Maintenance Standard **Gravel Road Maintenance**.

Works Item: Spot regravelling

The specification for **Spot regravelling** is contained under four Tabs, **General**, **Design**, **Intervention**, and **Costs**. The Tabs are revealed by double-clicking the **Spot regravelling** Works Item in the **Maintenance Standard** window.

■ Spot regravelling/General

The **General** Tab page for the Works Item **Spot regravelling** is shown below:

The screenshot shows a dialog box titled "Maintenance Works Item: Spot regravel @ depth<100mm". It has four tabs: General, Design, Intervention, and Costs. The General tab is active. The fields are as follows:

- Name: Spot regravel @ depth<100mm
- Short code: SPG100
- Surface: Unsealed
- Feature type: Carriageway (dropdown)
- Operation: Spot Regravelling (dropdown)
- Intervention type: ☐ Scheduled, ☒ Responsive

At the bottom are buttons for OK, Cancel, Apply, and Help. A status bar at the bottom says "The name of this works item".

This confirms the Name and Short Code of the Works Item. The operation type/works activity is **spot regravelling**, with intervention type **responsive**.

■ Spot regravelling/Design

Two design options for spot regravelling are given under the **Design** Tab.

The screenshot shows the same dialog box, but with the Design tab active. The fields are as follows:

- Material Properties:
 - Surface Material: Select a surface material to obtain default material properties (dropdown)
 - Max particle size: 0 mm
 - Plasticity index: 0 %
 - % passing 2.00mm sieve: 0 %
 - % passing 0.425mm sieve: 0 %
 - % passing 0.075mm sieve: 0 %
- Spot-regravelling:
 - ☐ Volume of material to add: 0 m³/km/yr
 - ☒ Annual material loss to replace: 30 %
- Gravel Thickness:
 - ☐ Final gravel thickness: 25 mm
 - ☒ Increase in gravel thickness: 0 mm
 - Compaction method: Mechanical (dropdown)

At the bottom are buttons for OK, Cancel, Apply, and Help. A status bar at the bottom says "General maintenance works parameters".

Either a specified volume of gravel can be added each year (specified as $\text{m}^3/\text{km}/\text{yr}$), or a specified percentage of the annual gravel loss is replaced (for example, 100% would replace all gravel lost). For this case study the latter design option is adopted, replacing 30% gravel lost each year. Note that the amount of gravel actually applied to the road section will be subject to user defined limits specified under **Intervention**.

■ Spot regravelling/Intervention

The **Intervention** Tab page gives the responsive criterion that has been specified for spot regravelling (that is, whenever gravel thickness falls below 100 mm). Note that with some works activities, more than one intervention criterion can be specified; however with spot regravelling, gravel thickness is the only parameter that is available.

The screenshot shows a software dialog box titled "Maintenance Works Item: Spot regravel @ depth<100mm". It has four tabs: "General", "Design", "Intervention" (which is selected), and "Costs".

Under the "Intervention" tab, there are two main sections:

- Responsive Criteria:** A list box containing the text "Gravel thickness <= 100 mm". To the right of the list box are three buttons: "Add New Criterion...", "Delete", and "Edit...".
- Limits:** A section with several input fields:
 - "Last year:" with a text box containing "2019" and a unit dropdown set to "year".
 - "Max. roughness:" with a text box containing "30" and a unit dropdown set to "IRI (m/km)".
 - "Max. quantity:" with a text box containing "100" and a unit dropdown set to " $\text{m}^3/\text{km}/\text{year}$ ".
 - "Interval:" with a text box containing "1" and a unit dropdown set to "year(s)".
 - "AADT:" with a text box containing "0".
 - "Minimum:" with a text box containing "1".
 - "Maximum:" with a text box containing "99".
 - "Maximum:" with a text box containing "100000".

At the bottom of the dialog box are four buttons: "OK", "Cancel", "Apply", and "Help". Below the dialog box, there is a status bar that reads "Add an intervention criterion to this improvement standard".

The **Intervention** Tab page also gives limits affecting the **spot regravelling** works:

❑ Last year

The last year in which spot regravelling should be considered. For this case study, the last year is set at 2019 (last year of analysis period).

❑ Maximum roughness

Sets the maximum roughness at which spot regravelling would be undertaken (30 m/km IRI).

❑ Maximum quantity

Sets an upper limit on the quantity of material to be used each year for spot regravelling, specified as $\text{m}^3/\text{km}/\text{year}$. This could reflect the upper limit considered practicable before complete resurfacing should be considered, but should also take account of availability of material and work capacity of the Roads Authority. For this case study, the quantity specified is $100 \text{ m}^3/\text{km}/\text{year}$.

❑ AADT

This allows the user to specify the AADT range over which the works are considered appropriate. For this case study, the range is specified as 0-100,000 to avoid elimination of grading based on this parameter.

■ **Spot regravelling/Costs**

The unit assigned to **spot regravelling** (expressed in US Dollars per cubic metre) can be reviewed via the **Costs** Tab.

The screenshot shows a dialog box titled "Maintenance Works Item: Spot regravel @ depth<100mm". It has four tabs: General, Design, Intervention, and Costs. The Costs tab is active. The dialog contains the following fields:

- Unit cost:** A table with two columns: Economic and Financial. The Economic column has a value of 12, and the Financial column has a value of 14.4. To the right of these columns is a dropdown menu showing "per m²".
- Unit Costs Of Preparatory Works:** A section containing four rows of input fields:

	Economic	Financial	Unit
Spot Regravelling:	0	0	per m²
Patching:	0	0	per m²
Edge-repair:	0	0	per m²
Crack sealing:	0	0	per m²
- Drainage:** A section containing a field for "Drainage maintenance cost factor" with a value of 1, and a label "0 < DMCF <= 1".

At the bottom of the dialog are four buttons: OK, Cancel, Apply, and Help. A status bar at the very bottom reads "Economic unit cost of works".

Works Item: Gravel Resurfacing

The third Works Item considered under the Maintenance Standard **Gravel Road Maintenance** is **Gravel Resurfacing**. The specification for this can be reviewed under the Tabs **General**, **Design**, **Intervention**, **Costs**, and **Effects**:

■ Gravel Resurfacing/General

The **General** Tab page, shown below, confirms that the Operation Type/Works Activity is Regravelling/Resurfacing and that the Intervention Type is responsive (the intervention criterion is defined under the **Intervention** Tab).

The screenshot shows a dialog box titled "Maintenance Works Item: Gravel Resurface at 50mm". It has five tabs: "General", "Design", "Intervention", "Costs", and "Effects". The "General" tab is selected. The form contains the following fields and options:

- Name:** Gravel Resurface at 50mm
- Short code:** RESURF
- Surface:** Unsealed
- Feature type:** Carriageway (dropdown menu)
- Operation:** Regravelling/Resurfacing (dropdown menu)
- Intervention type:** ☐ Scheduled, ☒ Responsive

At the bottom of the dialog are four buttons: "OK", "Cancel", "Apply", and "Help". Below the dialog box, a status bar reads "The name of this works item".

■ Gravel Resurfacing/Design

The **Design** Tab page for Gravel Resurfacing gives material properties (particle size distribution and Plasticity Index) of the proposed surfacing material. In addition, the final gravel thickness (or increase in gravel thickness) and compaction method are specified. For this case study, a final gravel thickness of 150 mm is specified, using mechanical compaction.

The screenshot shows a software dialog box titled "Maintenance Works Item: Gravel Resurface at 50mm". It has five tabs: "General", "Design" (which is selected), "Intervention", "Costs", and "Effects".

Under the "Design" tab, there are three main sections:

- Material Properties:** This section contains a dropdown menu for "Surface Material" with the text "Select a surface material to obtain default material properties". Below this are input fields for "Max particle size" (21.9 mm), "Plasticity index" (10.1 %), "% passing 2.00mm sieve" (51.1 %), "% passing 0.425mm sieve" (41.6 %), and "% passing 0.075mm sieve" (25.5 %).
- Spot-regravelling:** This section has two radio buttons. The first is "Volume of material to add:" with a value of 0 m³/km/yr. The second is "Annual material loss to replace:" with a value of 100 %.
- Gravel Thickness:** This section has two radio buttons. The first is "Final gravel thickness:" with a value of 150 mm. The second is "Increase in gravel thickness:" with a value of 0 mm. To the right of these is a "Compaction method:" dropdown menu set to "Mechanical".

At the bottom of the dialog box are four buttons: "OK", "Cancel", "Apply", and "Help". Below the dialog box, there is a label "Surface material".

■ Gravel Resurfacing/Intervention

For this case study, the intention is to resurface only if the gravel thickness falls below 50 mm. This is reflected in the intervention criterion adopted. The implementation of resurfacing is subject to the user-defined limits indicated under the **Intervention** Tab, shown below:

Maintenance Works Item: Gravel Resurface at 50mm

General | Design | **Intervention** | Costs | Effects

Responsive Criteria

Gravel thickness <= 50 mm

Add New Criterion...
Delete
Edit...

Limits

Last year: 2017 year
Max. roughness: 30 IRI (m/km)
Max. quantity: 5000 m²/km/year

Interval: 3 year(s)
ADT: 0

Minimum: 3 Maximum: 99
Minimum: 0 Maximum: 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Note that, for gravel resurfacing, roughness could also be used as the intervention criterion (for example, if the roughness is 15 m/km IRI or greater). Both parameters (layer thickness and roughness) could be used together. If more than one criterion is used with a given works item, a logical AND is assumed, and displayed on-screen (for example, if gravel thickness falls below 50 mm AND roughness is 15 m/km IRI or greater).

If the user wishes to trigger gravel resurfacing based on either layer thickness OR roughness (logical OR), then two separate Works Items for gravel resurfacing would be set up under the same Maintenance Standard with the appropriate intervention criteria.

To summarise, examples of intervention options for responsive **gravel resurfacing** are:

Intervention is based on:

■ **Gravel Thickness only**

Works Item (gravel resurfacing) with one intervention criterion:

- ☐ Gravel thickness ≤ 50 mm

■ **Roughness only**

Works Item (gravel resurfacing) with one intervention criterion:

- ☐ Roughness ≥ 15 IRI

■ **Gravel Thickness AND Roughness**

Works Item (gravel resurfacing)) with two intervention criteria:

- ☐ Gravel Thickness ≤ 50 mm

AND

- ☐ Roughness ≥ 15 IRI

■ **Either Gravel Thickness OR Roughness**

Works Item (gravel resurfacing) with thickness criterion:

- ☐ Gravel Thickness ≤ 50 mm

Works Item (gravel resurfacing) with roughness criterion:

- ☐ Roughness ≥ 15 IRI

The **Intervention** Tab page also shows the limits on the use of gravel resurfacing:

☐ Last year

The last year in which gravel resurfacing would be considered, set at 2017, year 18 of the 20 year analysis period.

☐ Maximum roughness

Sets the maximum roughness at which gravel resurfacing would be undertaken (30 m/km IRI).

☐ Interval

The minimum interval between successive **resurfacings** has been specified as three years for this case study.

The user must also specify the maximum interval between successive resurfacing operations, above which resurfacing will not be considered. As resurfacing is the highest ranking maintenance operation available under this Maintenance Standard, a value of 99 years is used to avoid elimination by this limit.

☐ AADT

For this case study, no limits on AADT are imposed (that is, minimum and maximum values set at 0 and 100,000 respectively). In this case the minimum interval specified above will override.

■ **Gravel Resurfacing/Costs**

The unit cost assigned to **gravel resurfacing** (expressed as US Dollars per cubic metre) can be reviewed under the **Costs** Tab.

The screenshot shows a software window titled "Maintenance Works Item: Gravel Resurface at 50mm". It has five tabs: General, Design, Intervention, Costs (selected), and Effects. The Costs tab contains the following fields:

- Economic** and **Financial** columns for unit costs.
- Unit cost:** Economic field contains "25", Financial field contains "30". A dropdown menu shows "per m³".
- Unit Costs Of Preparatory Works:**
 - Spot Regravelling: Economic "12", Financial "14.4", unit "per m²".
 - Patching: Economic "0", Financial "0", unit "per m²".
 - Edge-repair: Economic "0", Financial "0", unit "per m²".
 - Crack sealing: Economic "0", Financial "0", unit "per m²".
- Drainage:**
 - Drainage maintenance cost factor: "1", with a range "0 < DMCF <= 1".

At the bottom are buttons for OK, Cancel, Apply, and Help. A status bar at the very bottom reads "Economic unit cost of works".

■ Gravel Resurfacing/Effects

Note the **Effects** Tab with this Works Item, which gives the initial roughness value immediately after the **gravel resurfacing** works are completed.

A user defined value of 3 m/km IRI is used for this case study.

The screenshot shows a dialog box titled "Maintenance Works Item: Gravel Resurface at 50mm". It has five tabs: General, Design, Intervention, Costs, and Effects. The "Effects" tab is selected. Inside the dialog, under "Condition After Operation", there are two radio buttons: "User defined:" and "Derived". The "User defined:" radio button is selected. Below it, there is a text box labeled "Initial roughness:" containing the value "2", followed by the text "IRI (m/km)". Below this, there is a button labeled "Edit Detailed Calibration...". At the bottom of the dialog, there are four buttons: "OK", "Cancel", "Apply", and "Help". A status bar at the very bottom of the dialog reads "Maintenance work item costs".

The model calibration factors relating to gravel loss can be reviewed under **Section/Details**. Details of the modelling logic for unsealed roads are given in Chapter D4 of the [Analytical Framework and Model Descriptions](#).

3.8 Project Alternative: With Project

This Project Alternative includes the following **Road Works Standards**:

- M **Maintenance Standard for gravel road prior to upgrading**
- I **Improvement Standard representing the upgrading works**
- M **Maintenance Standard for paved road after upgrading**

3.8.1 Maintenance Standard: Maintenance Before Upgrading

Prior to upgrading, the existing gravel road will be given routine maintenance in the form of grading every six months and spot regravelling (if the gravel thickness falls below 100 mm). The works are specified on the Maintenance Standard window.

Maintenance Standard: PROJ1: Mtce before upgrading

General

Name: PROJ1: Mtce before upgrading

Short code: BEFORE

Surface class: Unsealed

OK

Cancel

Work Items

Grading before upgrade	GRADE6
Spot Reg before upgrade	SPG100

Add New Work Item...

Copy Work Item

Delete Work Item

Edit...

The name of this maintenance standard

Works Item: Grading

The specification for Grading within the Maintenance Standard **Maintenance Before Upgrading** is similar to that described under the Standard **Gravel Road Maintenance**. The Tab pages under **General**, **Design**, **Intervention** and **Costs** are shown below:

- **Grading/General**

Maintenance Works Item: Grading before upgrade

General | Design | Intervention | Costs

Name: Grading before upgrade

Short code: GRADE6

Surface: Unsealed

Feature type: Carriageway

Operation: Grading

Intervention type: ☒ Scheduled ☐ Responsive

OK

Cancel

Apply

Help

The name of this works item

■ Grading/Design

Maintenance Works Item: Grading before upgrade

General Design Intervention Costs

Material Properties:

Surface Material: Select a surface material to obtain default material properties

Max particle size: 0 mm Plasticity index: 0 %

% passing 2.00mm sieve: 0 % % passing 0.425mm sieve: 0 % % passing 0.075mm sieve: 0 %

Spot-regraveling:

☒ Volume of material to add: 0 m³/km/yr ☐ Annual material loss to replace: 100 %

Gravel Thickness:

☐ Final gravel thickness: 25 mm ☒ Increase in gravel thickness: 0 mm

Compaction method: Mechanical

OK Cancel Apply Help

Compaction method

■ Grading/Design

Maintenance Works Item: Grading before upgrade

General Design Intervention Costs

Intervention criteria:

☒ Time interval: 180 Days

☐ Traffic interval: 100 vehicles

Limits:

Last year: 2003 year

Max. roughness: 30 IRI (m/km)

Max. quantity: 5000 m³/km/year

Interval: 180 days

ΔADT: 0

Minimum: 10000

Maximum: 100000

OK Cancel Apply Help

Scheduled intervention criteria parameters

■ Grading/Costs

The screenshot shows a software window titled "Maintenance Works Item: Grading before upgrade" with a close button (X) in the top right corner. The window has four tabs: "General", "Design", "Intervention", and "Costs", with "Costs" being the active tab. Inside the "Costs" tab, there are two columns: "Economic" and "Financial". Below these columns is a "Unit cost:" label followed by two input fields: the first contains "120" and the second contains "144". To the right of these fields is a dropdown menu showing "per km". Below this section is a group box titled "Unit Costs Of Preparatory Works" containing four rows of input fields: "Spot Regravelling", "Patching", "Edge-repair", and "Crack sealing". Each row has two input fields (one under "Economic", one under "Financial") and a unit label "per m²". All input fields in this group are currently set to "0". Below the group box is another group box titled "Drainage" containing a single input field labeled "Drainage maintenance cost factor" with the value "1" and a range indicator "0 < DMCF <= 1". At the bottom of the window are four buttons: "OK", "Cancel", "Apply", and "Help". A status bar at the very bottom of the window displays the text "Economic unit cost of works".

The intervention limits associated with the grading works are given on the **Intervention** screen, and summarised in Table A1.3. Note that the **Last Year** is specified as 2003, the year before the start of the upgrading works.

Works Item: Spot regravelling before upgrading

The specification for spot regravelling under the Maintenance Standard **Maintenance Before Upgrading** is similar to that described under the Maintenance Standard **Gravel Road Maintenance**. Details of the Tab pages (**General**, **Design**, **Intervention**, and **Costs**) are shown below:

■ Spot regravelling/General

Maintenance Works Item: Spot Reg before upgrade

General | Design | Intervention | Costs

Name:

Short code:

Surface: Unsealed

Feature type:

Operation:

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

■ Spot regravelling/Design

Maintenance Works Item: Grading before upgrade

General | Design | Intervention | Costs

Material Properties:

Surface Material:

Max particle size: mm Elasticity index: %

% passing 2.00mm sieve: % % passing 0.425mm sieve: % % passing 0.075mm sieve: %

Spot-regravelling:

☒ Volume of material to add: m³/km/yr ☐ Annual material loss to replace: %

Gravel Thickness:

☐ Final gravel thickness: mm ☐ Increase in gravel thickness: mm

Compaction method:

OK Cancel Apply Help

Compaction method

■ Spot regravelling/Intervention

Maintenance Works Item: Grading before upgrade

General | Design | **Intervention** | Costs

Intervention criteria

☒ Time interval: 180 Days

☐ Traffic interval: 100 vehicles

Limits

Last year: 2003 year

Max. roughness: 30 IRI (m/km)

Max. quantity: 5000 m²/km/year

Interval: 180 days

ΔADT: 0

Minimum: 180 days

Maximum: 10000 days

100000

OK Cancel Apply Help

Scheduled intervention criteria parameters

■ Spot regravelling/Costs

Maintenance Works Item: Grading before upgrade

General | Design | Intervention | **Costs**

Economic Financial

Unit cost: 120 144 per km

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 1 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

The intervention limits associated with the spot regravelling works are given on the **Intervention** Tab page, and summarised in Table A1.3.

3.8.2 Improvement Standards

The specification for an Improvement Standard includes details of the improvement type (upgrading in this case study), and geometric characteristics of the road section after improvement. The **geometry** data set may be assigned by selecting a geometry class (for example, **mostly straight and gently undulating**), as defined under [HDM-4 Configuration](#) (see Chapter D1).

For this case study, the geometric characteristics are significantly different for each road section, as are the pavement design details and duration of the upgrading works. The different Improvement Standards for the three sections are reviewed below. A comparison of selected section details, before and after upgrading, is given in Table A1.4.

Table A1.4
Section details before and after upgrading works

Section ID		B001-01	B001-02	B001-03
Section Name		Town A - Town B	Town B - Town C	Town C - Town D
Speed flow type	B	Two lane road	Two lane road	Two lane road
	A	Two lane road	Two lane road	Two lane road
Traffic flow pattern	B	Seasonal	Seasonal	Seasonal
	A	Seasonal	Seasonal	Seasonal
Road class	B	Secondary/Main	Secondary/Main	Secondary/Main
	A	Primary/Trunk	Primary/Trunk	Primary/Trunk
Surface class	B	Unsealed	Unsealed	Unsealed
	A	Bituminous	Bituminous	Bituminous
Pavement type	B	Gravel	Gravel	Gravel
	A	Surface Treatment on Granular Base	Surface Treatment on Granular Base	Surface Treatment on STabilised Base
Carriageway width (m)	B	7	6	8
	A	7	6	8
Speed limit (km/h)	B	100	80	100
	A	120	100	120
Upgraded SN (excluding subgrade)		1.84	1.61	2.62
Upgrade financial cost (US dollars/km)		280000	300000	360000

Notes:

B = Before upgrading

A =After upgrading

3.8.3 Improvement Standard: Pave Section B001-01 in 2004

The specification for this improvement can be reviewed under the Tabs: **General**, **Design**, **Intervention**, **Costs**, **Pavement**, **Geometry**, and **Effects**. The information needed for each Tab page is reviewed below.

■ General

The screenshot shows the 'General' tab of a dialog box titled 'Improvement Standard: Pave Sect B001-01 in 2004'. The dialog has tabs for General, Design, Intervention, Costs, Pavement, Geometry, and Effects. The 'General' tab is active, showing the following fields:

- Name: Pave Sect B001-01 in 2004
- Short code: Pave01
- Feature type: Carriageway (dropdown)
- Existing Surface class: Unsealed (dropdown)
- Improvement type: Upgrading (dropdown)
- Duration: 2 years
- Intervention type: ☒ Scheduled, ☐ Responsive

At the bottom are buttons for OK, Cancel, and Help. A status bar at the very bottom reads: 'The name of this improvement standard'.

Confirms the improvement type **upgrading**, with duration of two years, and **scheduled** intervention.

■ Design

The screenshot shows the 'Design' tab of the same dialog box. The 'New Attributes' section contains the following fields:

- Speed Flow Type: Two Lane Road (dropdown)
- Road class: Primary or Trunk (dropdown)
- Pavement type: Surface Treatment on Granular Base (dropdown)
- Length adjustment factor: 1
- Increase in width: 0 m
- Additional number of lanes: 0
- Proportion of new construction: 0.5, with the formula $0 < P_{conew} < 1$

There is an 'Edit Construction Details...' button next to the 'Increase in width' field. At the bottom are buttons for OK, Cancel, and Help. A status bar at the very bottom reads: 'The speed flow type of the section following improvement'.

Confirms **Pavement Type** as Surface Treatment on a Granular base (STGB), as shown in Figure A1.4. The intention is to replace the existing gravel surfacing with imported granular material (150 mm) as sub-base, add 200 mm granular roadbase, with a double surface dressing on top. The design parameters required by HDM-4 are entered under the **Pavement** Tab below.

The length adjustment factor is specified as 1.0 as no change in length is planned (that is, no realignment). Similarly no increase in width is proposed.

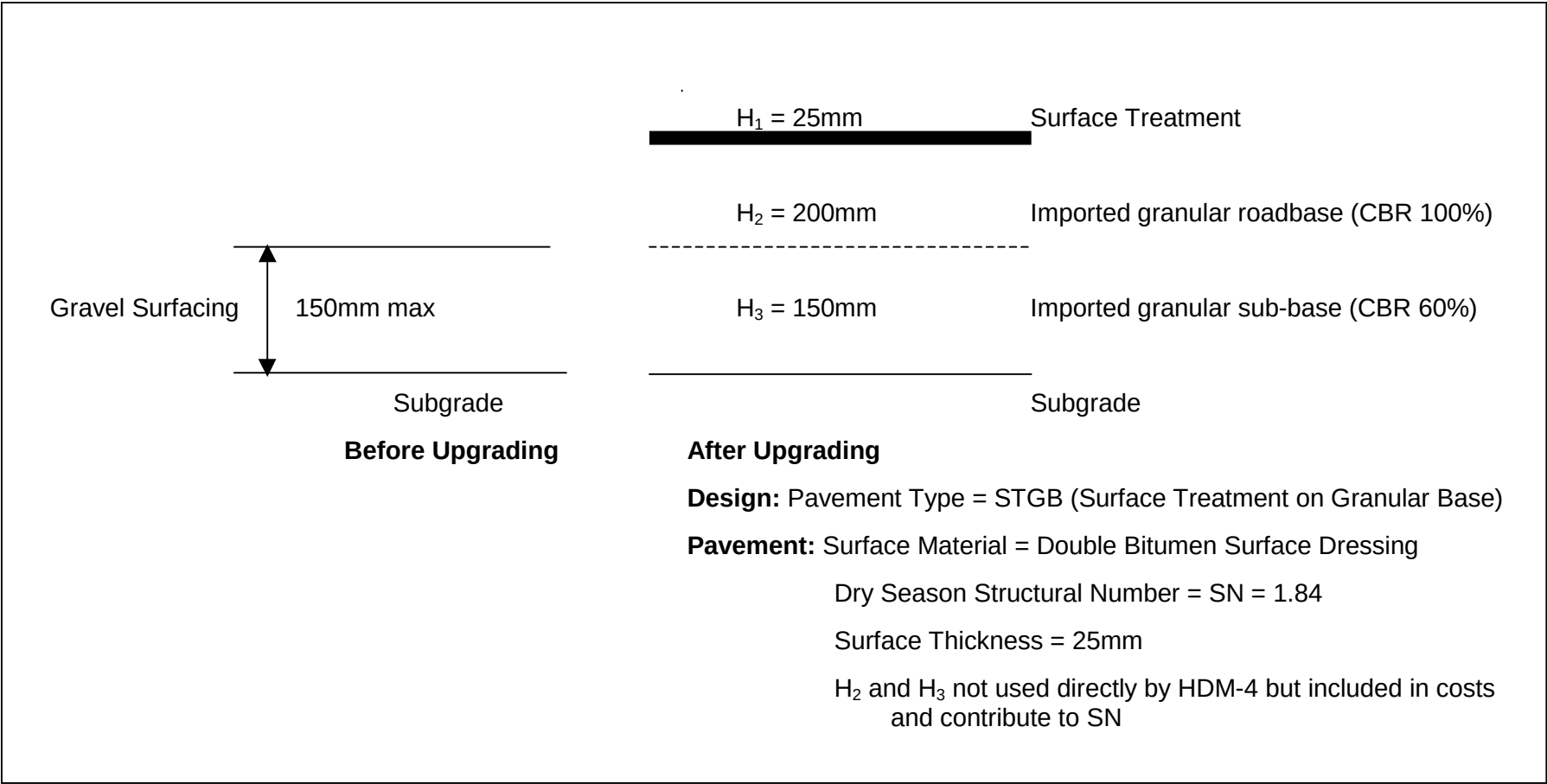


Figure A1.4 Case Study 1: Pavement design for Section B001-01 (Town A to Town B)

The Construction Quality Indicators (CDS and CDB) are specified under **Design/Edit Construction Details**. For this Improvement Standard, CDS = 1 and CDB = 0. Further details of these parameters are given in Chapters C2 and D2 of the [Analytical Framework and Model Descriptions](#).

■ Intervention

The screenshot shows a software dialog box titled "Improvement Standard: Pave Sect B001-01 in 2004". It has several tabs: General, Design, Intervention (selected), Costs, Pavement, Geometry, and Effects. Under the "Intervention" tab, there are two sections: "Scheduling" with a "Start in:" field set to "2004" followed by "year", and "Limits" with a "Last applicable year:" field set to "2004" followed by "year". At the bottom of the dialog are three buttons: "OK", "Cancel", and "Help". A status bar at the very bottom of the window contains the text "The year in which the improvement standard will commence".

This confirms the scheduled implementation date as the year 2004. The **Effective from** year assigned to the Improvement Standard (at project level) is also specified as 2004.

Note that the **Effective from** year could be specified differently to the implementation year. For example, if no maintenance was needed for the year prior to the start of improvement works, then the assignment (effective from) dates for standards would be:

M	Maintenance before upgrading	2000 (start of analysis period)
I	Pave Section B001-01 in 2004	2003

(Note: Improvement Standard effective from 2003, works implementation is 2004. The Improvement Standard takes precedence over the previous Maintenance Standard, so no works would be triggered in 2003).

M	Maintenance after upgrading	2006 (condition responsive)
----------	-----------------------------	-----------------------------

■ Costs

The screenshot shows the 'Costs' tab of a software interface for road improvement standards. The title bar reads 'Improvement Standard: Pave Sect B001-01 in 2004'. The interface has several tabs: General, Design, Intervention, Costs (selected), Pavement, Geometry, and Effects. Under the 'Costs' tab, there are two main sections: 'Economic' and 'Financial'. The 'Economic' section has a 'Unit cost' field set to 255000 and a unit dropdown set to 'per km'. The 'Financial' section has a 'Unit cost' field set to 300000. Below these, there is a table for 'Annual cost: stream' with five columns labeled 1 through 5, and a percentage column. The values are: 1: 50, 2: 50, 3: 33, 4: 0, 5: 0, and the percentage column is empty. Below this table is a section titled 'Unit Costs Of Preparatory Works' with a sub-table for 'Economic' and 'Financial' costs for four items: 'Re-surfacing of existing carriageway', 'Patching', 'Edge-repair', and 'Crack sealing'. All values in this sub-table are 0. At the bottom right are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads 'The economic unit cost of the improvement standard'.

The costs (expressed in US Dollars per km) are spread across the two-year construction period specified under the **General** Tab. 50% costs have been assigned to the year 2004, and 50% to 2005.

■ Pavement

This Tab page is used to specify details of the new pavement after upgrading.

The screenshot shows the 'Pavement' tab of the same software interface. The title bar and tabs are the same as in the previous screenshot. Under the 'Pavement' tab, there are several input fields: 'Surface material' is a dropdown menu set to 'Double Bitumen Surface Dressing'; 'Dry season Structural Number' is a text field set to 1.84; 'Surface thickness' is a text field set to 25 with a unit dropdown set to 'mm'; 'Relative compaction' is a text field set to 97 with a unit dropdown set to '%'. Below these is a section titled 'Road base (for stabilised base only)' with two sub-fields: 'Base thickness' set to 300 with a unit dropdown set to 'mm', and 'Resilient modulus' set to 15 with a unit dropdown set to 'GPa'. At the bottom right are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads 'Material for the new surface layer'.

- ❑ The proposed pavement design is included in Figure A1.4.
- ❑ Surface Material confirms the use of a double bitumen surface dressing. Note that the options given in the Surface Material **drop-down list** depend on the **Pavement Type** selected under the **Design** Tab (STGB for this road section).

- Dry Season Structural Number, SN, is derived for the pavement layers indicated in Figure A1.4 (that is, surface, granular base and sTabilised sub-base). Note that SN does **not** include a contribution from the subgrade.

For road section B001-01, the value of SN has been derived from layer thicknesses and coefficients as indicated below:

Layer	Layer type	Thickness (mm)	Layer coefficient
Surfacing	ST	25	0.2
Base	GB	200	0.14
Granular Sub-base	-	150	0.11

$$SN_s = SNBASU_s + SNSUBA_s = 1.84$$

where:

$$\begin{aligned} SNBASU_s &= \text{contribution of surfacing and base layers for season, } s \\ &= 0.0394 * [(25 * 0.2) + (200 * 0.14)] = 1.30 \end{aligned}$$

$$\begin{aligned} SNSUBA_s &= \text{contribution of sub-base for season, } s \\ &= 0.54 \text{ from Equation 3.3 in Chapter C2 of the } \a href="#">Analytical Framework and Model Descriptions \end{aligned}$$

Recommended values for layer coefficients are given in Chapters C2 and D2 of the [Analytical Framework and Model Descriptions](#), which also gives full details of the HDM-4 definitions relating to Structural Number.

Note that roadbase details are not requested as the proposed roadbase is not sTabilised.

- The surface thickness refers to the thickness of the new surfacing that is 25 mm for this road section (Double bitumen surface dressing).
- Relative compaction of pavement layers, taken as 97% on average.

■ Geometry

The screenshot shows the 'Geometry' tab of a software dialog box titled 'Improvement Standard: Pave Sect B001-01 in 2004'. The dialog has several tabs: General, Design, Intervention, Costs, Pavement, Geometry, and Effects. The 'Geometry' tab is active. It contains a 'Geometry Class' dropdown menu with the text 'Select a geometry class to obtain default geometry values'. Below this is a 'Details' section with two columns of input fields. The left column includes 'Rise + fall' (10 m/km), 'No. of rises + falls' (1 no./km), 'Avg horiz curvature' (50 deg/km), 'Superelevation' (3 %), 'Gradat' (0.1 m/s²), 'Speed limit' (120 km/h), and 'Speed limit enforcement' (1.1). The right column is titled 'Speed Reduction Factors' and includes 'NMT' (1, 0.6 ≤ XNMT ≤ 1), 'MT' (1, 0.6 ≤ XMT ≤ 1), and 'Road side friction' (1, 0.6 ≤ XFRI ≤ 1). At the bottom are 'OK', 'Cancel', and 'Help' buttons. A status bar at the very bottom reads 'Average road Rise plus Fall'.

In this case study, the effect of the upgrading works on **Geometry** is summarised in Table A1.4. There are no changes to the horizontal or vertical alignment. Note that the posted speed limit (after upgrading) is 120 km/h, compared with 100 km/h before upgrading.

These data were based on the data held for the gravel road prior to upgrading. Note that the geometry data **before upgrading** is held in the **Road Network** folder under **Section/Geometry** and **Section/Details/Alignment**.

■ Effects

The screenshot shows the 'Effects' tab of the same software dialog box. The 'Effects' tab is active. It contains a 'Condition After Works' section with a 'Road use' dropdown menu set to '<undefined>' and an 'Edit Detailed Calibration...' button. Below this are two sections, each with a checkbox and input fields. The first section, 'User defined Roughness And Rutting', has a checked checkbox and fields for 'Roughness' (2 IRI (m/km)) and 'Mean rut depth' (0 mm). The second section, 'User defined Surface Characteristics', also has a checked checkbox and fields for 'Skid resistance' (0.55 SCRIM (at 50km/h)) and 'Surface texture' (0.7 mm). At the bottom are 'OK', 'Cancel', and 'Help' buttons. A status bar at the very bottom reads 'Use user defined roughness and rutting values'.

The screen confirms that the road condition after the improvement works will be derived by the HDM-4 **Works Effects** model. The calibration factors can be reviewed under the

Edit Detailed Calibration button. Further details of calibration of **Road Works Effects** are given in Part D of the [Analytical Framework and Model Descriptions](#).

3.8.4 Improvement Standard: Pave Section B001-02 in 2004

The Improvement Standard for this section has a different pavement design (detailed under **Pavement** below) to that specified for Section B001-01. The construction period for Section B001-02 is one year. The specification for this improvement can be reviewed under the Tabs: **General**, **Design**, **Intervention**, **Costs**, **Pavement**, **Geometry**, and **Effects**. The information needed for each Tab page is reviewed below.

■ General

The screenshot shows a software dialog box titled "Improvement Standard: Pave Sect B001-02 in 2004". It has seven tabs: General, Design, Intervention, Costs, Pavement, Geometry, and Effects. The "General" tab is active. The fields are as follows:

- Name: Pave Sect B001-02 in 2004
- Short code: Pave02
- Feature type: Carriageway (dropdown)
- Existing Surface class: Unsealed (dropdown)
- Improvement type: Upgrading (dropdown)
- Duration: 1 years
- Intervention type: ☒ Scheduled, ☐ Responsive

At the bottom right are buttons for OK, Cancel, and Help. At the bottom left, there is a text label: "The name of this improvement standard".

Confirms the improvement type **upgrading**, with duration of one year, and **scheduled** intervention.

■ Design

The screenshot shows the 'Design' tab of the 'Improvement Standard: Pave Sect B001-02 in 2004' dialog box. The 'New Attributes' section contains the following fields:

- Speed Flow Type: Two Lane Road (dropdown)
- Road class: Primary or Trunk (dropdown)
- Pavement type: Surface Treatment on Granular Base (dropdown)
- Length adjustment factor: 1 (text box)
- Increase in width: 0 m (text box)
- Additional number of lanes: 0 (text box)
- Proportion of new construction: 0.5 (text box) with the note $0 < P_{conew} < 1$

There is an 'Edit Construction Details...' button next to the 'Increase in width' field. At the bottom right are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads: 'The speed flow type of the section following improvement'.

Confirms the Pavement Type after improvement as Surface Treatment on Granular Base (STGB). The pavement design is described under **Pavement** below. The factors CDS and CDB can be reviewed under **Design/Edit Construction Details**.

■ Intervention

The improvement has been scheduled to start in 2004.

■ Costs

The screenshot shows the 'Costs' tab of the 'Improvement Standard: Pave Sect B001-02 in 2004' dialog box. It features two columns: 'Economic' and 'Financial'.

Unit cost: Economic: 238000, Financial: 280000, per km (dropdown)

Annual cost stream: 1: 100, 2: 0, 3: 0, 4: 0, 5: 0 %

Unit Costs Of Preparatory Works:

	Economic	Financial	
Re-surfacing of existing carriageway:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

At the bottom right are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads: 'The economic unit cost of the improvement standard'.

The construction period for this section is one year, and so all works costs are allocated to the year 2004.

■ Pavement

This screen confirms details of the pavement after upgrading.

Improvement Standard: Pave Sect B001-02 in 2004

General Design Intervention Costs **Pavement** Geometry Effects

Surface material: Double Bitumen Surface Dressing

Dry season Structural Number: 1.61

Surface thickness: 25 mm

Relative compaction: 97 %

Road base (for stabilised base only)

Base thickness: 300 mm

Resilient modulus: 0.5 GPa

OK Cancel Help

Material for the new surface layer

The pavement design for the upgrading of Section B001-02 is given in Figure A1.5. For this road section, the intention is to remove the existing gravel surfacing to formation level, then stabilise the existing subgrade material (to a depth of 150 mm) below formation. This improved subgrade will be considered (for HDM-4 purposes) as a stabilised sub-base on top of which a granular roadbase layer (150 mm thick) will be placed. A double bitumen surface dressing (thickness 25 mm) will be applied as surfacing.

This design may be appropriate where the existing subgrade material is generally weak but is suitable (in terms of plasticity and particle size distribution) for strengthening by the addition of a lime or cement stabiliser. Note that stabilisation would normally be considered only if the cost of the process is less than the cost of replacing the existing weak material with imported suitable material.

The structural number, SN is derived for the pavement layers indicated in Figure A1.5 (that is, surface, granular base and stabilised sub-base). Note that SN does **not** include a contribution from the subgrade.

For road section B001-02, the value of SN has been derived from layer thicknesses and coefficients as indicated below:

Layer	Layer type	Thickness (mm)	Layer coefficient
Surfacing	ST	25	0.2
Base	GB	150	0.14
Stabilised sub-base	-	150	0.11

$$SN_s = SNBASU_s + SNSUBA_s = 1.61$$

where:

$$\begin{aligned} SNBASU_s &= \text{contribution of surfacing and base layers for season, } s \\ &= 0.0394 * [(25 * 0.2) + (150 * 0.14)] = 1.02 \end{aligned}$$

$$\begin{aligned} SNSUBA_s &= \text{contribution of sub-base for season, } s \\ &= 0.59 \text{ from Equation 3.3 in Chapter C2 of the } \text{Analytical Framework and Model Descriptions} \end{aligned}$$

Recommended values for layer coefficients are given in Chapters C2 and D2 of the [Analytical Framework and Model Descriptions](#), which also gives full details of the HDM-4 definitions relating to Structural Number.

■ Geometry

Improvement Standard: Pave Sect B001-02 in 2004

General | Design | Intervention | Costs | Pavement | **Geometry** | Effects

Geometry Class: Select a geometry class to obtain default geometry values

Details

Rise + fall:	15	m/km
No. of rises + falls:	1	no./km
Avg horiz curvature:	300	deg/km
Superelevation:	3	%
Gradat:	0.1	m/s ²
Speed limit:	100	km/h
Speed limit enforcement:	1.1	

Speed Reduction Factors

NMT:	1	0.6 <= XNMT <= 1
MT:	1	0.6 <= XMT <= 1
Road side friction:	1	0.6 <= XFRI <= 1

OK Cancel Help

Average road Rise plus Fall

The geometric characteristics of road section B001-02 after upgrading are shown on the **Geometry** Tab page. Note that the posted speed limit is changed from 80 km/h (before upgrading) to 100 km/h.

■ Effects

The condition after the upgrading works is derived by the HDM-4 **Works Effects** models.

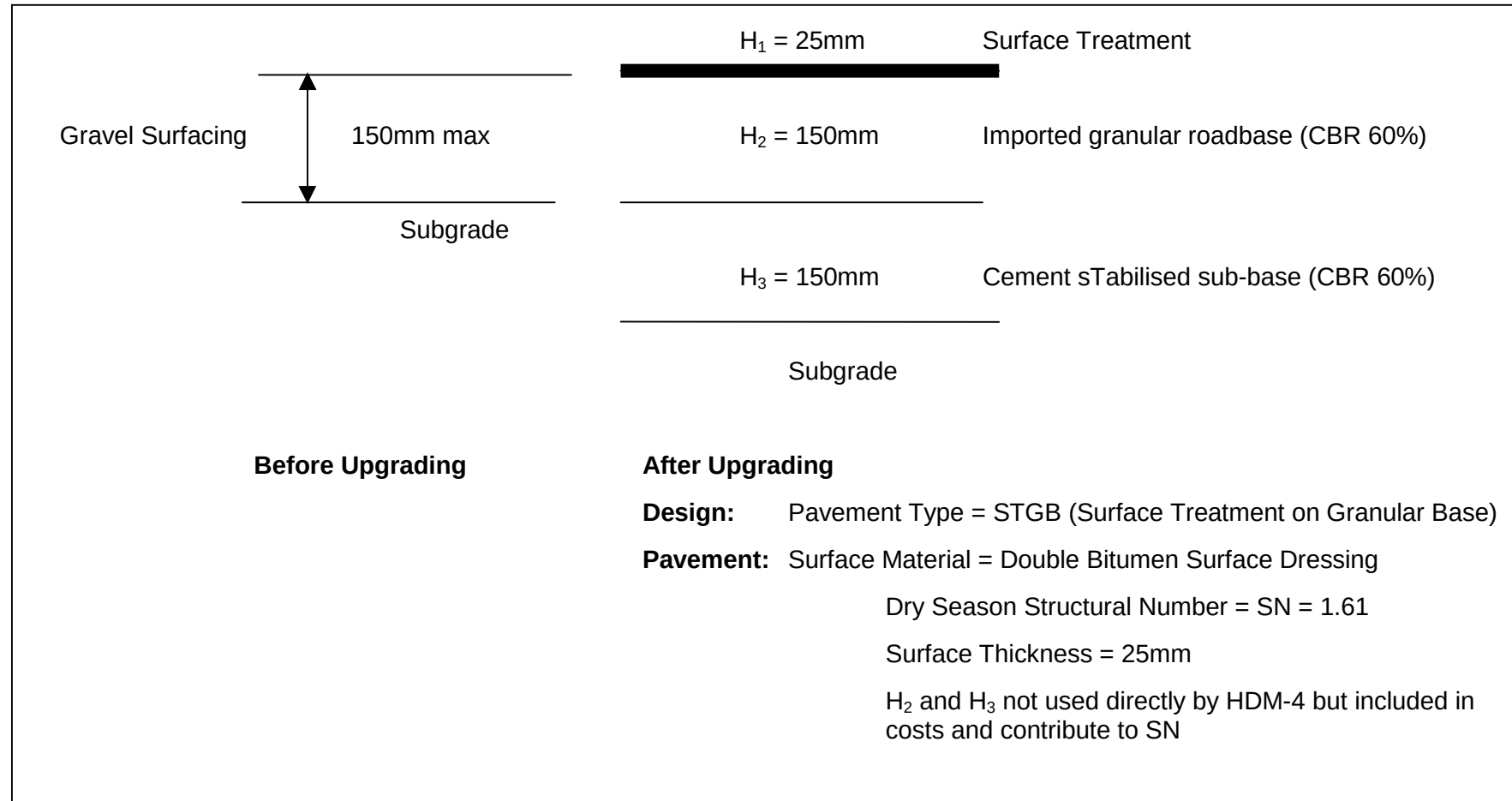


Figure A1.5 Case Study 1: Pavement Design for Section B001-02 (Town B to Town C)

3.8.5 Improvement Standard: Pave Section B001-03 in 2004

The Improvement Standard for this road section has a different pavement design (detailed under **Pavement** below) to the previous sections. The construction period for Section B001-03 is two years. The specification for this improvement can be reviewed under the Tabs: **General**, **Design**, **Intervention**, **Costs**, **Pavement**, **Geometry**, and **Effects**. The information needed for each Tab page is reviewed below.

- **General**

Name and **Short Code** refer to Section B001-03. The construction period for the proposed improvement works is two years.

- **Design**

The screenshot shows a software dialog box titled "Improvement Standard: Pave Sect B001-03 in 2004". It has several tabs: General, Design, Intervention, Costs, Pavement, Geometry, and Effects. The "Design" tab is currently selected. Inside the dialog, there is a section labeled "New Attributes" which contains the following fields and values:

- Speed Flow Type: Two Lane Road (dropdown menu)
- Road class: Primary or Trunk (dropdown menu)
- Pavement type: Surface Treatment on Stabilised Base (dropdown menu)
- Length adjustment factor: 1 (text input)
- Increase in width: 0 m (text input with unit)
- Additional number of lanes: 0 (text input)
- Proportion of new construction: 0.5 (text input)

Below the "Proportion of new construction" field, there is a formula: $0 < P_{conew} < 1$. To the right of the "Increase in width" field is a button labeled "Edit Construction Details...". At the bottom of the dialog are three buttons: "OK", "Cancel", and "Help". A status bar at the very bottom of the window reads "The speed flow type of the section following improvement".

Confirms the pavement type after improvement as Surface Treatment on STabilised Base (STSB). The pavement design is described under **Pavement** below.

- **Intervention**

Scheduled for 2004, as with the other road sections.

- **Costs**

As specified with 50% works costs incurred in each of the construction years 2004 and 2005.

■ Pavement

For this road section, the proposed pavement design comprises a 150 mm sTabilised sub-base placed on existing formation level with a 200 mm sTabilised road base and a double surface dressing (Figure A1.6).

For road section B001-03, the value of SN has been derived from layer thicknesses and coefficients as indicated below:

Layer	Layer type	Thickness (mm)	Layer coefficient
Surfacing	ST	25	0.2
sTabilised base	SB	200	0.22
sTabilised sub-base	-	150	0.14

$$SN_s = SNBASU_s + SNSUBA_s = 2.62$$

where:

$$\begin{aligned} SNBASU_s &= \text{contribution of surfacing and base layers for season, } s \\ &= 0.0394 * [(25 * 0.2) + (200 * 0.22)] = 1.93 \end{aligned}$$

$$\begin{aligned} SNSUBA_s &= \text{contribution of sub-base for season, } s \\ &= 0.69 \text{ from Equation 3.3 in Chapter C2 of the } \text{Analytical Framework and Model Descriptions} \end{aligned}$$

Recommended values for layer coefficients are given in Chapters C2 and D2 of the [Analytical Framework and Model Descriptions](#), which also gives full details of the HDM-4 definitions relating to Structural Number.

The layer coefficients are representative of the **as new** condition of the pavement. It is recognised that the sTabilised layers will develop cracking. Details of the deterioration models for sTabilised layers are given in the [Analytical Framework and Model Descriptions](#).

The proposed design includes a sTabilised sub-base and sTabilised base. As the roadbase is sTabilised, its thickness (200 mm) and Resilient Modulus (15 GPa) are requested. HDM-4 does not require these parameters for the sTabilised sub-base.

Note that use of a sTabilised roadbase gives a significant risk of reflection cracking through the surfacing. The risk is effectively reduced if a granular layer is placed above the sTabilised layer (for example, as for section B001-02 using a sTabilised sub-base and granular roadbase).

■ Geometry

Improvement Standard: Pave Sect B001-03 in 2004

General | Design | Intervention | Costs | Pavement | **Geometry** | Effects

Geometry Class: Select a geometry class to obtain default geometry values

Details

Rise + fall:	30	m/km	Speed Reduction Factors	
No. of rises + falls:	1	no./km	NMT:	1 0.6 ≤ XNMT ≤ 1
Avg horiz curvature:	15	deg/km	MT:	1 0.6 ≤ XMT ≤ 1
Superelevation:	3	%	Road side:	1 0.6 ≤ XFRI ≤ 1
σ _{adral} :	0.1	m/s ²	friction	
Speed limit:	120	km/h		
Speed limit enforcement:	1.1			

OK Cancel Help

Average road Rise plus Fall

The geometric characteristics of road section B001-03 after upgrading are shown on the **Geometry** Tab page. Note that the posted speed limit is changed from 100 km/h (before upgrading) to 120 km/h.

■ Effects

The condition after the upgrading works is derived by the HDM-4 **Works Effects** models.

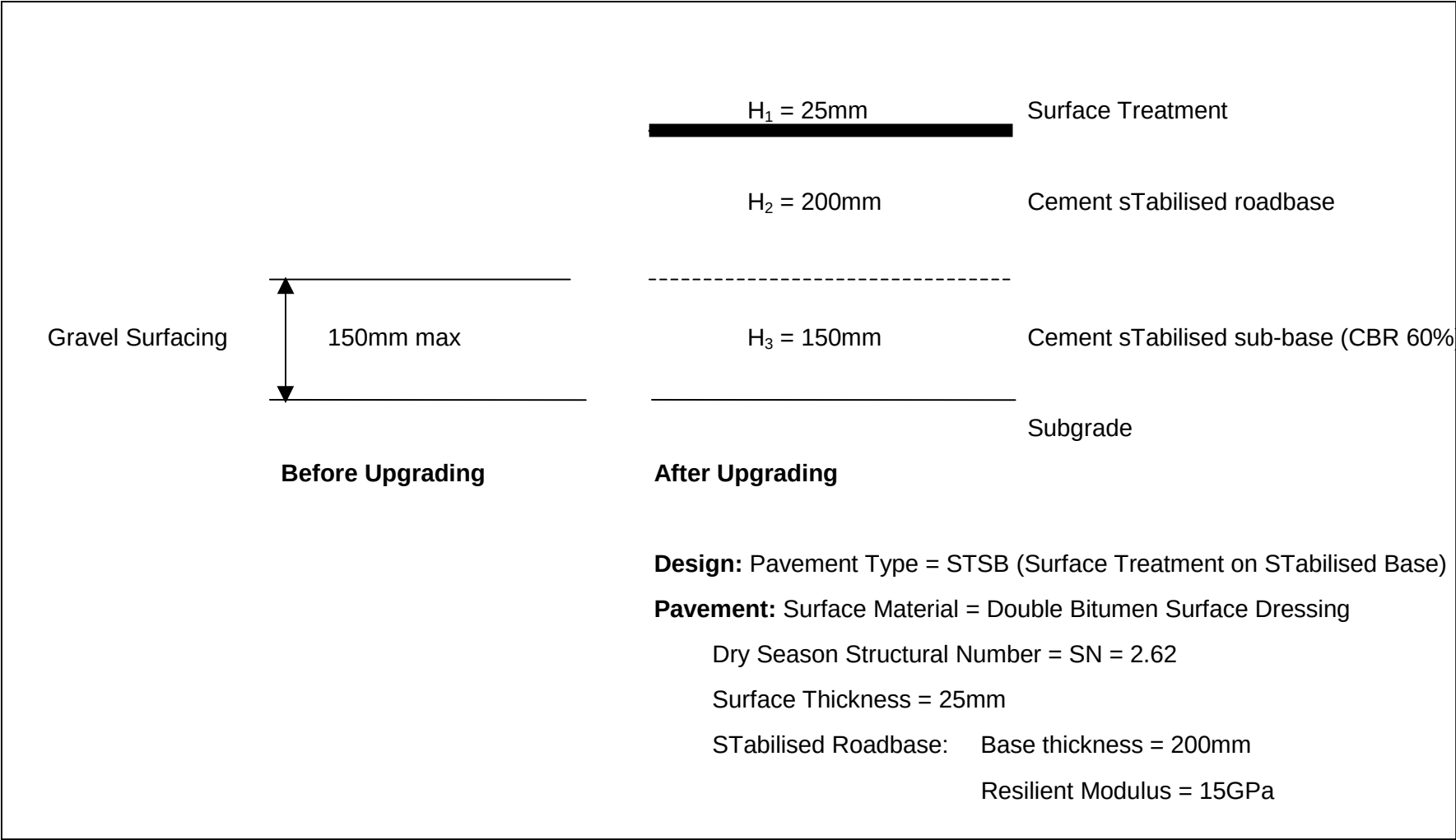


Figure A1.6 Case Study 1: Pavement design for Section B001-03 (Town C to Town D)

3.8.6 Maintenance Standard: Crack Sealing and Patching Paved Road

After upgrading, the paved road is to receive routine maintenance, defined under the Maintenance Standard **Crack sealing and patching Paved Road**. As summarised in Table A1.2, this includes two works items:

- 1 Crack Sealing if wide structural cracking reaches 5%
- 2 Patching if the severely damaged area reaches 5%

Both works are condition responsive (as defined above) and are effective from the year following the completion of the upgrading works (that is, effective from 2006 for sections B001-01 and B001-03, and 2005 for section B001-02.) The effective start date is specified at project level (under **Alternatives**).

The opening screen of this Maintenance Standard confirms the two work items **Crack Sealing** and **Patching**.

Maintenance Standard: PROJ1:Crack seal+Patch paved rd

General

Name: PROJ1:Crack seal+Patch paved rd

Short code: SEAPAT

Surface class: Bituminous

Work Items

Work Item	Code
Patching	PATCH
Crack Sealing	SEAL

Buttons: Add New Work Item..., Copy Work Item, Delete Work Item, Edit...

Buttons: OK, Cancel

The name of this maintenance standard

Works Item: Crack Sealing

The specification for Crack Sealing is held under four Tabs: **General**, **Intervention**, **Costs**, and **Effects**. The details required for each Tab page are reviewed below:

■ General

The screenshot shows the 'Maintenance Works Item: Crack Sealing' dialog box with the 'General' tab selected. The fields are as follows:

- Name: Crack Sealing
- Short code: SEAL
- Surface: Bituminous
- Feature type: Carriageway
- Operation: Crack Sealing
- Intervention type: ☐ Scheduled, ☒ Responsive

Buttons at the bottom: OK, Cancel, Apply, Help.

Footer text: The name of this works item

This Tab page confirms the Name and Code assigned to this **Works Item**. The works activity is specified as **crack sealing** and intervention type is responsive.

■ Intervention

The screenshot shows the 'Maintenance Works Item: Crack Sealing' dialog box with the 'Intervention' tab selected. The fields are as follows:

- Responsive Criteria: Wide Structural Cracking >= 5 %
- Buttons: Add New Criterion..., Delete, Edit...
- Limits:
 - Last year: 2019 year
 - Max. roughness: 12.5 IRI (m/km)
 - Max. quantity: 1500 m²/km/year
 - Interval: 1 year(s)
 - ΔADT: 0
 - Minimum: 1
 - Maximum: 99

Buttons at the bottom: OK, Cancel, Apply, Help.

Footer text: Add an intervention criterion to this improvement standard

The **Intervention** Tab page confirms the responsive criterion (Wide structural cracking affecting 5% or more of carriageway area over the section). The other criterion available is based on the number of transverse thermal cracks per km).

The user defined limits are:

Last Year	2019, the last year of the analysis period.
Maximum Roughness	12.5 m/km IRI as the road is now bituminous.
Maximum Quantity	The upper limit on crack sealing works, expressed in m ² /km/year. For this Works Item , the upper limit is set at 1500 m ² /km/year. This reflects the capacity of the road authority to undertake crack sealing.
AADT	Applicable range set to 0-100,000 to avoid elimination based on traffic volume.

■ Costs

The **Costs** Tab confirms the unit cost of the crack sealing works, (expressed in US Dollars per square metre).

■ Effects

The **Effects** Tab confirms the percentage of distress to be repaired in terms of **Transverse thermal cracking** and **Wide Structural Cracking** (that is, set at 100% for each). This is subject to the limits defined under **Intervention** above.

Works Item: Patching

The specification for **Patching** can be reviewed under the Tabs: **General**, **Intervention**, **Costs** and **Effects**. The details required for each Tab page are reviewed below:

■ General

Maintenance Works Item: Patching

General | Intervention | Costs | Effects

Name: Patching

Short code: PATCH

Surface: Bituminous

Feature type: Carriageway

Operation: Patching

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

This Tab page confirms the Name and Code assigned to this **Works Item**. The works activity is specified as **patching** and intervention type is responsive.

■ Intervention

The **Intervention** Tab page confirms the responsive criterion (when 5% of section carriageway area is severely damaged). Note that other criteria could be defined, based on potholing, ravelling, or wide structural cracking.

The user defined limits are:

Last Year	2019, the last year of the analysis period.
Maximum Roughness	12.5 m/km IRI as the road is now bituminous.
Maximum Quantity	The upper limit on patching works, expressed in m ² /km/year. For this works item , the upper limit is set at 1500 m ² /km/year. This reflects the capacity of the road authority to undertake patching.
AADT	Applicable range set to 0-100,000 to avoid elimination based on traffic volume.

■ Costs

The **Costs** Tab confirms the unit cost of the patching works, (expressed in US Dollars per square metre)

■ Effects

The screenshot shows a dialog box titled "Maintenance Works Item: Patching" with a close button (X) in the top right corner. The dialog has four tabs: "General", "Intervention", "Costs", and "Effects". The "Effects" tab is currently selected. Inside the "Effects" tab, there is a section titled "Percentage Of Distress Repaired". Below this title, there are four radio buttons, each followed by a text input field and a percentage sign (%). The first radio button, "Severely damaged area:", is selected, and its input field contains the value "100". The other three radio buttons, "Potholing only:", "Wide structural cracking only:", and "Ravelling only:", are not selected, and their input fields also contain the value "100". At the bottom of the dialog, there are four buttons: "OK", "Cancel", "Apply", and "Help". Below the buttons, there is a status bar that reads "Patch severely damaged area".

The **Effects** Tab confirms the percentage of distress to be repaired in terms of **Potholing** (set at 100%). This is subject to the limits defined under **Intervention** above.

Note that the user may select one of the radio buttons shown. For this case study, the responsive criterion is based on the number of potholes per kilometre, and this is intended to trigger pothole patching.

Further details of road works effects are given in Chapter D2 of the [Analytical Framework and Model Descriptions](#).

4 Run HDM-4 and examine the results

The analysis setup is defined under the **Setup Run** Tab.

4.1 Setup Run

The **Setup Run** Tab is accessed via the **Analyse Projects** button.

This screen confirms the base project alternative for economic analysis (maintain gravel road) and the discount rate (6%).

The costs and benefits of the **Upgrade gravel road** alternative will be compared with those for the **Maintain gravel road** alternative, as defined under **Specify Alternatives/Alternatives** (Section 3.5). Note that the **HDM-4 Workspace** may hold data for any number of projects with associated network, fleet and work standards data. The run that has been set up will consider only those project alternatives defined under 'Alternatives', with the specified selected sections and vehicles.

For this case study, accident costs, energy balance analysis, emissions calculations, and acceleration effects are not included in the analysis.

4.2 Run Analysis

When **Start** is selected, the analysis commences and produces the output necessary for report generation. Any serious errors or omissions in the input data will stop the analysis, and an appropriate error message given. Otherwise, missing data (for example, certain costs information) may be advised as a **Warning** message on screen.

A status bar at the bottom of the screen indicates progress and the current analysis stage. When the analysis is finished, the dialogue stops scrolling and reports **Analysis completed successfully**. The status message will read:

Analysis Stage: Completed - total analysis time = hh : mm : ss

The user may then **select Generate Reports** to display the folders holding pre-defined report options. The pre-defined report categories available with the current program release are:

- **Deterioration/Work Effects**
- **Road User Effects**
- **Environmental Effects**
- **Cost Streams**

The full list of standard reports is displayed under Project/Generate Reports.

Note that reports generally refer to data presented in Tabular form. Certain reports are also available in graphical form (Charts). In addition, users may set up customised reports using a report writer facility.

The results of the HDM-4 run for Case Study 1 are examined by reference to the reports and (where available) the associated charts produced under the **Generate Reports** button:

- **Deterioration/Works Effects**
- **Cost Streams**

4.3 Deterioration and works effects

The **Timing of Works** report allows the user to check the works that would be implemented in each analysis year for each project alternative. Two variants are available:

1 Timing of Works (by year)

For each project alternative, this report lists, by analysis year, the works description, quantity and costs for each road section. A summary of total annual economic costs is provided for each project alternative.

2 Timing of Works (by section)

This report lists similar details by section together with the summary of total annual economic costs.

For this case study, the Timing of Works (by section) report is included in the Reports section at the end of this chapter.

Considering the **Without Project** alternative, the report indicates that, at Section B001-01 (Town A-Town B), gravel resurfacing would be implemented in the years 2003, 2007, 2010, 2013 and 2016 (with associated preparatory spot regraveling). Spot regraveling would be implemented every year except the first year and each year immediately following gravel resurfacing. Grading is implemented every six months as scheduled. These works are specified under the Maintenance Standard **Gravel Road Maintenance** described previously.

The **With Project** alternative for the same road section (B001-01) confirms that the upgrade to paved standard would be implemented during 2004 and 2005. Before upgrading, the existing road is graded at six monthly intervals between 2000 and 2003 with spot regraveling. After upgrading, the paved road has no works specified until 2016, when patching is listed each year between 2016 and 2019 (end of analysis period). These works are specified under the three works standards:

-  **Maintenance before upgrading**
-  **Pave Section B001-01 in 2004**
-  **Crack sealing and patching paved road**

The effect of these works on roughness is indicated in the Deterioration report and associated chart showing roughness at the end of the year.

Case study 1: Plot of Roughness vs Time for 1 section (B001-01)

insert 1 page plot of roughness for each project alternative.

This plot shows the roughness progression during the analysis period for each project alternative. Comparison with the Timing of Works (by section) report indicates that, before upgrading, the regravelling works (listed as gravel resurfacing) have a significant impact on roughness. Regravelling has been triggered when the roughness reaches a level of about 16 or 17 IRI. Note that intervention (regravelling) was requested if the thickness of gravel surfacing was reduced to 50 mm.

After upgrading section B001-01, the roughness increases less dramatically. Patching is triggered between 2016 to 2019, towards the end of the analysis period.

Note that the Improvement Standard specification included the option to define the effects of the upgrading works in terms of roughness, mean rut depth, skid resistance and surface texture (under the **Effects** Tab). For this case study, these values are derived by the HDM-4 Works Effects model. An overview of the HDM-4 modelling logic is given in Chapter A1; full details are given in Chapter D4 of the [Analytical Framework and Model Descriptions](#).

4.4 Cost streams

The economic analysis reports are accessed via **Generate Reports/Cost Streams**.

The Economic Analysis Summary (By Alternative) report gives a summary of costs, discounted Net Present Value (NPV) and Internal Rate of Return (IRR) by project alternative. Cost and NPV details are presented by road section in the Economic Analysis Summary (By Section) report.

For this case study, the overall NPV is reported as 1.51 (millions of US Dollars). The breakdown by section indicates that two sections give a positive NPV.

Section ID	Section description	Discounted NPV (millions of US Dollars)
B001-01	Town A to Town B	0.62
B001-02	Town B to Town C	- 0.57
B001-03	Town C to Town D	1.46
All sections		1.51

Note that the traffic levels are lowest for Section B001-02 (negative NPV), also that maximum benefits are derived for Section B001-03 which has highest traffic flow. A revised project comprising sections B001-01 and B001-03 only would give an overall NPV of 2.08 (millions of US Dollars).

Appendix A2 - Project Analysis Case Study 2

1 Optimum Rehabilitation Standards for a Paved Road

This case study presents the economic analysis of alternative rehabilitation standards for a 50 km long paved road. The existing road carries 8000 vehicles per day (AADT) and exhibits significant levels of roughness and surface distress. This case study evaluates several possible rehabilitation alternatives including overlay, mill and replace, and inlay.

The objective of the case study is to present the definition of section alternatives appropriate to this type of analysis and demonstrate the HDM-4 deterioration model for paved roads. The definition and timing of the Maintenance [Road Works Standards](#) are discussed, and the results are examined.

To demonstrate this case study the following steps are followed:

- **Locate the case study data** (see Section 1.1)
- **Review the case study input data** (see Section 1.2)
- **Run HDM-4 and examine the results** (see Section 1.3)

1.1 Locate the case study data

The case study data are included in the default database installed with the HDM-4 software. The data for this case study are located in the **Projects folder** in the [Case Studies Workspace](#). The name of the case study is **Case Study 2 Rehabilitation of paved roads (by section)**.

To locate the data for this case study:

- Go to the [Case Studies Workspace](#)
- Open the **Projects folder**

Double-click on the case study named **Case Study 2a. Rehabilitation of paved roads (by section)**

1.2 Review the case study input data

The data can be reviewed under the HDM-4 **Project** tabs listed below:

- **Define Project Details**

The following Tab pages may be displayed:

- ☐ General (see Section 1.2.1)
- ☐ Select Sections (see Section 1.2.2)
- ☐ Select Vehicles (see Section 1.2.3)
- ☐ Define Normal Traffic (see Section 1.2.4)

■ Specify Alternatives

The following Tab pages may be displayed:

- ❑ Alternatives (see Section 1.2.5)

■ Analyse Projects

The following Tab pages may be displayed:

- ❑ Setup Run (see Section 1.3.1)
- ❑ Run Analysis (see Section 1.3.2)

1.2.1 General

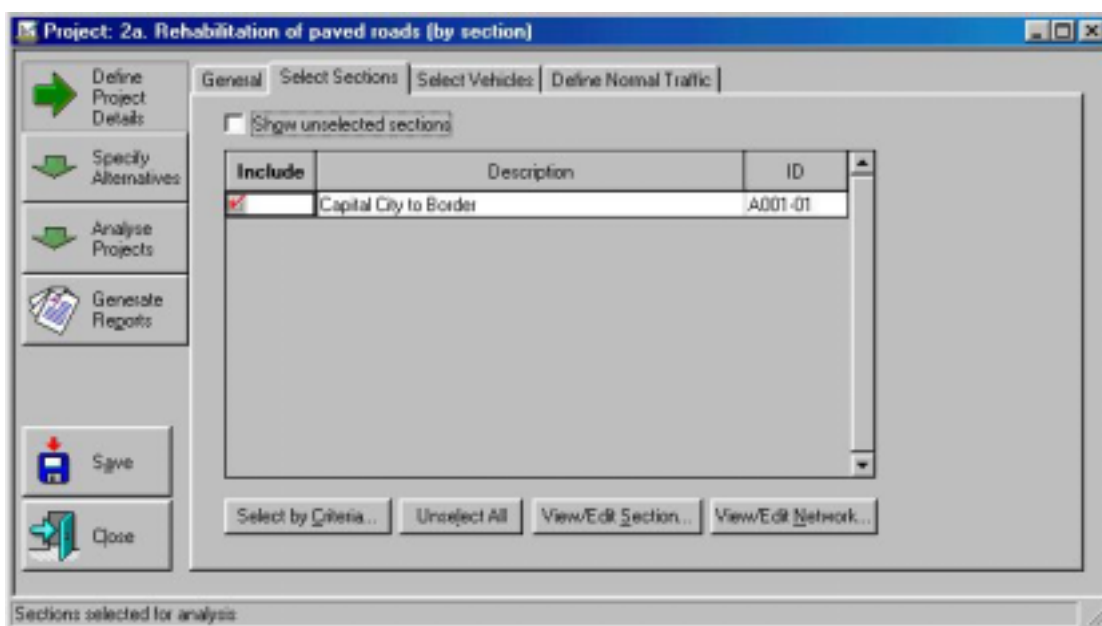
This Tab page confirms the project description, analysis type, analysis period and the pre-defined **Road Network** and **Vehicle Fleet**.

This case study is presented as a section analysis. The road under study is represented by one section, and the different rehabilitation proposals represent section alternatives. The road section will be selected from the **Northern Province** network, stored in the **Road Network** folder. The vehicles using the road will be selected from the Northern Province vehicle fleet, stored in the **Vehicle Fleet** folder.

It is also possible to conduct the analysis by **Project** although we have only one road section. The analysis by **Project** option would combine the results of selected sections into one **Project**.

The analysis period is defined by a start year 2000, and a duration 20 years, (that is, 2000 - 2019).

1.2.2 Select Sections



This Tab page indicates that one section will be included in the analysis. By double-clicking on the Section Description reveals the **Definition/Geometry/Pavement/Condition** Tabs where the section details are held.

■ Definition

The **Definition** Tab gives details of basic section characteristics including road class, speed-flow type and traffic flow pattern. The Pavement Type has been specified as **Surface Treatment on Asphalt Pavement** (STAP). It is important that the correct Pavement Type is specified, as the deterioration relationships held in HDM-4 are defined by Pavement Type (see Table A2.1).

Details of the existing pavement are discussed under **Pavement** below:

■ Geometry

Section: Capital City to Border

Definition | **Geometry** | Pavement | Condition

Rise + Fall: m/km

Avg horiz curvature: deg/km

Speed limit: km/h

Altitude: m

Drain type:

Average road Rise plus Fall

Calibration Parameters

Alignment | Drainage, Shoulders, and NMT Lanes | History | Surface Distress | Surface Texture | Structural Defects

No. of rises + falls: no./km

Superelevation: %

σ_{adral} : m/s²

Speed limit enforcement:

Speed Reduction Factors

NMT: $0.6 \leq X_{\text{NMT}} \leq 1$

MT: $0.6 \leq X_{\text{MT}} \leq 1$

Road side friction: $0.6 \leq X_{\text{FRI}} \leq 1$

Average number of Rises and Falls per km

Details of the road section's geometry, including horizontal and vertical alignment, are held under **Section/Geometry** and **Section/Details/Alignment**. The data indicates that the road section under study is in rolling terrain (Rise + Fall = 20 m/km) and subject to a speed limit of 100 kph.

Alignment data can be assessed from the existing mapping, if available, or measured by a topographical survey.

■ Pavement

The existing pavement construction (Figure A2.1) consists of an asphaltic concrete surfacing (total thickness 125 mm), over a 200 mm thick granular roadbase and 150 mm thick granular sub-base. The in situ subgrade CBR is 8%. The pavement details required by HDM-4 for this pavement type (Surface Treatment on Asphalt Pavement) are indicated on the **Pavement** Tab page below:

Note that the specification of Pavement Type (on the **Section/Definition** page) refers to the current pavement construction and should be compatible with the **Previous Works** details specified on the **Pavement** Tab page.

The Pavement Type is automatically updated immediately after any maintenance works. The definition of bituminous Pavement Types based on surface and base types is given in Table A2.1. (Table A2.2 contains descriptions of surface and base materials.) A summary of Pavement Type resets after maintenance works is given in Table A2.3.

For the road section under study, the changes in Pavement Type since the date of the last construction (1988) are summarised in Figure A2.1.

The last reconstruction was 1988, given by pavement type AMGB. After the application of overlay in 1992, Table A2.3 shows that:

$$\text{AMGB} + \text{Overlay} = \text{AMAP} \quad \dots(1.1)$$

After surface dressing (reseal) in 1995 (last resurfacing), Table A2.3 gives:

$$\text{AMAP} + \text{Surface Dressing} = \text{STAP} \quad \dots(1.2)$$

(specified for this case study under Definition)

Note that within HDM-4, the dates of the Previous Works shown under Section/Pavement are recognised as Age1, Age2, Age3 and Age4 as summarised below:

Age4 = Date of last reconstruction or new construction = 1988

Age3 = Date of last rehabilitation (overlay) = 1992

Age2 = Date of last resurfacing = 1995

Age1 = Date of last preventative treatment = 1995

Note that the dates refer to HDM-4 Works Types. The classification of Works Activities, by Work Type, is summarised in Table A2.4.

The dates Age2, Age3 and Age4 are used as a base line for the timing of scheduled treatments (Age2 for surface treatments, Age3 for rehabilitation, Age4 for reconstruction). For example, if surface dressing was requested every 4 years within a Maintenance Standard effective from the year 2000, the first possible application would be done immediately based on:

$$\text{Date of last resurfacing (1995) + 4} < 2000 \quad \dots(1.3)$$

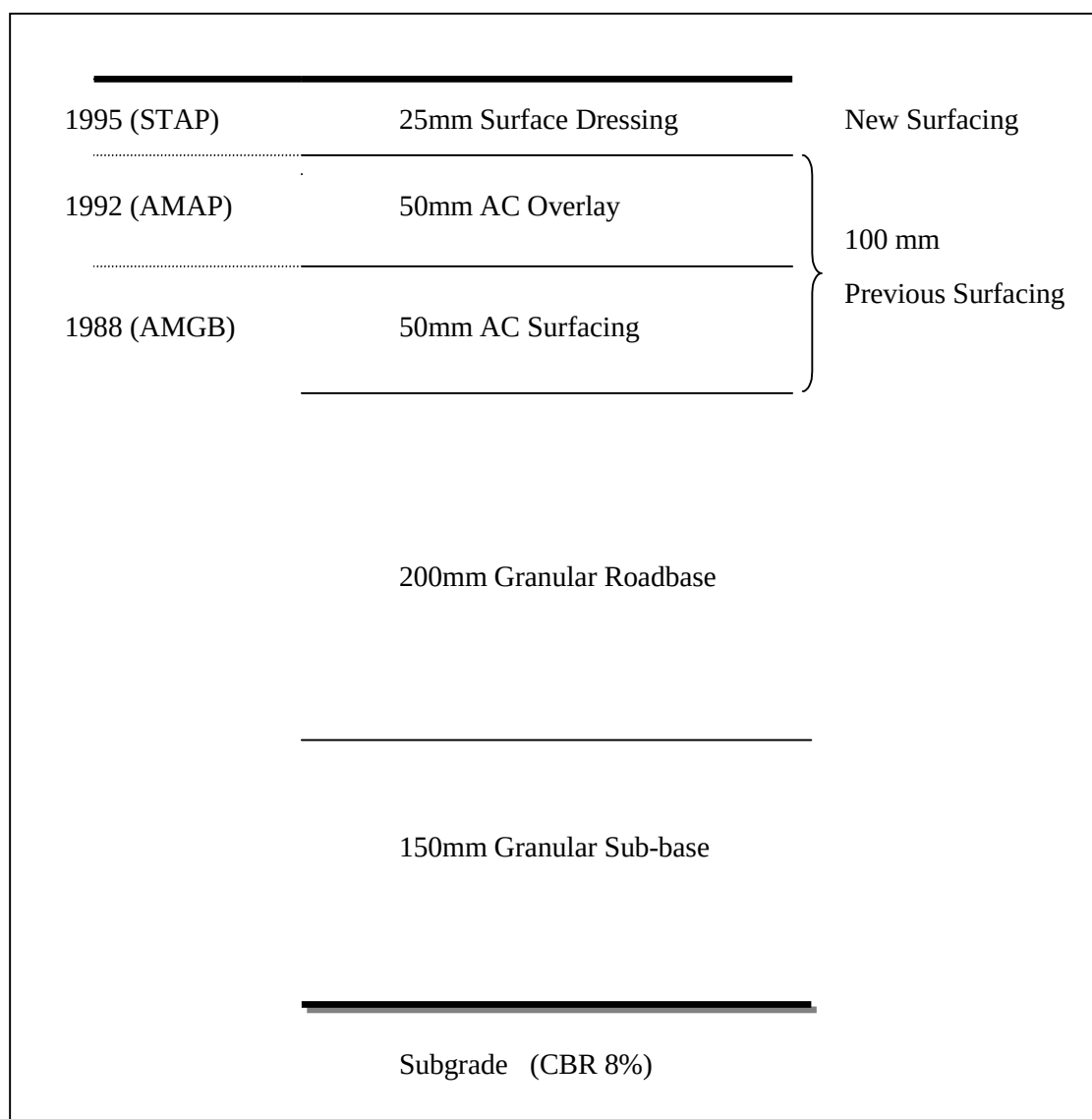


Figure A2.1 Case Study 2: Details of existing pavements and pavement type resets

Table A2.1
Generic HDM-4 bituminous pavement types

Surface type	Surface material	Base type	Base material	Pavement type
AM	AC	GB	CRS	AMGB
	HRA		GM	
	PMA	AB	AB	AMAB
	RAC	SB	CS	AMSB
	CM		LS	
	PA	AP	TNA	AMAP
	SMA		FDA	
	xx			
ST	CAPE	GB	CRS	STGB
	DBSD		GM	
	SBSD	AB	AB	STAB
	SL	SB	CS	STSB
	PM		LS	
	xx	AP	TNA	STAP
			FDA	

Note: AM and ST surfacings on concrete pavements, that is, AMCP and STCP, are modelled in HDM-4 as concrete pavement types in the rigid pavement sub-model.

The abbreviations in Table A2.1 are described in Table A2.2.

Table A2.2
Descriptions of surface and base materials

Surface type		Surface materials	
Abbreviation	Description	Abbreviation	Description
AM	Asphalt Mix	AC	Asphaltic Concrete
		CM	Soft Bitumen Mix (Cold Mix)
		HRA	Hot Rolled Asphalt
		PA	Porous Asphalt
		PMA	Polymer Modified Asphalt
		RAC	Rubberised Asphalt Concrete
		SMA	Stone Mastic
ST	Surface Treatment	CAPE	Cape Seal
		DBSD	Double Bituminous Surface Dressing
		PM	Penetration Macadam
		SBSD	Single Bituminous Surface Dressing
		SL	Slurry Seal
Base types		Base materials	
Abbreviation	Description	Abbreviation	Description
AB	Asphalt Base	CRS	Crushed Stone
AP	Asphalt Pavement	GM	Natural Gravel
GB	Granular Base	CS	Cement Stabilisation
SB	Stabilised Base	LS	Lime Stabilisation
		TNA	Thin Asphalt Surfacing
		FDA	Full Depth Asphalt

Table A2.3
Pavement type resets after maintenance works

Works activity	Existing pavement type							
	AMGB	AMSB	AMAB	AMAP	STGB	STSB	STAB	STAP
Routine works	AMGB	AMSB	AMAB	AMAP	STGB	STSB	STAB	STAP
Preventive Treatment	AMGB	AMSB	AMAB	AMAP	STGB	STSB	STAB	STAP
Reseal	STAP	STAP / STSB ¹	STAP	STAP	STGB	STSB	STAB	STAP
Overlay	AMAP	AMAP / AMSB ¹	AMAP	AMAP	AMGB	AMSB	AMAB	AMAP
Inlay	AMGB	AMSB	AMAB	AMAP	STGB	STSB	STAB	STAP
Mill & replace to intermediate surface layer	**AP	**AP	**AP	**AP	N/A	**SB	**AB	**AP
Mill & replace to base	**GB	**SB	**AB	**AP	**GB	**SB	**AB	**AP

Source: *NDLI (1995)*

Notes:

- 1 The pavement type depends on the critical thickness (Hmin) of the existing bituminous surfacing that is user-definable in [HDM Configuration](#).
- ** Indicates that these two characters are dependent on the specific works activity (or operation) and the surface material.
- N/A Not applicable.

The **Pavement** Tab page shows the current pavement strength, defined by the Adjusted Structural Number of the pavement (Adjusted SNP). The Adjusted SNP incorporates a weighting factor that reduces the contribution from the sub-base and subgrade, as described in the [Analytical Framework and Model Descriptions](#). (Note that the Adjusted SNP is not the same as the Modified Structural Number used in HDM-III).

The following HDM-4 conventions should be noted:

- 1 SNP refers to the full structural number including contributions from the sub-base and subgrade.
- 2 SN refers to the structural number comprising contributions from the surfacing, roadbase and sub-base only.

For this case study, the Structural Number (SN) was entered directly together with the subgrade CBR using Option (1) on the Section/Pavement screen. By clicking Option 1, the calculated SNP value is displayed at the top of the screen.

Table A2.4
Ranking of road works applicable to the carriageway

Works type	Works activity / operation	Ranking	Unit cost
New section	Dualisation of an existing section	1	per km
Upgrading	Upgrading to a new surface class	2	per km
Realignment	Geometric realignment	3	per km
Widening	Lane addition	4	per m ² or per km
	Partial widening	5	per m ² or per km
Reconstruction	Pavement reconstruction	6	per m ² or per km
Rehabilitation	Mill and replace	7	per m ²
	Overlay rubberised asphalt	8	per m ²
	Overlay dense-graded asphalt	9	per m ²
	Overlay open-graded asphalt	10	per m ²
	Inlay	11	per m ²
	Thin overlay	12	per m ²
Resurfacing (Resealing)	Cape seal with shape correction	13	per m ²
	Cape seal	14	per m ²
	Double surface dressing with shape correction	15	per m ²
	Double surface dressing	16	per m ²
	Single surface dressing with shape correction	17	per m ²
	Single surface dressing	18	per m ²
	Slurry seal	19	per m ²
Preventive Treatment	Fog sealing	20	per m ²
	Rejuvenation	21	per m ²
Routine Pavement	Edge-repair ¹	22	per m ²
	Patching ¹	22	per m ²
	Crack sealing ¹	22	per m ²

Note that Option (2) allows the user to calculate the SNP value with the help of the **SNP Calculation Wizard**, which can calculate SNP based on deflections or layer thickness and coefficients.

■ **Condition**

The condition of the road in 1998 can be reviewed under the **Condition** Tab page.

Section: Capital City to Border

Definition | Geometry | Pavement | **Condition**

Condition at end of year	1998
Roughness (IRI - m/km)	6.00
Total area of cracking (%)	10.00
Ravelled area (%)	10.00
Number of Potholes (No./km)	2.00
Edge break area (m ² /km)	10.00
Mean rut depth (mm)	5.00
Texture depth (mm)	0.30
Skid resistance (SCRIM 50km/h)	0.40
Drainage	Excellent

Add a column for a new year of condition data

For project analysis, this data should preferably be collected as part of a detailed condition survey. However, users can set up aggregate data specifying detailed values under **HDM Configuration**. Default values are shown in Chapter D2.

1.2.3 Select Vehicles

This Tab page confirms the selection of vehicles (from the pre-defined **Northern Province** fleet. Individual vehicle attributes can be reviewed by double-clicking on the appropriate vehicle type description.

Project: Case Study 2: analysis by project

General | Select Sections | **Select Vehicles** | Define Normal Traffic

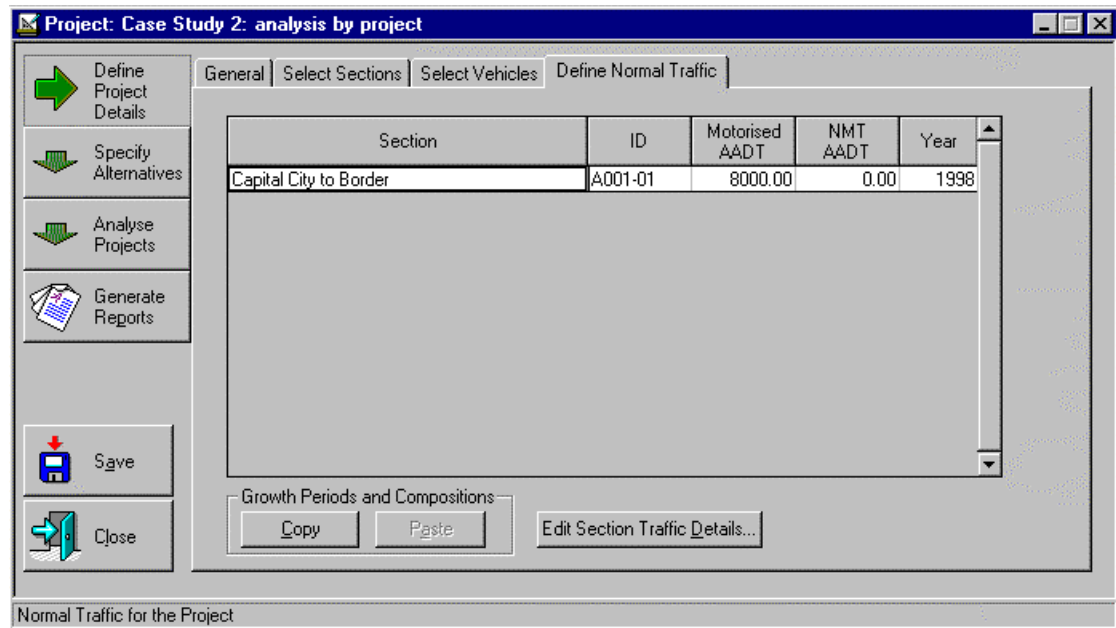
☐ Show unselected vehicles
 ☐ Include NMT in analysis

Include	Vehicle	Category
☒	Artic Truck	Motorised
☒	Heavy truck	Motorised
☒	Light Goods Vehicle	Motorised
☒	Light Truck	Motorised
☒	Medium Bus	Motorised
☒	Medium Car	Motorised
☒	Medium Truck	Motorised

Show/hide any unselected vehicles in the selected fleet

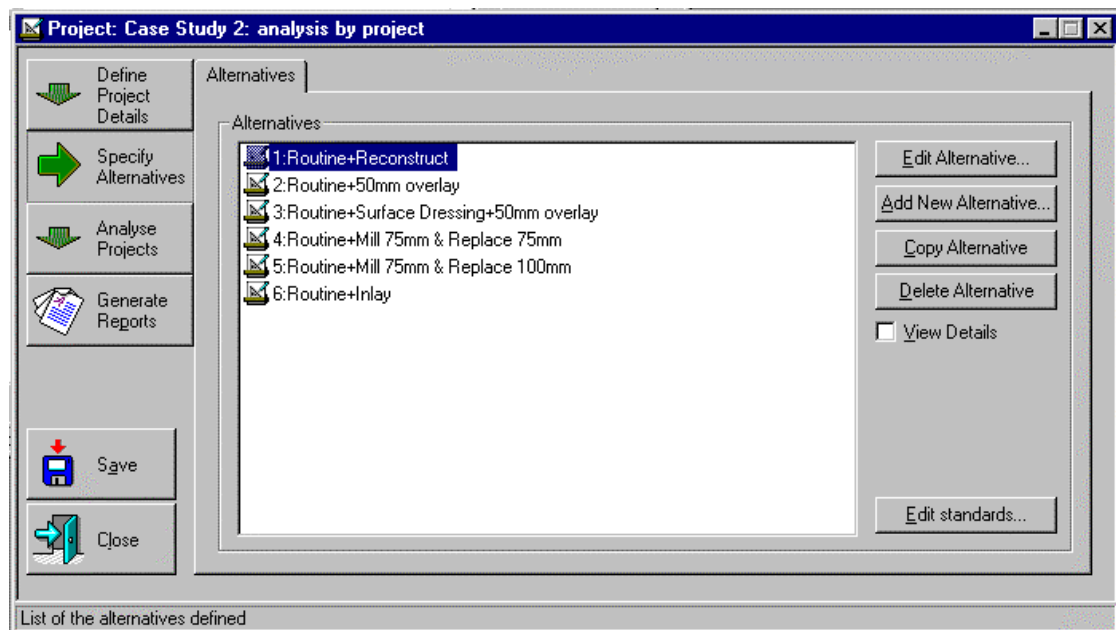
1.2.4 Define Normal Traffic

This Tab page confirms the volume of traffic (Motorised and Non-motorised AADT) using the selected road section in the given year. The initial traffic composition and growth rates (by vehicle type) can be reviewed by double-clicking the appropriate line.



1.2.5 Alternatives

The **Alternatives** Tab is displayed after selecting the **Specify Alternatives** button. The **Alternatives** screen is split into two boxes. The upper box shows the names of the six section alternatives that have been set up for this case study.



The bottom box shows details of the **Road Works Standards** (assignments) associated with each section alternative.

The six section alternatives considered for this case study are defined below. The analysis period is 20 years (from year 2000 - 2019).

Alternative	Description
1	This is the do-minimum alternative. Routine pavement maintenance is undertaken each year, as necessary, based on the pavement condition. No rehabilitation works are undertaken until the road condition reaches a poor condition, that is, reconstruction when IRI ≥ 12 AND total damaged area $\geq 40\%$ of carriageway area.
2	With this alternative, a 50 mm asphaltic concrete overlay is applied when the roughness level reaches 6 IRI OR when structural cracking affects 15% of the carriageway area. As the roughness was at IRI 6 in 1998, this should trigger the first overlay in the year 2000. This alternative includes routine pavement maintenance (condition responsive) over the full analysis period.
3	This alternative is similar to Alternative 2, with an extra work item, surface dressing, undertaken to restore skidding resistance when the Sideways Force Coefficient (SFC) falls below 0.4. As with Alternative 2, an overlay should be triggered in the year 2000. Thereafter this alternative will show what effect the introduction of regular surface dressing will have on the overlay frequency.
4	With this alternative the upper 75 mm of the bituminous surfacing is milled out and replaced whenever the roughness value reaches 6 IRI AND when structural cracking affects 20% of the carriageway area. Routine pavement maintenance (condition responsive) is applied throughout the analysis period.
5	This alternative is similar to Alternative 4 except that after 75 mm surfacing is milled, a thickness of 100 mm asphaltic material is placed, that is, effectively replacing the existing upper surfacing and applying a 25 mm overlay. Routine pavement maintenance (condition responsive) is applied throughout the analysis period.
6	With this alternative, an inlay is recommended, based on the mean rut depth reaching 20 mm. It is anticipated that the outer wheelpaths would be inlaid, representing some 25% of the carriageway area. Routine pavement maintenance (condition responsive) is applied throughout the analysis period.

The Section alternatives can be summarised as follows:

Alternative	Proposed maintenance works
1	Do-minimum alternative Reconstruction if Roughness ≥ 12 IRI AND Total Damaged Area $\geq 40\%$ carriageway area
2	50 mm overlay if Roughness ≥ 6 IRI OR Cracking $\geq 15\%$
3	Surface dressing if SFC ≤ 0.4 50 mm overlay if Roughness ≥ 6 IRI OR Cracking $\geq 15\%$
4	Mill 75 mm and replace 75 mm if Roughness ≥ 6 IRI AND Cracking $\geq 20\%$ carriageway area
5	Mill 75 mm and replace 100 mm if Roughness ≥ 6 IRI AND Cracking $\geq 20\%$ carriageway area
6	Inlay to wheel paths if Mean Rut Depth ≥ 20 mm

Note that each alternative includes routine pavement maintenance in the form of:

- Crack sealing if area of wide structural cracking $\geq 5\%$
- Patching if Severely Damaged Area $\geq 10\%$

The **Road Works Standards** and associated works assigned to each alternative are summarised in Table A2.5. In this case study, each alternative is defined by one Maintenance Standard. Note that an identical set of routine pavement works (crack sealing and patching) is common to each Maintenance Standard. To be considered for implementation during any analysis year, the routine works and rehabilitation works must be specified within the same Maintenance Standard. (If routine works and rehabilitation works are specified as two separate Maintenance Standards with the same **effective from** date, the software accepts only one standard, that is, that which is assigned to the alternative first).

The intervention limits for the Works Items are summarised in Table A2.6.

Table A2.5
Case Study 2: Details of road works standards for each project alternative

Project alternative		Road Works Standard	Effective from year	Maintenance works
Alternative 1 Routine + Reconstruct	M	Routine + Reconstruct (Case Study 2) (R&REC)	2000	Reconstruct at IRI 12 AND 40% Damage (RECON)
				Crack Sealing (CRKSL)
				Patching (PATCH)
Alternative 2 Routine + 50 mm overlay	M	Routine + 50 mm Overlay (R&OV50)	2000	Overlay 50 mm at IRI 6 (OVL50)
				Overlay 50 mm at 15% Cracking (OV50)
				Crack Sealing (CRKSL)
				Patching (PATCH)
Alternative 3 Routine + Surface Dressing + 50 mm Overlay	M	Routine + SD + 50 mm Overlay (RSDOV)	2000	Overlay 50 mm at IRI 6 (OVL50)
				Overlay 50 mm at 15% Cracking (OV50)
				Surface Dressing at SFC < 0.4 (SDRESS)
				Crack Sealing (CRKSL)
				Patching (PATCH)
Alternative 4 Routine + Mill 75mm & Replace 75 mm	M	Routine + Mill 75 mm & Replace 75 mm (MR75)	2000	Mill 75 mm + Replace 75 mm at 6 IRI (MILREP)
				Crack Sealing (CRKSL)
				Patching (PATCH)
Alternative 5 Routine + Mill 75 mm & Replace 100 mm	M	Routine + Mill 75 mm & Replace 100 mm (MR100)	2000	Mill 75 mm + Replace 100 mm at 6 IRI (MILREP)
				Crack Sealing (CRKSL)
				Patching (PATCH)
Alternative 6 Routine + Inlay	M	Routine + Inlay at Rut Depth 20 mm	2000	Inlay at Rut Depth 20 mm (INLAY)
				Crack Sealing (CRKSL)
				Patching (PATCH)

Table A2.6
Case Study 2: Intervention limits for maintenance works

	Maintenance Standard	Effective from year	Maintenance Works	*S/R	Last Year	Max (IRI)	Max Quantity (m ² /km/yr)	Interval		AADT	
								Min	Max	Min	Max
1	Routine + Reconstruct	2000	Reconstruct	R	2099	n/a	n/a	n/a	n/a	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000
2	Routine + 50 mm Overlay	2000	Overlay	R	2099	12.5	n/a	3 yrs	99 yrs	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000
3	Routine + SD + 50 mm Overlay	2000	Surface Dressing	R	2099	12.5	n/a	1 yrs	99 yrs	0	100,000
			Overlay	R	2099	12.5	n/a	3 yrs	99 yrs	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000
4	Routine + Mill 75 mm & Replace 75 mm	2000	Mill75&Replace75	R	2099	n/a	n/a	n/a	n/a	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000
5	Routine + Mill 75 mm & Replace 100 mm	2000	Mill75&Replace100	R	2099	n/a	n/a	n/a	n/a	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000
6	Routine + Inlay	2000	Inlay	R	2099	12.5	n/a	n/a	n/a	0	100,000
			Crack Sealing	R	2099	12.5	1500	n/a	n/a	0	100,000
			Patching	R	2099	12.5	100	n/a	n/a	0	100,000

Notes:

n/a = not applicable; * S = Scheduled intervention, R = Responsive intervention

Note that for certain works items, the unit costs of preparatory works are entered separately on the **Works Item/Costs** Tab page. In Case Study 2, this applies to the following works items.

Alternative	Maintenance works item	Preparatory costs
2	50 mm overlay	Patching
		Edge repair
3	50 mm overlay	Patching
		Edge repair
	Surface dressing	Patching
		Edge repair
		Crack Sealing
6	Inlay	Patching
		Edge repair
		Crack Sealing

Where HDM-4 does not offer a separate facility for entry of preparatory costs, such costs (if any) should be included in the costs of the Works Item.

Issues relating to the Maintenance Works Standard for each alternative are discussed below:

■ Alternative 1 - Maintenance Standard: Routine + Reconstruct

The works included in this Maintenance Standard are the routine pavement works (crack sealing and patching) and reconstruction. The works are listed in the opening screen for the Maintenance Standard:

Maintenance Standard: PROJ2: Routine & reconstruct	
General	
Name:	PROJ2: Routine & reconstruct
Short code:	R&REC
Surface class:	Bituminous
Work Items	
Recon at IRI 12 and 40% damage	RECON
Patching	PATCH
Crack Sealing	CRKSL
Add New Work Item... Copy Work Item Delete Work Item Edit...	
The name of this maintenance standard	

The specification for each of these works can be reviewed by double-clicking the appropriate works item name.

❑ Works Item: Crack Sealing and Patching

The specifications for these works, included in all alternatives for this case study, are similar to those described for Case Study 1 within the Maintenance Standard **Crack Sealing and Patching Paved Road** (see Appendix A1).

❑ Works Item: Reconstruct at IRI 12 and 40% damage

The specification for Reconstruction is held under the Tabs:

General/Design/Intervention/Costs/Effects

Note that the Works Activity (Pavement Reconstruction) is specified under **General**. The **Design** Tab shows the proposed new Pavement Type as Asphalt Mix on Granular Base (AMGB) together with pavement details required by HDM-4.

Note that the details under **Design** refer to the new pavement construction. The requested structural number refers to the layers above the subgrade.

The reconstruction **Costs** to be specified include all costs associated with the works.

The Tab pages specifying the **Reconstruction** Works Item are shown below:

Maintenance Works Item: Recon at IRI 12 and 40% damage [X]

General | Design | Intervention | Costs | Effects

Name:

Short code:

Surface: Bituminous

Feature type:

Operation:

Intervention type: ☐ Scheduled
☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Recon at IRI 12 and 40% damage [X]

General | Design | Intervention | Costs | Effects

Pavement type:

Surface material:

Dry season Structural Number:

Surface thickness: mm

Relative compaction: %

Road base (for stabilised base only)

Base thickness: mm Resilient modulus: GPa

Construction Defect Indicators

Bituminous surfacing: 0.5 <= CDS <= 1.5

Base: 0 <= CDB <= 1.5

OK Cancel Apply Help

Pavement type

Maintenance Works Item: Recon at IRI 12 and 40% damage

General | Design | Intervention | Costs | Effects

Responsive Criteria

Roughness ≥ 12 IRI AND Total damaged area $\geq 40\%$

Add New Criterion...
Delete
Edit...

Limits

Last year: 2099 year Minimum Maximum
Max. roughness: 12.5 IRI (m/km) Interval: 1 99 year(s)
Max. quantity: 5000 m²/km/year AADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Recon at IRI 12 and 40% damage

General | Design | Intervention | Costs | Effects

Economic Financial
Unit cost: 40 48 per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 1 0 < DMCF $\leq$ 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Recon at IRI 12 and 40% damage

General | Design | Intervention | Costs | **Effects**

Condition After Works

Roughness And Rutting

☒ **Derived**

☐ User defined: Roughness: 2 IRI (m/km)
Mean rut depth: 0 mm

☐ User defined Surface Characteristics

Surface texture: 0.7 mm
Skid resistance: 0.55 SCRIM (at 50km/h)

Edit Calibration Factors...

OK Cancel Apply Help

Derive roughness and rutting following works

■ **Alternative 2 - Maintenance Standard: Routine + 50 mm overlay**

The works included in the Maintenance Standard are the routine pavement works (crack sealing and patching) and a 50 mm thick overlay (condition responsive, based on IRI). The works are listed in the opening screen for the Maintenance Standard:

Maintenance Standard: P2P3P4: Rout + 50 mm overlay

General

Name: P2P3P4: Rout + 50 mm overlay

Short code: R&OV50

Surface class: Bituminous

OK Cancel

Work Items

Overlay 50mm at IRI 6	OVL50
Overlay 50mm at 15% cracking	OV50
Patching	PATCH
Crack Sealing	CRKSL

Add New Work Item...
Copy Work Item
Delete Work Item
Edit...

The name of this maintenance standard

The specification for the overlay works can be reviewed by double-clicking the appropriate works item name.

Note that two works items for overlaying are specified. Both are condition responsive, one based on roughness, the other on structural cracking. This means that either roughness OR structural cracking can trigger overlaying. The Works Items are identical

except for the **Intervention** Tab page. The Works Item based on roughness is reviewed below:

- ❑ Maintenance Works: Overlay 50 mm at IRI 6

The specification for the Overlay works can be reviewed under the Tabs:

General/Design/Intervention/Costs/Effects

The Works Activity (overlay dense-graded asphalt) is specified under **General**. The overlay material, thickness and dry season strength coefficient are specified under **Design**. The Construction defect indicators for the bituminous overlay (CDS) is also specified.

Note that if overlay is implemented, the Pavement Type will be automatically reset according to Table A2.3:

$$\text{STAP} + \text{Overlay} = \text{AMAP} \quad \dots(1.4)$$

The Tab pages specifying the Overlay Works are reviewed below:

Maintenance Works Item: Overlay 50mm at IRI 6

General | Design | Intervention | Costs | Effects

Name: Overlay 50mm at IRI 6

Short code: OVL50

Surface: Bituminous

Feature type: Carriageway

Operation: Overlay dense-graded asphalt

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Overlay 50mm at IRI 6

General Design Intervention Costs Effects

Surface material: **Asphaltic Concrete**

Thickness of new surfacing: **50** mm

Dry season Strength coefficient: **0.4**

Depth of milling: **0** mm

Area of carriageway to inlay: **50** %

Construction Defect Indicators

Bituminous surfacing: **1** 0.5 <= CDS <= 1.5

OK Cancel Apply Help

Material for the new surface layer

Maintenance Works Item: Overlay 50mm at IRI 6

General Design Intervention Costs Effects

Responsive Criteria

Roughness >= 6 IRI

Add New Criterion...

Delete

Edit...

Limits

Last year: **2099** year

Max. roughness: **12.5** IRI (m/km)

Max. quantity: **5000** m²/km/year

Interval: **3** **99** year(s)

ΔADT: **0** **100000**

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Overlay 50mm at IRI 6

General Design Intervention **Costs** Effects

Economic Financial

Unit cost: per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	12	14.4	per m ²
Edge-repair:	14	16.8	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Overlay 50mm at IRI 6

General Design Intervention Costs **Effects**

Condition After Works

Roughness And Rutting

☒ Derived:

☐ User defined: Roughness: IRI (m/km)

Mean rut depth: mm

☐ User defined Surface Characteristics

Surface texture: mm

Skid resistance: SCRIM (at 50km/h)

OK Cancel Apply Help

Derive roughness and rutting following works

■ **Alternative 3 - Maintenance Standard: Routine + SD + 50 mm overlay**

Maintenance Standard: PROJ2: Rout+SD+50mm overlay

General

Name: PROJ2: Rout+SD+50mm overlay

Short code: RSDOV

Surface class: Bituminous

Work Items

Overlay 50mm at 6 IRI	OVL50
Overlay 50mm at 15% cracking	OV50
Surface Dressing at SFC<=0.4	SDRESS
Patching	PATCH
Crack Sealing	CRKSL

Add New Work Item...

Copy Work Item

Delete Work Item

Edit...

The name of this maintenance standard

The works included in this Maintenance Standard are the same as for Alternative 2 plus an extra works item, Surface Dressing (SD).

The specification for the Surface Dressing Works Item can be reviewed by double-clicking the appropriate works item name:

- Maintenance Works: Surface Dressing at SFC <= 0.4

The specification for Surface Dressing can be reviewed under the Tabs:

General/Design/Intervention/Costs/Effects

The Works Activity is specified as **Surface Dressing Single (General Tab)**, with responsive intervention based on the skid resistance (**Intervention Tab**). Note that a double surface dressing is also available, and shape correction may be applied to either single or double surface dressings. Shape correction adds to the cost, but has a more beneficial effect in terms of roughness (details are given in Chapter D2 of the [Analytical Framework and Model Descriptions](#)).

For this case study, intervention is based on the skid resistance. Additional criteria that may be used with surface dressing are cracking, ravelling, texture depth, and total damaged area.

Note that the unit costs of surface dressing works and preparatory works are specified under **Costs**.

The pavement type (STAP) will be reset automatically according to Table A2.3 if either the surface dressing or overlay works is implemented.

For example:

$$\text{STAP} + \text{Surface Dressing} = \text{STAP} \quad \dots(1.5)$$

$$\text{STAP} + \text{Overlay} = \text{AMAP} \quad \dots(1.6)$$

AMAP + Surface Dressing = STAP ... (1.7)

AMAP + Overlay = AMAP ... (1.8)

The Tab pages specifying the Surface Dressing Works are given below:

Maintenance Works Item: Surface Dressing at SFC<=0.4

General | Design | Intervention | Costs | Effects

Name: Surface Dressing at SFC<=0.4

Short code: SDRESS

Surface: Bituminous

Feature type: Carriageway

Operation: Surface dressing single

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Surface Dressing at SFC<=0.4

General | Design | Intervention | Costs | Effects

Surface material: Single Bitumen Surface Dressing

Thickness of new surfacing: 15 mm

Dry season Strength coefficient: 0.2

Depth of milling: 0 mm

Area of carriageway to inlay: 50 %

Construction Defect Indicators

Bituminous surfacing: 1 0.5 <= CDS <= 1.5

OK Cancel Apply Help

Material for the new surface layer

Maintenance Works Item: Surface Dressing at SFC<=0.4

General | Design | Intervention | Costs | Effects

Responsive Criteria

Skid resistance <= 0.4 SFC

Add New Criterion...

Delete

Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 5000 m²/km/year

Interval: 1 Minimum 99 Maximum year(s)

ADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Surface Dressing at SFC<=0.4

General | Design | Intervention | Costs | Effects

Economic Financial

Unit cost: 5 6 per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	12	14.4	per m ²
Edge-repair:	14	16.8	per m ²
Crack sealing:	5	6	per m ²

Drainage

Drainage maintenance cost factor: 1 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Surface Dressing at SFC≤0.4

General | Design | Intervention | Costs | **Effects**

Condition After Works

Roughness And Rutting

☒ **Derived:** [Edit Detailed Calibration...](#)

☐ **User defined:** Roughness: 3 IRI (m/km)
Mean rut depth: 0 mm

☐ **User defined Surface Characteristics**

Surface texture: 0.7 mm
Skid resistance: 0.55 SCRIM (at 50km/h)

OK Cancel Apply Help

Derive roughness and rutting following works

■ **Alternative 4 - Maintenance Standard: Routine + Mill 75 mm & Replace 75 mm**

Maintenance Standard: PROJ2: Rout+Mill 75+Rep 75

General

Name: PROJ2: Rout+Mill 75+Rep 75

Short code: MR75

Surface class: Bituminous

OK Cancel

Work Items

Mill 75mm+Replace 75mm at 6 IRI	MILREP
Patching	PATCH
Crack sealing	CRKSL

Add New Work Item...
Copy Work Item
Delete Work Item
Edit...

The name of this maintenance standard

In addition to routine pavement works (crack sealing and patching) this Maintenance Standard includes **Mill and Replace** (condition responsive based on IRI).

The specification for **Mill and Replace** can be reviewed under the Tabs:

General/Design/Intervention/Costs/Effects

- ❑ Works Activity: Mill 75 mm + Replace 75 mm at 6 IRI

The Works Activity **Mill and Replace** is specified under **General**, with details given under **Design**. In this alternative, the depth of milling and thickness of new

surfacing are both specified as 75 mm which represents the replacement of the existing upper layers (25 mm surface dressing over a 50 mm overlay). Note that the depth of milling will normally not exceed the total thickness of the existing surfacing (Specified by road section under Section/Pavement). The software does not check this, as Maintenance Works are not defined at section level.

The intervention criterion for **Mill and Replace** is based on IRI and cracking. Other available intervention criteria are:

- Cumulative ESAL
- Mean rut depth
- Potholing
- Ravelling
- Rut depth standard deviation
- Total damaged area

Note that the unit cost of the works should include all associated preparatory works.

If the **Mill and Replace** works is implemented as specified, the existing Pavement Type (STAP) will be reset according to Table A2.3.

$$\text{STAP} + \text{Mill \& Replace} = \text{AMAP} \quad \dots(1.9)$$

The Tab pages specifying the **Mill and Replace** Works for Alternative 4 are reviewed below:

Maintenance Works Item: Mill 75mm+Replace 75mm at 6 IRI

General | Design | Intervention | Costs | Effects

Name: Mill 75mm+Replace 75mm at 6 IRI

Short code: MILREP

Surface: Bituminous

Feature type: Carriageway

Operation: Mill and Replace

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Mill 75mm+Replace 75mm at 6 IRI

General Design Intervention Costs Effects

Surface material: **Asphaltic Concrete**

Thickness of new surfacing: **75** mm

Dry season Strength coefficient: **0.4**

Depth of milling: **75** mm

Area of carriageway to inlay: **50** %

Construction Defect Indicators

Bituminous surfacing: **1** $0.5 \leq \text{CDS} \leq 1.5$

OK Cancel Apply Help

Material for the new surface layer

Maintenance Works Item: Mill 75mm+Replace 75mm at 6 IRI

General Design Intervention Costs Effects

Responsive Criteria

Roughness ≥ 6 IRI AND Add New Criterion...

Cracking ≥ 20 % Delete

Edit...

Limits

Last year: **2099** year Minimum Maximum

Max. roughness: **12.5** IRI (m/km) Interval: **1** **99** year(s)

Max. quantity: **5000** m²/km/year AADT: **0** **100000**

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Mill 75mm+Replace 75mm at 6 IRI

General | Design | Intervention | **Costs** | Effects

Economic Financial

Unit cost: per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Mill 75mm+Replace 75mm at 6 IRI

General | Design | Intervention | **Costs** | **Effects**

Condition After Works

Roughness And Rutting

☒ Derived:

☐ User defined: Roughness: IRI (m/km)
Mean rut depth: mm

☐ User defined Surface Characteristics

Surface texture: mm
Skid resistance: SCRIM (at 50km/h)

OK Cancel Apply Help

Derive roughness and rutting following works

■ **Alternative 5 - Maintenance Standard: Routine + Mill 75 mm & Replace 100 mm**

Maintenance Standard: PROJ2: Rout+Mill 75+Rep 100

General

Name: PROJ2: Rout+Mill 75+Rep 100

Short code: MR100

Surface class: Bituminous

Work Items

Mill 75mm+Replace 100mm at 6 IRI	MILREP
Patching	PATCH
Crack sealing	CRKSL

Add New Work Item...

Copy Work Item

Delete Work Item

Edit...

The name of this maintenance standard

This Maintenance Standard is similar to that specified for Alternative 4, except in this case a 100 mm thick surfacing is added after milling out 75 mm.

As with Alternative 4, the Pavement Type after implementation of the specified **Mill & Replace** works would be given by:

$$\text{STAP} + \text{Mill \& Replace} = \text{AMAP} \quad \dots(1.10)$$

The Tab pages specifying the **Mill and Replace** Works for Alternative 5 are reviewed below:

Maintenance Works Item: Mill 75mm+Replace 100mm at 6 IRI

General | Design | Intervention | Costs | Effects

Name: Mill 75mm+Replace 100mm at 6 IRI

Short code: MILREP

Surface: Bituminous

Feature type: Carriageway

Operation: Mill and Replace

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Mill 75mm+Replace 100mm at 6 IRI

General | Design | Intervention | Costs | Effects

Surface material: Asphaltic Concrete

Thickness of new surfacing: 100 mm

Dry season Strength coefficient: 0.4

Depth of milling: 75 mm

Area of carriageway to inlay: 50 %

Construction Defect Indicators:

Bituminous surfacing: 1 0.5 <= CDS <= 1.5

OK Cancel Apply Help

Material for the new surface layer

Maintenance Works Item: Mill 75mm+Replace 100mm at 6 IRI

General | Design | Intervention | Costs | Effects

Responsive Criteria

Roughness ≥ 6 IRI AND
Cracking ≥ 20 %

Add New Criterion...
Delete
Edit...

Limits

Last year: 2099 year
Max. roughness: 12.5 IRI (m/km)
Max. quantity: 5000 m²/km/year

Interval: 1
AADT: 0

Minimum: 1
Maximum: 99 year(s)
100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Mill 75mm+Replace 100mm at 6 IRI

General | Design | Intervention | Costs | Effects

Economic Financial
Unit cost: 30 36 per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	0	0	per m ²
Edge-repair:	0	0	per m ²
Crack sealing:	0	0	per m ²

Drainage

Drainage maintenance cost factor: 1 0 < DMCF ≤ 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Mill 75mm+Replace 100mm at 6 IRI

General | Design | Intervention | Costs | Effects

Condition After Works

Roughness And Rutting

☒ **Derived:** [Edit Detailed Calibration...](#)

☐ **User defined:** Roughness: 2 IRI (m/km)
Mean rut depth: 0 mm

☐ **User defined Surface Characteristics**

Surface texture: 0.7 mm
Skid resistance: 0.55 SCRIM (at 50km/h)

OK Cancel Apply Help

Derive roughness and rutting following works

■ **Alternative 6 - Maintenance Standard: Routine + Inlay**

Maintenance Standard: PROJ2: Rout+Inlay

General

Name: PROJ2: Rout+Inlay

Short code: INLAY

Surface class: Bituminous

OK Cancel

Work Items

Inlay at rut depth 20mm	INLAY
Patching	PATCH
Crack Sealing	CRKSL

Add New Work Item...
Copy Work Item
Delete Work Item
Edit...

The name of this maintenance standard

In addition to the routine pavement works (crack sealing and patching), this Maintenance Standard includes **Inlay**, that is condition responsive, based on the mean rut depth.

The specification for the **Inlay** works can be reviewed under the Tabs:

General/Design/Intervention/Costs/Effects

❑ Maintenance Works: Inlay at Rut Depth 20 mm

The Works Activity **Inlay** is specified under the **General** Tab. The percentage of the total carriageway surface area to be inlaid is specified under the **Design** Tab. A value of 25% has been specified, representing the outer wheel tracks, each taken as 1 metre wide. The carriageway width is 8 m.

For this case study, mean rut depth is adopted as the intervention criterion. Additional criteria available for use with responsive **Inlay** are **Rut depth standard deviation** and **Total damaged area**.

The unit costs of the **Inlay** works and associated preparatory works are specified under the **Costs** Tab.

If inlay works are implemented the Pavement Type will remain unchanged as **STAP**. Table A2.3 indicates that inlay does not affect Pavement Type.

The Tab pages specifying the **Inlay** Works are reviewed below:

Maintenance Works Item: Inlay at rut depth 20mm

General | Design | Intervention | Costs | Effects

Name: Inlay at rut depth 20mm

Short code: INLAY

Surface: Bituminous

Feature type: Carriageway

Operation: Inlay

Intervention type: ☐ Scheduled ☒ Responsive

OK Cancel Apply Help

The name of this works item

Maintenance Works Item: Inlay at rut depth 20mm [X]

General Design Intervention Costs Effects

Surface material: Asphaltic Concrete

Thickness of new surfacing: 25 mm

Dry season Strength coefficient: 0

Depth of milling: 0 mm

Area of carriageway to inlay: 25 %

Construction Defect Indicators

Bituminous surfacing: 1 0.5 <= CDS <= 1.5

OK Cancel Apply Help

The percentage of the total carriageway surface area to inlay

Maintenance Works Item: Inlay at rut depth 20mm [X]

General Design Intervention Costs Effects

Responsive Criteria

Mean rut depth >= 20 mm

Add New Criterion...

Delete

Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 5000 m²/km/year

Interval: 1

ΔADT: 0

Minimum: 99 year(s)

Maximum: 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Inlay at rut depth 20mm

General | Design | Intervention | **Costs** | Effects

Economic Financial

Unit cost: per m²

Unit Costs Of Preparatory Works

Spot Regravelling:	0	0	per m ²
Patching:	12	14.4	per m ²
Edge-repair:	14	16.8	per m ²
Crack sealing:	5	6	per m ²

Drainage

Drainage maintenance cost factor: 0 < DMCF <= 1

OK Cancel Apply Help

Economic unit cost of works

Maintenance Works Item: Inlay at rut depth 20mm

General | Design | Intervention | **Costs** | **Effects**

Condition After Works

Roughness And Rutting

☒ Derived:

☐ User defined: Roughness: IRI (m/km)

Mean rut depth: mm

☐ User defined Surface Characteristics

Surface texture: mm

Skid resistance: SCRIM (at 50km/h)

OK Cancel Apply Help

Derive roughness and rutting following works

1.3 Run HDM-4 and examine the results

1.3.1 Setup Run

The **Setup Run** Tab is accessed via the **Analyse Projects** button.

This screen confirms the base alternative for economic analysis (that is, the first alternative entered), and that the discount rate is 6%.

For this case study, accident costs, energy balance emissions and acceleration effects are not included in the analysis.

1.3.2 Run analysis

Starts the analysis and produces the output necessary for report generation.

1.4 Generate reports

The output from the HDM-4 analysis are produced under **Generate Reports/Select Reports**.

1.4.1 Deterioration/works effects

The **Timing of Works** report lists the maintenance works items (and associated costs) by section or by year. The works report by section is useful when identifying the works triggered by a particular maintenance standard. This report will often be read in conjunction with the **Deterioration Summary** report that indicates the progression of certain defects associated with condition responsive intervention criteria. For example, the progression of roughness and Total Damaged Area (defined below) can be tracked to check that reconstruction works are correctly triggered according to the specified intervention criteria.

The works resulting from the maintenance standard specified by section alternative are summarised below. Observations are made regarding the works triggered, and the need, where appropriate, to investigate the progression of defects and intervention levels associated with the condition responsive works.

■ **Alternative 1: Routine and Reconstruct**

2000 - 2007	Routine pavement works
2008	Reconstruction
2015 - 2019	Routine pavement works
Observations	The absence of any routine pavement works between 2009 - 2014 gives cause for concern. Intervention levels for routine works would need to be reviewed in association with progression of defects on the Deterioration Summary report.

■ **Alternative 2: Routine and 50 mm overlay**

2000	Overlay at 15% cracking including preparatory patching and edge repair
2007	Overlay at 15% cracking
2015	Overlay at IRI 6
Observations	The absence of routine pavement works between overlay applications should be investigated by comparing the intervention criteria with the progression of defects on the Deterioration Summary report.

■ **Alternative 3: Routine + SD + 50 mm overlay**

2000	Overlay at IRI 6
2001 - 2007	Surface dressing each year (SFC < 0.4)
2008	Overlay at IRI 6
2009 - 2013	Surface dressing each year (SFC < 0.4)
2014	Overlay at IRI 6
2015 - 2018	Surface dressing each year (SFC < 0.4)
2019	Overlay at IRI 6
Observations	The introduction of surface dressing has the effect of delaying the development of structural cracking, resulting in overlays triggered by roughness. (With Alternative 2, overlays in 2000 and 2007 were triggered by structural cracking.) The intervention level for surface dressing should be reviewed in association with the Summary Deterioration report.

■ **Alternative 4: Routine + Mill 75 mm and Replace 75 mm and
Alternative 5: Routine + Mill 75 mm and Replace 75 mm**

The works proposed by these alternatives are summarised below. Note the different thickness of new surfacing (after milling) for alternatives 4 and 5.

2000	Mill and Replace
2008	
2010	
2011	Routine pavement works
2012	
2013	Mill and Replace
Observations	Mill and Replace is condition responsive based on roughness (IRI ≥ 6) and structural cracking ($\geq 20\%$ carriageway area). The levels are reached in the same year. The intervention levels for routine pavement works should be reviewed by reference to the Deterioration report. Alternative 5 shows less benefit than Alternative 4 (see Section 1.4.2).

■ **Alternative 6: Routine + Inlay**

2000 - 2001	Routine pavement works
2002	Inlay at rut depth 20 mm
2003 - 2006	Routine pavement works
2007	Inlay at rut depth 20 mm
2008 - 2010	Routine pavement works
2011	Inlay at rut depth 20 mm
2012 - 2013	Routine pavement works
2014	Inlay at rut depth 20 mm
2015 - 2016	Routine pavement works
2017	Inlay at rut depth 20 mm
2018 - 2019	Routine pavement works
Observations	Inlay is triggered five times during the analysis period, based on the mean rut depth reaching 20 mm. Note that inlay is specified for 25% carriageway area.

1.4.2 Cost streams

The economic analysis summary (by section) report gives the discounted NPV for each section alternative when compared against the base alternative (Alternative 1).

Alternative	Description	Economic Works Costs discounted (US\$ millions)	NPV
1	Routine and Reconstruct	18.8	(base)
2	Routine + 50 mm overlay	18.0	78.1
3	Routine + SD + 50 mm overlay	56.0	50.6
4	Routine + Mill 75 mm & Replace 75 mm	21.2	92.5
5	Routine + Mill 75 mm & Replace 100 mm	24.8	86.5
6	Routine + Inlay	10.2	86.9

1.5 Selected reports

This section contains reports discussed in Section 1.4. The following section contains details of parameters listed in the Summary report.

The **Deterioration Summary Report** shows the following condition data by section for each analysis year:

IRI_{av} = Average roughness m/km IRI

ACA_{av} = Area of All Structural Cracking (expressed as % of carriageway area)

NPT_{av} = Number of potholes

Note:

Area of Potholes = $APOT = 0.1 NPT_{av}$

and % Potholes = $100 \times 0.1 NPT_{av} / \text{Carriageway area for section}$

ARV_{av} = Area of Ravelling (expressed as % of carriageway area)

Note that the significant condition parameters available as intervention criteria for Case Study 2 are:

- **Roughness (IRI)**

May be used as an intervention criteria for Reconstruction, Overlay, and Mill & Replace

- **Total (unpatched) damaged area (ADAMR)**

May be used as an intervention criterion for Reconstruction and Surface Dressing, and defined by:

$$ADAMR = AVEB + APOT + ACRA + ARV \quad \dots(1.11)$$

where:

AVEB	=	area of edge break
APOT	=	area of potholes
ACRA*	=	total area of cracking
ARV	=	area of ravelling

* Note :

$$ACRA = ACA + ACT \quad \dots(1.12)$$

where:

ACA	=	area of all structural cracking
ACT	=	area of transverse thermal cracking

■ Area of Wide Structural Cracking (ACW)

May be used as an intervention criterion for Crack Sealing.

Note that ACW is a component of Area of All Structural Cracking (ACA)

$$ACA = ACW + ACN \quad \dots(1.13)$$

where:

ACN	=	area of narrow structural cracking
-----	---	------------------------------------

Note also that **transverse thermal cracks** (n°/km) may be adopted as an intervention criterion for crack sealing

■ Severely Damaged Area (ADAMS)

May be used as an intervention criterion for patching and is defined as:

$$ADAMS = ACW + ARV + APOT \quad \dots(1.14)$$

where: ACW, ARV, APOT are defined above.

Examination of the terms included under total unpatched damaged area (ADAMR) and Severely damaged area (ADAMS) indicates that:

$$ADAMR = ADAMS + AVEB + ACT + ACN \quad \dots(1.15)$$

Appendix A3 - Project Analysis Case Study 3

1 Traffic Volume Capacity Improvements

This case study presents the economic analysis of widening a paved road. The existing road is 7 m wide, with an AADT of 15,000 in 1998. Non-motorised transport contributed an extra 400 **vehicles** in 1998, comprising pedestrians, animal carts, and bicycles. The analysis assumes that routine pavement maintenance is undertaken on a condition responsive basis for all alternatives. Three widening alternatives are considered, widening by 1m, widening by 3m, and adding two extra lanes. It is recognised that a 1m widening is an impractical proposition; this alternative is included to observe the resulting economic indicators.

The objective of this case study is to demonstrate the congestion analysis of HDM-4, with particular reference to specification of the input data and examination of the results.

To demonstrate this case study the following steps are followed:

- **Locate the case study data** (see Section 1.1)
- **Review the case study input data** (see Section 1.2)
- **Run HDM-4 and examine the results** (see Section 1.3)

1.1 Locate the case study data

The case study data are included in the default database installed with the HDM-4 software. The data for this case study are located in the **Projects folder** in the **Case Studies Workspace**. The name of the case study is **Case Study 3 Capacity improvements**.

To locate the data for this case study:

- Go to the **Case Studies Workspace**
- Open the **Projects folder**

Double-click on the case study named **Case Study 3. Capacity improvements**

1.2 Review the case study input data

The case study run data is held in the **Projects** folder under Case Study 3.

The data can be reviewed under the HDM-4 **Project** Tabs listed below:

- **Define Project Details**

The following Tab pages may be displayed:

- ☐ General (see Section 1.2.1)
- ☐ Select Sections (see Section 1.2.2)
- ☐ Select Vehicles (see Section 1.2.3)
- ☐ Define Normal Traffic (see Section 1.2.4)

■ Specify Alternatives

The following Tab page may be displayed:

- ❑ Alternatives (see Section 1.2.5)

■ Analyse Projects

The following Tab pages may be displayed:

- ❑ Setup Run (see Section 1.3.1)
- ❑ Run Analysis (see Section 1.3.2)

1.2.1 General

The screenshot shows a software window titled "Project: 3. Capacity improvements". On the left is a vertical toolbar with icons and labels: "Define Project Details", "Specify Alternatives", "Analyse Projects", "Generate Reports", "Save", and "Close". The main area has four tabs: "General", "Select Sections", "Select Vehicles", and "Define Normal Traffic". The "General" tab is active and contains the following fields:

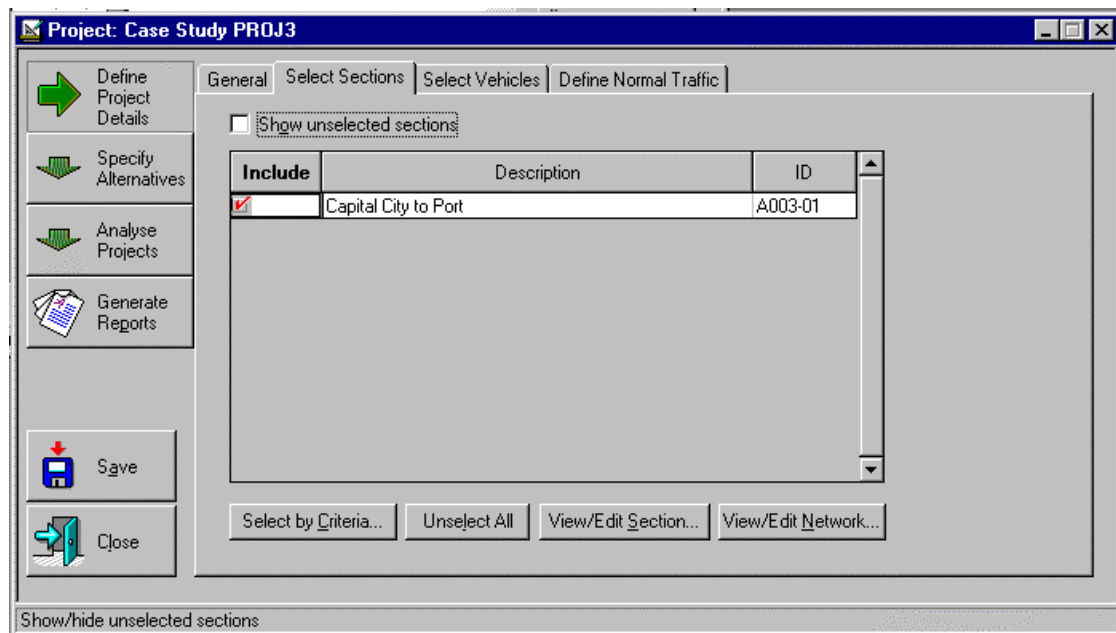
- Description:** Project Case Study 3: Traffic volume capacity improvements
- Analyse by:** Radio buttons for "Section" and "Project" (selected).
- Start year:** 2000
- Duration:** 20 years
- Road Network:** Northern Province Project Road Sections (dropdown menu)
- Vehicle Fleet:** Northern Province Vehicles (dropdown menu)
- Currencies:**
 - Fleet: US Dollar × 1 = output currency
 - Works: US Dollar × 1 = output currency
 - Output: US Dollar (dropdown menu)

At the bottom left, it says "For Help, press F1".

This screen confirms the project description, analysis type, analysis period and the pre-defined **Road Network** and **Vehicle Fleet**.

This case study is presented as a project analysis. The road under study is represented by one section, 10 km long, and the different widening proposals represent project alternatives. The analysis period is defined by the start year 2000 and duration 20 years (that is, 2000 - 2019).

1.2.2 Select Sections



The screen confirms that only one road section will be included in the analysis. By double-clicking on the section description reveals the **Definition/Geometry/Pavement/Condition** Tabs which gives access to the section details.

The following details have particular relevance to this case study; they are concerned with the effects of road widening.

Section: Capital City to Port	
Description: Capital City to Port	Length: 10 km
ID: A003-01	Carriageway width: 7 m
Speed flow type: Two Lane Road	Shoulder width: 1 m
Traffic flow pattern: Commuter	Number of Lanes: 2
Road class: Primary or Trunk	Traffic AADT: 15000 Year: 1998 Flow direction: Two-way
Climate zone: North Region	
Surface class: Bituminous	
Pavement Type: Asphalt Mix on Granular Base	

■ Definition

The speed-flow type of the existing road is specified as a two lane road. The existing carriageway width is 7 m, with two traffic lanes. If lane addition is implemented (alternative 4 in this case study), the width of each additional lane will be calculated as

the existing carriageway width divided by the existing number of traffic lanes (that is, $7/2 = 3.5$ m for this case study).

The speed-flow type defines the capacity of the existing road. The capacity may be increased by Improvement Works such as **partial widening** (by up to 3 m) or **lane addition** (by 1 - 9 lanes). The capacity of the road after improvement is defined by the speed-flow type specified within the Improvement Standard. Default values of the ultimate capacity for several speed-flow types are given in Table A3.1. Further details of speed-flow model parameters are given in Chapter D5, and in Part B of the [Analytical Framework and Model Descriptions](#).

Table A3.1
Ultimate capacity for selected speed-flow types

Speed-flow type	Carriageway width (m)	Ultimate capacity Q_{ult} (PCSE/lane/hr)
Single lane road	<4	600
Intermediate road	4 to 5.5	900
Two lane road	5.5 to 9	1400
Wide two lane road	9 to 12	1600
Four lane road	>12	2000

The traffic flow (AADT) on the road section is confirmed as 15,000 in 1998. Details of AADT, traffic composition and growth rate, as defined for this project case study, will be reviewed under **Define Normal Traffic**.

Note also that for this road section there is some speed reduction due to roadside activities or motorised/non-motorised transport. The speed reduction factors are specified under **Section/Details/Alignment**.

Calibration Parameters

Alignment | Drainage, Shoulders, and NMT Lanes | History | Surface Distress | Surface Texture | Structural Defects

No. of rises + falls: no./km

Superelevation: %

Gradat: m/s²

Speed limit enforcement:

Speed Reduction Factors

NMT: $0.6 \leq X_{NMT} \leq 1$

MT: $0.6 \leq X_{MT} \leq 1$

Road side friction: $0.6 \leq X_{FRI} \leq 1$

OK Cancel Help

Average number of Rises and Falls per km

The speed reduction factors are defined as follows:

XNMT Speed reduction: effect of NMT on MT

1 = no reduction

0.6 = significant reduction

XMT Speed reduction: effect of MT on NMT

1 = no reduction

0.6 = significant reduction

Road side friction Speed reduction due to roadside activities (effect on MT only), where:

1 = no roadside friction

0.6 = significant friction

■ Pavement

Section: Capital City to Port

Definition | Geometry | **Pavement** | Condition

Surfacing

Material type: Asphaltic Concrete

Most recent surfacing thickness: 100 mm

Previous/old surfacing thickness: 0 mm

Previous works (HDM-4 Work Types)

Last reconstruction or new construction: 1988 year

Last rehabilitation (overlay): 1988 year

Last resurfacing (resealing): 1988 year

Last preventative treatment: 1988 year

Strength

Calculated Wet season model parameters

SNP: 4.35 DEF: 0.62 mm

[1] ☒ Structural Number: 3.3

Subgrade CBR: 8 %

☐ Dry Season ☒ Wet Season

[2] ☐ Calculated SNP: Calculate SNP...

Road base (for stabilised base only)

Base thickness: mm

Resilient modulus: GPa

Details... OK Cancel Help

Surface material

The pavement type has been defined as **Asphalt Mix on Granular Base (AMGB)** under the **General** Tab. The dates of Previous works listed under the **Pavement** Tab indicates that the pavement type is unchanged since the original construction in 1988.

The Structural Number (SN), estimated as 3.3, represents the existing pavement layers above the subgrade. The SNP value includes the subgrade (8% CBR) contribution. Note that for the improvement works (partial widening or lane addition), a value of 3.3 has also been specified for the Structural Number, SN, (under Improvement/Pavement).

■ Condition

Section: Capital City to Port

Definition | Geometry | Pavement | **Condition**

Condition at end of year	1998	Add New Year Delete Year Sort Years
Roughness (IRI - m/km)	6.00	
Total area of cracking (%)	10.00	
Ravelled area (%)	12.00	
Number of Potholes (No./km)	0.00	
Edge break area (m²/km)	10.00	
Mean rut depth (mm)	10.00	
Texture depth (mm)	0.50	
Skid resistance (SCRIM 50km/h)	0.40	
Drainage	Excellent	

Yearly condition data

This confirms the condition levels defined for 1998. For project appraisal, this data would normally be collected as part of a detailed condition survey.

1.2.3 Select Vehicles

Project: Case Study PROJ3

General | **Select Sections** | **Select Vehicles** | Define Normal Traffic

☐ Show unselected vehicles ☒ Include NMT in analysis

Include	Vehicle	Category
☒	Animal cart	NMT
☒	Artic Truck	Motorised
☒	Heavy truck	Motorised
☒	Light Goods Vehicle	Motorised
☒	Light Truck	Motorised
☒	Medium Bus	Motorised
☒	Medium Car	Motorised
☒	Medium Truck	Motorised
☒	Person	NMT
☒	Raleigh Bicycle	NMT

Show/hide any unselected vehicles in the selected fleet

The vehicles selected from the pre-defined **Northern Province** fleet are shown on this screen. Note that Non-Motorised Transport (NMT) is included in the analysis (NMT box ticked at top of **Select Vehicles** Tab page). Individual vehicle attributes can be reviewed by double-clicking on the appropriate vehicle type description.

1.2.4 Define Normal Traffic

This confirms the volume of traffic using the selected road section in the given year. The initial traffic composition and growth rates (by vehicle category and type) can be reviewed by double-clicking the section description. The Normal Traffic details are given on two Tab pages **Motorised** and **NMT**:

Normal Traffic Details

Motorised | **NMT**

Details for section: Capital City to Port

Total AADT: Year:

Growth Periods:

Vehicle	Initial Composition (%)	Annual % increase from 2000
Medium Bus	5.00	3.00
Artic Truck	2.50	3.00
Heavy truck	7.50	3.00
Medium Truck	10.00	3.00
Light Truck	5.00	3.00
Light Goods Vehicle	20.00	4.00
Medium Car	50.00	4.00

The annual average daily traffic on the road section

Normal Traffic Details

Motorised NMT

Details for section: Capital City to Port

Total AADT: Year:

Growth Periods

Vehicle	Initial Composition (%)	Annual % increase from 2000
Animal cart	25.00	2.00
Person	50.00	5.00
Raleigh Bicycle	25.00	3.00

OK Cancel Help

The annual average daily traffic on the road section

1.2.5 Alternatives

Project: 3. Capacity improvements

Define Project Details

Specify Alternatives

Analyse Projects

Generate Reports

Save

Close

Alternatives

Alternatives

- 1: Base case without widening
- 2: Widening by 1m
- 3: Widening by 3m
- 4: Lane addition (2 lanes)

☒ View Details

Details for alternative 1: Base case without widening

Section	Assignments	Year
Capital City to Port	P2P3P4: Rout + 50 mm overlap	2000

List of the alternatives defined

The four different alternatives considered for this case study are defined below. The analysis period is 20 years (from years 2000 to 2019).

Alternative	Description
1	This is the do-minimum alternative. Routine pavement maintenance is undertaken each year, as necessary, based on the pavement condition. In addition, a 50 mm overlay is applied when the roughness level reaches 6 IRI OR when structural cracking affects 15% of the carriageway area.
2	With this alternative, the existing road is widened by 1 m during the period (2000-2001). The maintenance regime of Alternative 1 (Routine + 50 mm overlay), which is condition responsive, is effective from year 3 (2002).
3	With this alternative, the existing road is widened by 3 m during the period (2000-2001). The condition responsive maintenance regime of Alternative 1 is effective from year 3 (2002).
4	With this alternative, the existing road is widened by adding two lanes during the period (2000-2002). The condition responsive maintenance regime of Alternative 1 (Routine + 50 mm Overlay) is effective from year 4 (2003).

The key data relating to the alternative widening scenarios are given in Table A3.2 and Table A3.3.

Table A3.2
Summary of widening alternatives

Alternative	Widening	Duration (years)	Economic cost of widening (US\$/km)	After Widening	
				Carriageway width (m)	Speed-flow relationship
1	None	-	-	(7)	(Two lane road)
2	+1 m	2	51,000	8	Two lane road
3	+3 m	2	102,000	10	Wide two lane
4	+2 lanes	3	238,000	14	Four lane road

Notes:

- 1 All data for Section ID A003-01, length 10 km
- 2 Alternative 1 represents the base case (7 m wide two-lane commuter road)

As noted previously, the speed-flow type determines the capacity of the road section. The base case (Alternative 1) has been assigned the speed-flow type for a two-lane road. The same speed-flow type has been assigned to Alternative 2. With Alternatives 3 and 4, the capacity can be expected to increase due to the revised speed-flow types specified under the respective Improvement Standards.

The various widening alternatives have an impact on the speed reduction due to NMT, MT, and roadside friction. For the existing road section, these data are specified under

Section/Details/Alignment. For the widening alternatives, the data are specified under **Improvement/Geometry**. The values assigned for each alternative are given in Table A3.3.

Table A3.3
Speed limit and speed reduction factors

Alternative	Widening	Speed limit km/h	Speed reduction factors		
			NMT	MT	Roadside friction
1	None	100	0.7	0.9	0.7
2	+1 m	100	0.75	0.9	0.7
3	+3 m	100	0.9	1	0.8
4	+2 lanes	120	1	1	1

The **Road Works Standards** and associated works assigned to each alternative are summarised in Table A3.4. Each **widening** assignment is scheduled to start in year 1 of the analysis period (year 2000), with a construction period of two or three years.

Each widening alternative includes a Maintenance Standard (Routine + 50 mm overlay) providing condition responsive routine pavement works and condition responsive overlay, effective from the year following completion of the widening works. This Maintenance Standard includes the following works items.

- 1 Crack Sealing if Area of Wide Structural Cracking $\geq 5\%$ Carriageway Area
- 2 Patching if Severely Damaged Area $\geq 5\%$ Carriageway Area
- 3 50 mm overlay if Roughness > 6 IRI OR if Area of Structural Cracking $\geq 15\%$ carriageway area

These maintenance works define the base alternative.

Table A3.4
Details of road works standards for each project alternative

Project alternative		Road Works Standard	Effective from year	Maintenance Works/Improvement Type
Alternative 1 Base case Without widening road	M	Routine + 50 mm Overlay (R&OV50)	2000	Crack Sealing (CRKSL)
				Patching (PATCH)
				Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 2 Widening by 1m	I	Partial Widening by 1 m (WIDE1m)	2000	Partial Widening
	M	Routine + 50 mm Overlay (R&OV50)	2002	Crack Sealing (CRKSL)
				Patching (PATCH)
				Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 3 Widening by 3 m	I	Partial Widening by 3 m (WIDE3m)	2000	Partial Widening
	M	Routine + 50 mm Overlay (R&OV50)	2002	Crack Sealing (CRKSL)
				Patching (PATCH)
				Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 4 Widening by 2 lanes	I	Lane addition (ADD2L)	2000	Lane addition
	M	Routine + 50 mm Overlay (R&OV50)	2003	Crack Sealing (CRKSL)
				Patching (PATCH)
				Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)

This case study is concerned with the effects of road widening. Note that the **base case** road geometry can be reviewed under Section/General, as discussed previously under **Select Sections**. Relevant details of each alternative are discussed below:

Alternative 1: Base Case Without Widening

- **M** Routine + 50 mm overlay (Effective from year 2000)

This project alternative includes one Maintenance Standard **Routine + 50 mm overlay** which is effective from the first year of the analysis period. As noted in Table A3.4, this Maintenance Standard comprises three works activities (Crack Sealing, Patching, and Overlay), each of which is condition responsive.

Details of this Maintenance Standard can be reviewed by double-clicking the above description under **Maintenance Standards**. The details are similar to those described for Case Study 2, Alternative 2.

Maintenance Standard: P2P3P4: Rout + 50 mm overlay

General

Name: P2P3P4: Rout + 50 mm overlay

Short code: R&OV50

Surface class: Bituminous

OK Cancel

Work Items

Work Item	Code
Overlay 50mm at IRI 6	OVL50
Overlay 50mm at 15% cracking	OV50
Patching	PATCH
Crack Sealing	CRKSL

Add New Work Item...
Copy Work Item
Delete Work Item
Edit...

The name of this maintenance standard

Details of each works activity can be reviewed by double-clicking the appropriate works name. The details are similar to those described for Case Study 2, Alternative 2.

The intervention criteria can be reviewed via the **Intervention** Tab associated with each works activity.

Maintenance Works Item: Crack Sealing

General Intervention Costs Effects

Responsive Criteria

Wide Structural Cracking >= 10 %

Add New Criterion...
Delete
Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 1500 m²/km/year

Interval: 1 year(s)

AADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Patching

General Intervention Costs Effects

Responsive Criteria

Severely damaged area $\geq 5\%$

Add New Criterion...

Delete

Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 100 m²/km/year

Interval: 1

Minimum Maximum

99 year(s)

ADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Overlay 50mm at IRI 6

General Design Intervention Costs Effects

Responsive Criteria

Roughness ≥ 6 IRI

Add New Criterion...

Delete

Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 5000 m²/km/year

Interval: 3

Minimum Maximum

99 year(s)

ADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Maintenance Works Item: Overlay 50mm at 15% cracking [X]

General | Design | Intervention | Costs | Effects

Responsive Criteria

Cracking >= 15 %

Add New Criterion...

Delete

Edit...

Limits

Last year: 2099 year

Max. roughness: 12.5 IRI (m/km)

Max. quantity: 5000 m²/km/year

Interval: 3 99 year(s)

Minimum Maximum

ADT: 0 100000

OK Cancel Apply Help

Add an intervention criterion to this improvement standard

Alternative 2: Widening by 1 m

- **I** Partial Widening by 1 m (Effective from year 2000)
- **M** Routine + 50 mm overlay (Effective from year 2002)

Details of the Improvement Standard (partial widening) are given under seven Tabs:

General/Design/Intervention/Costs/Pavement/Geometry/Effects

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

Name:

Short code:

Feature type:

Existing Surface class:

Improvement type:

Duration: years

Intervention type: ☒ Scheduled
☐ Responsive

OK Cancel Help

The name of this improvement standard

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

New Attributes

Speed Flow Type:

Road class:

Pavement type:

Length adjustment factor:

Increase in width: m

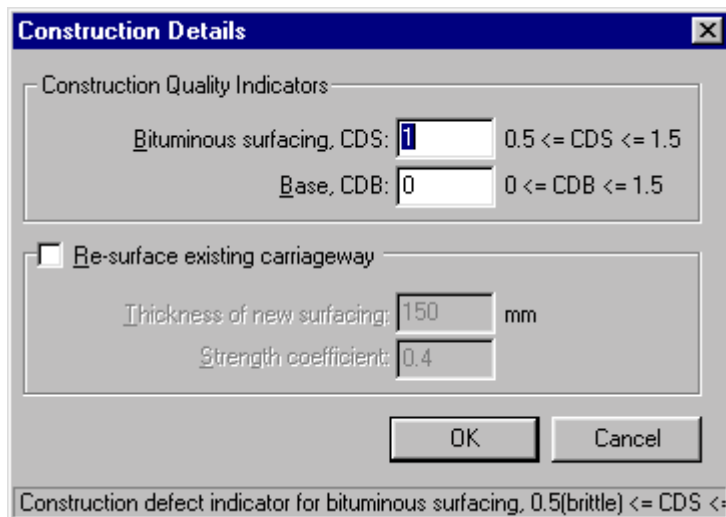
Additional number of lanes:

Proportion of new construction: $0 < P_{conew} < 1$

Edit Construction Details...

OK Cancel Help

The speed flow type of the section following improvement



Construction Details

Construction Quality Indicators

Bituminous surfacing, CDS: 0.5 <= CDS <= 1.5

Base, CDB: 0 <= CDB <= 1.5

☐ Re-surface existing carriageway

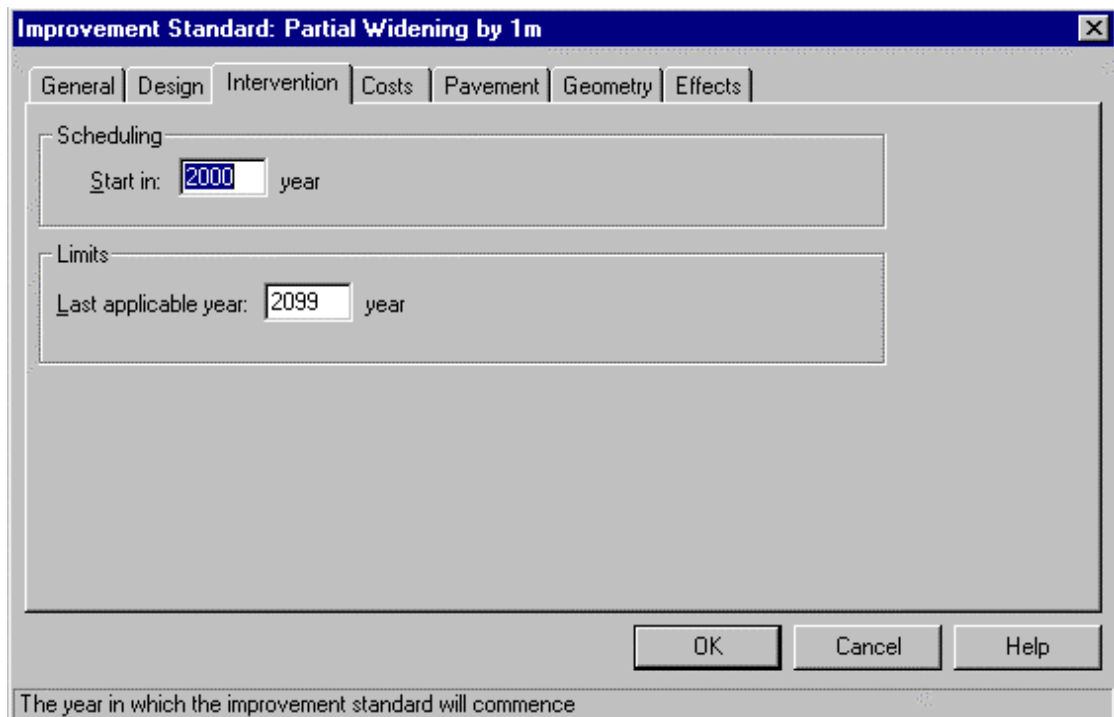
Thickness of new surfacing: mm

Strength coefficient:

OK Cancel

Construction defect indicator for bituminous surfacing, 0.5(brittle) <= CDS <= 1.5

Note: 'Re-surface existing carriageway' is **not** selected



Improvement Standard: Partial Widening by 1m

General Design Intervention Costs Pavement Geometry Effects

Scheduling

Start in: year

Limits

Last applicable year: year

OK Cancel Help

The year in which the improvement standard will commence

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

Economic Financial

Unit cost: 34000 40000 per km

Annual cost: 1 2 3 4 5
stream 50 50 0 0 0 %

Unit Costs Of Preparatory Works

	Economic	Financial	
Re-surfacing of existing carriageway:	0	0	per m ²
Patching:	12	14.4	per m ²
Edge-repair:	14	16.8	per m ²
Crack sealing:	5	6	per m ²

OK Cancel Help

The economic unit cost of the improvement standard

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

Surface material: Asphaltic Concrete

Dry season Structural Number: 3.3

Surface thickness: 100 mm

Relative compaction: 97 %

Road base (for stabilised base only)

Base thickness: 25 mm

Resilient modulus: 15 GPa

OK Cancel Help

Material for the new surface layer

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

Geometry Class: Select a geometry class to obtain default geometry values

Details

Rise + fall:	0	m/km	Speed Reduction Factors
No. of rises + falls:	0	no./km	
Avg horiz curvature:	0	deg/km	
Superelevation:	0	%	
σ _{adral} :	0.1	m/s ²	
Speed limit:	100	km/h	
Speed limit enforcement:	1.1		

NMT: 0.75 0.6 ≤ XNMT ≤ 1
MT: 0.9 0.6 ≤ XMT ≤ 1
Road side friction: 0.7 0.6 ≤ XFRI ≤ 1

OK Cancel Help

Acceleration Noise due to driver and road alignment, in m/s²

Improvement Standard: Partial Widening by 1m

General | Design | Intervention | Costs | Pavement | Geometry | Effects

Condition After Works

Road user: <undefined> [Edit Detailed Calibration...](#)

☐ User defined Roughness And Rutting

Roughness:	2	IRI (m/km)
Mean rut depth:	0	mm

☐ User defined Surface Characteristics

Skid resistance:	0.55	SCRIM (at 50km/h)
Surface texture:	0.7	mm

OK Cancel Help

Use user defined roughness and rutting values

The improvement type (partial widening) and duration are specified under the **General** Tab.

The increase in width, together with the pavement type (of the partial widening) and speed-flow type for the widened road is specified under **Design**. For this alternative the speed-flow type after widening has been specified as a two-lane road (the same as the base case), as the widening is only 1m. The pavement type (of the partial widening) is Asphalt Mix on Granular Base (AMGB), which is the same as the existing pavement (**Section/Pavement**). Note that the construction quality indicators (CDS and CDB) of the partial widening can be reviewed via **Design/Edit Construction Details**. The main works costs and associated preparatory costs are held under the **Costs** Tab, and are spread across the two-year construction period.

Details of the surfacing used for widening (material and thickness) and the Structural Number (SN) of the partial widening are specified under **Pavement**. For improvements, SN will normally be assessed using design layer thicknesses and **as new** layer coefficients.

Alternative 3: Widening by 3 m

- **I** Partial Widening by 3 m (Effective from year 2000)
- **M** Routine + 50 mm overlay (Effective from year 2002)

This alternative is similar to Alternative 2, except that the widening is 3 m, specified under Improvement Standard/Design. The speed-flow type after widening is a **wide two-lane road**, which has increased capacity over the original **two-lane road** (see Table A3.1). In addition, speed reduction due to NMT is less significant (that is, higher factor in Table A3.3).

Alternative 4: Lane addition (2 lanes)

- **I** Lane addition (2 lanes) - Effective from year 2000
- **M** Routine + 50 mm overlay - Effective from year 2003

This alternative proposes the addition of two lanes, specified as Improvement Type **Lane Addition** with a construction period of three years. The additional number of lanes is specified as 2. As noted earlier, the assumed lane width is 3.5 m (based on an existing two-lane carriageway width of 7 m), giving a carriageway width of 14 m after the addition of the two lanes. (Exogenous benefits and costs have been included in this assignment, and may be reviewed via Specify Alternatives/Edit Alternatives/Edit Assignment).

The **General** Tab page is shown below:

The screenshot shows a software dialog box titled "Improvement Standard: Lane Addition (2 lanes)". It has several tabs: General, Design, Intervention, Costs, Pavement, Geometry, and Effects. The "General" tab is selected. The form contains the following fields and options:

- Name:** Lane Addition (2 lanes)
- Short code:** ADD2L
- Feature type:** Carriageway (dropdown menu)
- Existing Surface class:** Bituminous (dropdown menu)
- Improvement type:** Lane addition (dropdown menu)
- Duration:** 3 years
- Intervention type:**
 - ☒ Scheduled
 - ☐ Responsive

At the bottom right are buttons for "OK", "Cancel", and "Help". At the bottom left, there is a text label: "The name of this improvement standard".

Note that on the **Design** Tab page the speed-flow type after improvement has been specified as **four lane road**, which has improved capacity over the **two lane road** specified for the base case (see Table A3.1). In addition, there is no speed reduction due to NMT, MT, or roadside friction.

The screenshot shows the 'Design' tab of the 'Improvement Standard: Lane Addition (2 lanes)' dialog box. The 'New Attributes' section contains the following fields:

- Speed Flow Type: Four Lane Road (dropdown)
- Road class: Primary or Trunk (dropdown)
- Pavement type: Asphalt Mix on Granular Base (dropdown)
- Length adjustment factor: 0 (text box)
- Increase in width: 0 m (text box)
- Additional number of lanes: 2 (text box)
- Proportion of new construction: 0.5 (text box) with the note $0 < P_{conew} < 1$

There is an 'Edit Construction Details...' button next to the 'Increase in width' field. At the bottom are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads: 'The speed flow type of the section following improvement'.

The cost of the improvement works and associated preparatory works are specified on the **Costs** Tab page.

The screenshot shows the 'Costs' tab of the 'Improvement Standard: Lane Addition (2 lanes)' dialog box. The 'Unit cost' section has two input fields: 'Economic' (238000) and 'Financial' (280000), with a unit dropdown set to 'per km'. Below this is the 'Annual cost: stream' section with five input boxes (1: 34, 2: 33, 3: 33, 4: 0, 5: 0) and a '%' symbol. The 'Unit Costs Of Preparatory Works' section contains a table:

	Economic	Financial	
Re-surfacing of existing carriageway:	0	0	per m ²
Patching:	12	14.4	per m ²
Edge-repair:	14	16.8	per m ²
Crack sealing:	5	6	per m ²

At the bottom are 'OK', 'Cancel', and 'Help' buttons. A status bar at the bottom reads: 'The economic unit cost of the improvement standard'.

1.3 Run HDM-4 and examine the results

1.3.1 Setup Run

The **Setup Run** Tab is accessed via the **Analyse Projects** button.

The Setup Run screen confirms that the base alternative for economic analysis is Alternative 1, and that the discount rate is 6%.

For this case study, accident costs, energy balance emissions and acceleration effects are not included in the analysis.

1.3.2 Run Analysis

This starts the analysis and produces the output necessary for report generation.

1.4 Generate reports

The output reports from the HDM-4 analysis are produced under **Generate Reports/Select Reports**.

1.4.1 Deterioration/works effects

The **Timing of Works** (by section) report lists, by alternative, the works that would be implemented by the specified works standards in each analysis year. The report confirms the dates of the improvement works for alternatives 2, 3 and 4.

Alternative	1	2	3	4
Widening	None	1 m	3 m	2 lanes
Widening Works date	-	2000 - 2001	2000 - 2001	2000 - 2002
Baseline IRI (year)	6.14 (2000)	5.98 (2002)	5.38 (2002)	4.64 (2003)

Note that with Alternatives 2-4, the roughness value immediately following the widening works is based on the full carriageway width, and hence the **baseline** IRI value decreases as the widening (with no defects immediately after construction) increases.

The **Timing of Works** report should be read in conjunction with the **Deterioration Summary** report, which indicates the progression of certain defects throughout the analysis period.

1.4.2 Road user effects

The impact of the widening alternatives can be assessed by examination of the **Volume-Capacity Ratio** report which tabulates the volume-capacity ratio (VCR) by time period and calendar year for each project alternative and road section.

The long-term effectiveness of the road widening is indicated by the time taken for the VCR to reach a value of 1.0 (that is, when volume equals capacity).

The effect of widening on vehicle speeds is demonstrated by the **Vehicle Speed** report.

1.4.3 Cost streams

The Economic Analysis Summary indicates that, for the range of widening represented by alternatives 2-4, the NPV increases significantly with width of carriageway.

Alternative	Widening	Carriageway width after widening (m)	NPV (millions of US\$)
2	1m	8	
3	3m	10	
4	2 lanes	14	

Appendix A4 - Project Analysis Case Study 4

1 New Bypass Construction (traffic diversion effects)

This case study presents the economic analysis of a project to construct a bypass around a town centre. The objective is to demonstrate the specification of the bypass (as a section alternative within a project alternative), and to examine the resulting traffic diversion effects.

The roads under study are shown schematically in Figure A4.1. Road sections A, B, C and D represent the network at a town centre. The network shown, models traffic movements between Zone 1 and Zone 2. The proposed project is the construction of a bypass, represented by one section (Section E), 10 km long.

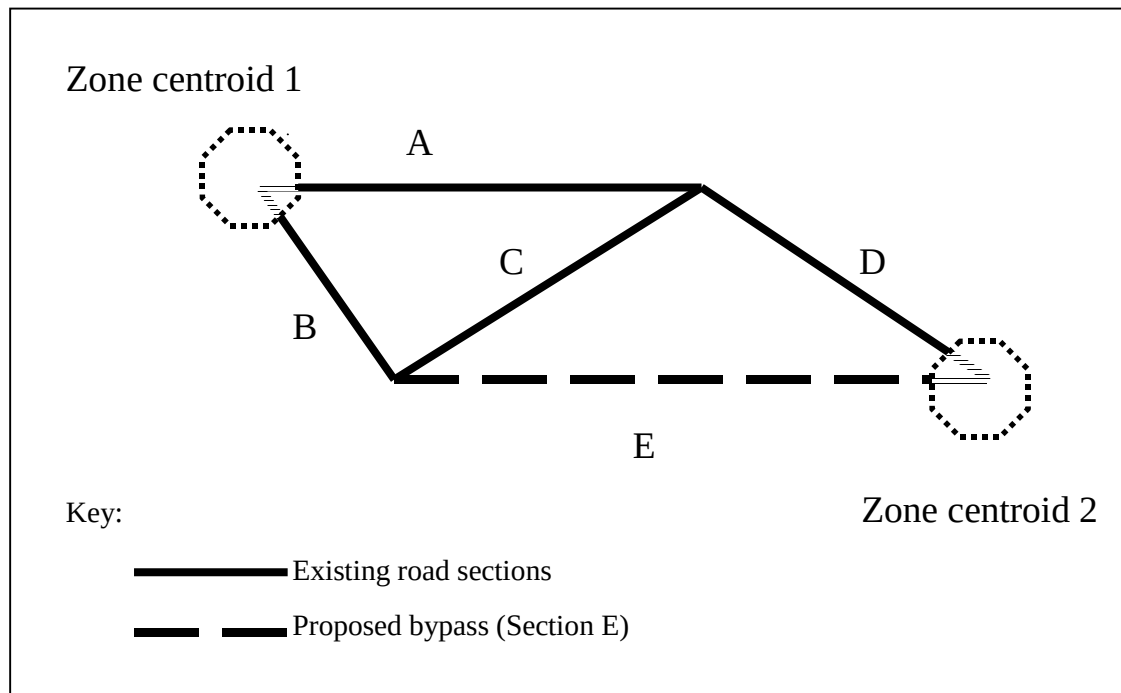


Figure A4.1 Case Study 4: Construction of a new bypass

The case study considers four project alternatives as defined below:

Alternative 1: Base case without the bypass

Alternative 2: Bypass width 8m, constructed between 2000-2001, opening in 2002.

Alternative 3: Bypass width 10m, constructed between 2000-2002, opening in 2003.

Alternative 4: Bypass width 14m, constructed between 2000-2003, opening in 2004.

The traffic flows (AADT) on each section after bypass construction will normally be derived using an external traffic demand model (that is, derived outside HDM-4). The AADT data used with this case study are summarised in Table A4.1. The vehicle composition data (Table

A4.2) predict a transfer of heavy vehicles from Sections A, C and D to Section E once the bypass (Section E) is constructed.

Table A4.1
AADT values by project alternative

Road Section	Alternative 1	Alternative 2		Alternative 3		Alternative 4	
	Base case Without bypass 1998	Without bypass 2002	With bypass 2002	Without bypass 2003	With bypass 2003	Without bypass 2004	With bypass 2004
A	5600	6062	2062	6183	2103	6306	2144
B	3200	3464	7464	3533	7613	3604	7766
C	3200	3464	464	3533	473	3604	483
D	8800	9526	2526	9716	2576	9910	2627
E (bypass)	n/a	n/a	7000	n/a	7140	n/a	7283

Notes:

- 1 Bypass opening year is 2002 for Alternative 2, 2003 for Alternative 3, and 2004 for Alternative 4.
- 2 Data for Sections A-D may be reviewed via Specify Alternatives/Edit Alternative/Diverted Traffic.
- 3 Data for Section E may be reviewed via Specify Alternatives/Edit Alternative/Edit New Section.
- 4 Shaded data included for information, and is not entered to HDM-4.
- 5 n/a = not applicable

Table A4.2
Traffic composition and growth rates

Vehicle type	Alternative 1		Alternatives 2-4: With bypass (opening year)			
	Without bypass:		Sections A,C,D		Sections B,E	
	Sections A-D (1998)		Compo- sition (%)	Annual growth rate (%)	Compo- sition (%)	Annual growth rate (%)
Medium bus	5	2	5	2	5	2
Heavy truck	5	2	3	2	10	2
Medium truck	10	2	7	2	14	2
Light goods vehicle	10	2	10	2	10	2
Medium car	60	2	70	2	50	2
Mini-bus	5	2	5	2	5	2
Artic truck	5	2	0	0	6	2

To demonstrate this case study the following steps are followed:

- **Locate the case study data** (see Section 1.1)
- **Review the case study input data** (see Section 1.2)
- **Run HDM-4 and examine the results** (see Section 1.3)

1.1 Locate the case study data

The case study data are included in the default database installed with the HDM-4 software. The data for this case study are located in the **Projects folder** in the **Case Studies Workspace**. The name of the case study is **Project Case Study 4: Construction of new bypass**.

To locate the data for this case study:

- Go to the **Case Studies Workspace**
- Open the **Projects folder**

Double-click on the case study named **Case Study 4. Construction of a new by-pass**

1.2 Review the case study input data

The case study input data can be reviewed under the HDM-4 **Projects** work flow buttons and associated Tab pages/screens as follows:

- **Define Project Details** (see Section 1.2.1)

The following Tab pages are reviewed:

- ☐ General
- ☐ Select Sections

- ❑ Select Vehicles
- ❑ Define Normal Traffic
- **Specify Alternatives** (see Section 1.2.2)

The following screens are reviewed:

- ❑ Alternatives

Alternative Details

New Construction Section Option

Normal Traffic

Section Details (Definition/Geometry/Pavement/Condition)

Diverted Traffic

Diverted Traffic Details

- **Analyse Projects**

The following Tab pages are reviewed:

- ❑ Setup Run (see Section 1.3.1)
- ❑ Run Analysis (see Section 1.3.2)

1.2.1 Define Project Details

The project details are held under four Tabs:

General/Select Sections/Select Vehicles/Define Normal Traffic

These screens are shown on the following pages.

- **General**

The screenshot shows a software window titled "Project: 4. Construction of new bypass". The window has a sidebar on the left with icons for "Define Project Details", "Specify Alternatives", "Analyse Projects", "Generate Reports", "Save", and "Close". The main area has four tabs: "General", "Select Sections", "Select Vehicles", and "Define Normal Traffic". The "General" tab is active, showing the following fields:

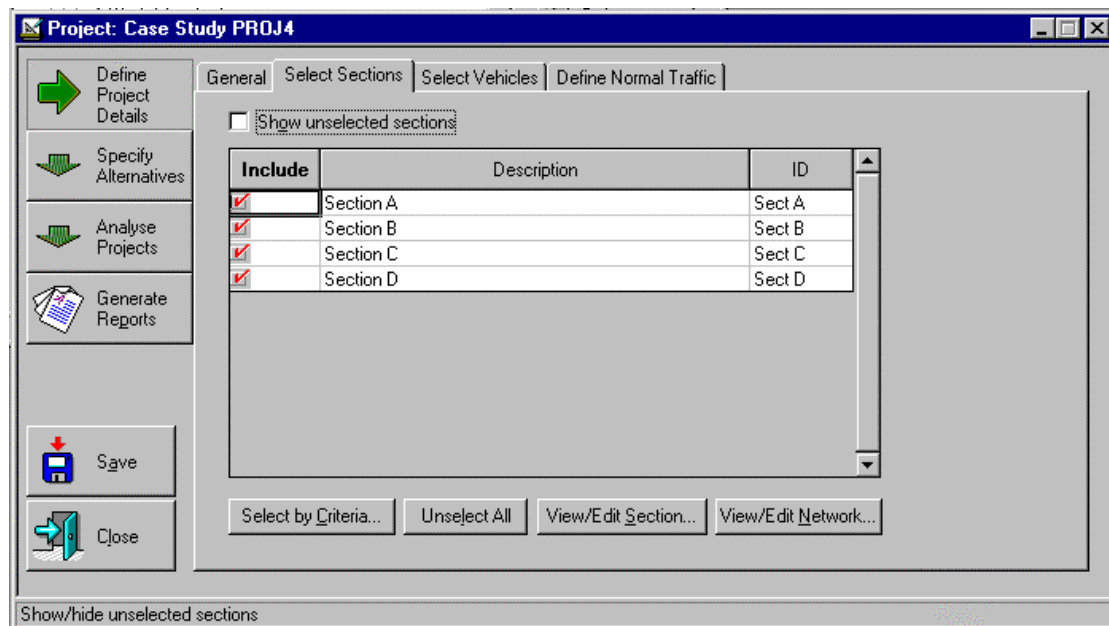
- Description:** Project Case Study 4: Construction of a new bypass.
- Analyse by:** ☐ Section ☒ Project
- Start year:** 2000 **Duration:** 5 years
- Road Network:** Northern Province Project Road Sections (dropdown)
- Vehicle Fleet:** Northern Province Vehicles (dropdown)
- Currencies:**
 - Fleet: US Dollar × 1 = output currency
 - Work: US Dollar × 1 = output currency
 - Output: US Dollar (dropdown)

At the bottom of the window, a status bar reads: "Start of year of analysis period (otherwise referred to as the Project Start Year)".

The **General** screen confirms the project description, analysis type, analysis period (2000-2019) and the pre-defined **Road Network** and **Vehicle Fleet**. Note that the

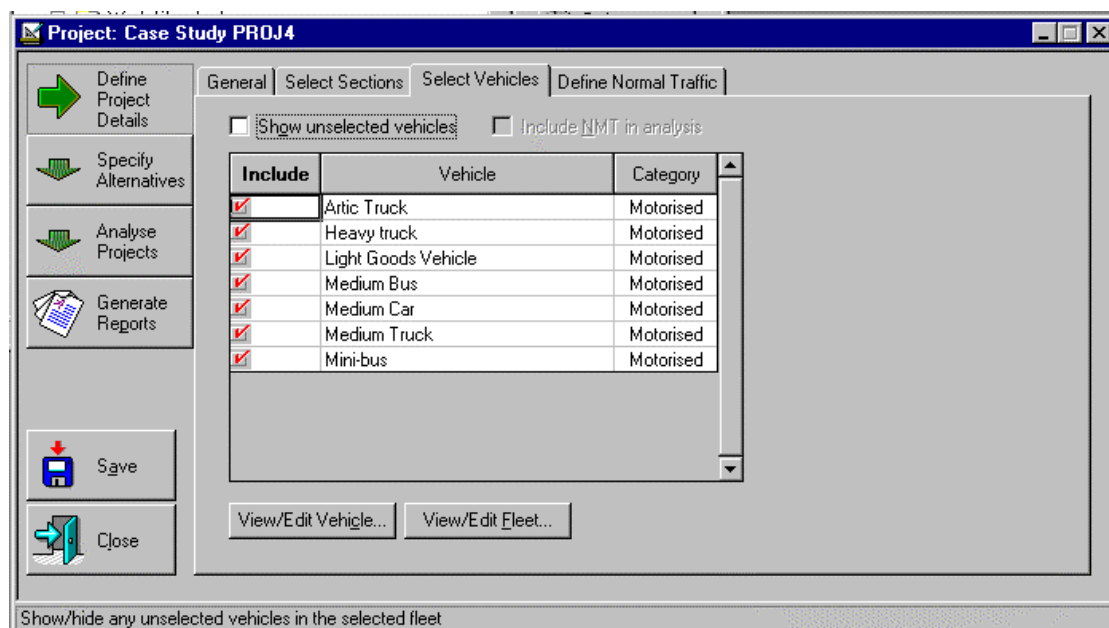
construction of a new road section can only be modelled as a section alternative within a project alternative (analysis by project).

■ Select Sections



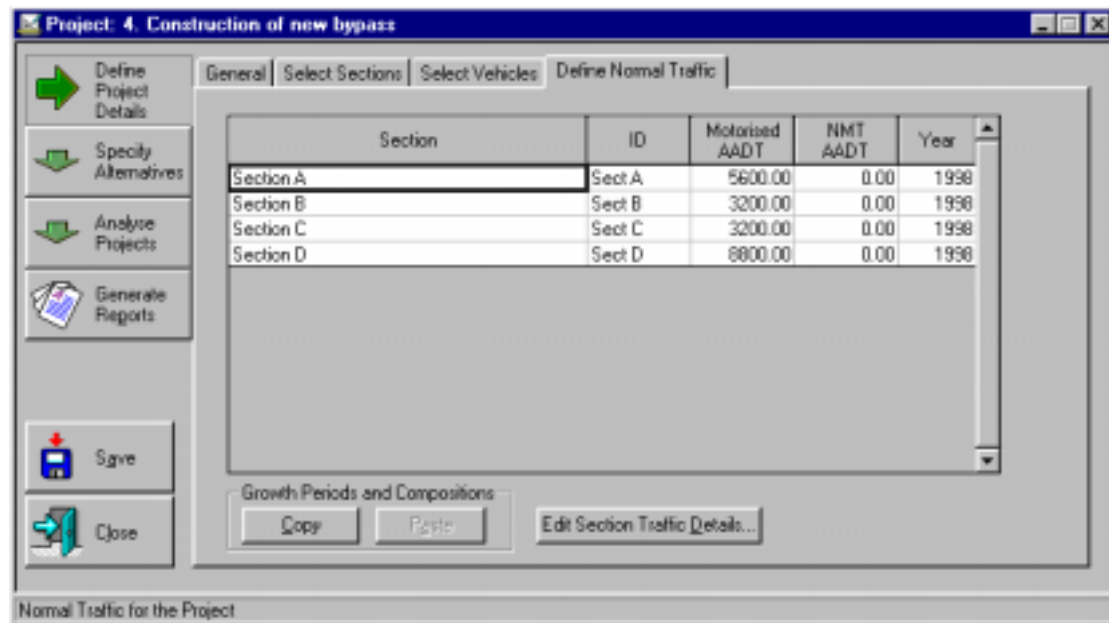
The four road sections (Sections A-D) included in the analysis are shown in the **Select Sections** screen. Section details can be accessed by double-clicking the section description. The section representing the bypass (Section E) is not included. Section E represents a proposed new section and is defined under specific project alternatives (see later). Note that, even after the analysis, Section E is not automatically added to the **Road Network** folder, as the section does not actually exist; however the user may add it to the appropriate Network after construction.

■ Select Vehicles



The vehicle types that may be assigned to the sections (including the new section) are given under **Select Vehicles**.

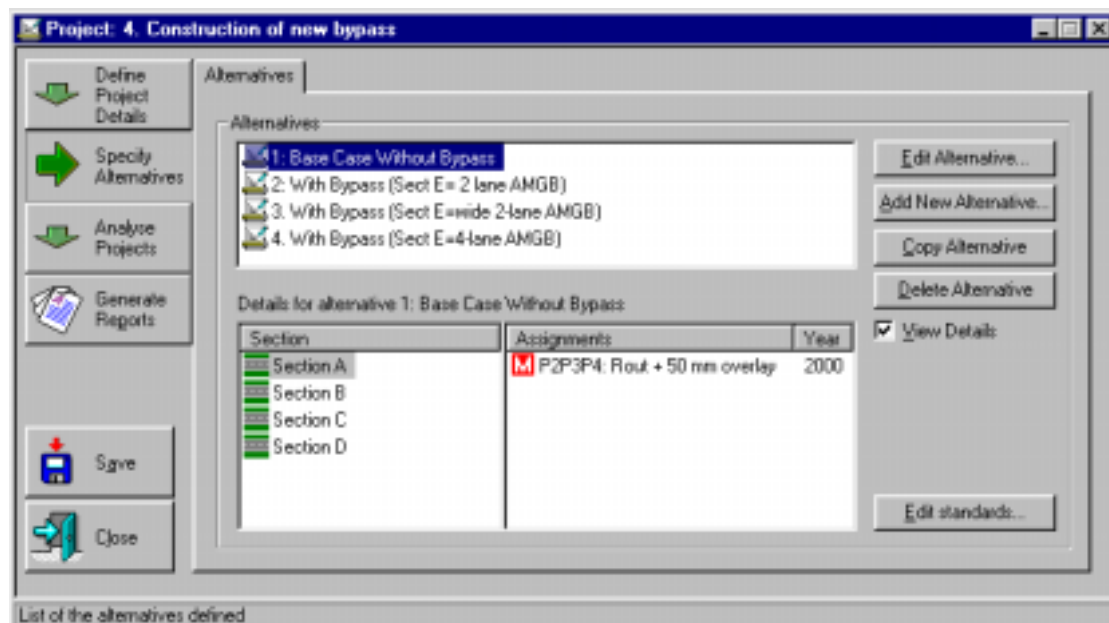
■ Define Normal Traffic



Traffic data (AADT, vehicle composition and growth rates) for Sections A, B, C and D are defined under **Define Normal Traffic**. The vehicle composition and growth rates for a particular section can be reviewed by double-clicking the appropriate row on the above Tab page. Note that traffic data for the proposed bypass (Section E) are assigned separately under the appropriate Project Alternatives.

1.2.2 Specify Alternatives

■ Alternatives



The four project alternatives considered for this case study are displayed above.

Alternative 1 represents the base case, that is, existing road sections A-D without the bypass. Alternatives 2, 3 and 4 include the bypass (Section E), with carriageway width and pavement

construction as defined in Table A4.3. The screen above indicates the maintenance standard assigned to Section A under Alternative 1 (base case).

Table A4.3
Details of proposed bypass by project alternative

Section details	Project alternative		
	2	3	4
Length (m)	10	10	10
Carriageway width (m)	8	10	14
Pavement type	AMGB	AMGB	AMGB
Construction start year	2000	2000	2000
Works duration (years)	2	3	4
Opening year	2002	2003	2004
Total economic cost (millions US Dollars)	2.735	3.418	4.786
Total financial cost (millions US Dollars)	3.2	4.0	5.6
Number of lanes	2	2	4
Speed flow type	2-lane	wide 2-lane	4-lane
Traffic flow pattern	Commuter	Commuter	Commuter
Speed limit (kph)	100	100	120

The maintenance standards associated with each project alternative are summarised in Table A4.4. For each section, the same maintenance standard has been assigned to each constituent section alternative. This allows the effect of different bypass carriageway widths to be assessed directly.

All alternatives include condition responsive routine maintenance (crack sealing and patching), as well as a condition responsive overlay.

Table A4.4
Summary of maintenance standards by project alternative

Road section	Project alternatives			
	1 (base case)	2	3	4
Section A	$RM_A + OVL_A$	$RM_A + OVL_A$	$RM_A + OVL_A$	$RM_A + OVL_A$
Section B	$RM_B + OVL_B$	$RM_B + OVL_B$	$RM_B + OVL_B$	$RM_B + OVL_B$
Section C	$RM_C + OVL_C$	$RM_C + OVL_C$	$RM_C + OVL_C$	$RM_C + OVL_C$
Section D	$RM_D + OVL_D$	$RM_D + OVL_D$	$RM_D + OVL_D$	$RM_D + OVL_D$
Section E (New section)		AMGB (2-lane) RM + OVL	AMGB (wide 2-lane) RM + OVL	AMGB (4-lane) RM + OVL

Notes:

- 1 RM = condition responsive Routine Pavement Maintenance: includes patching and crack sealing (section denoted by subscript).
- 2 OVL = condition responsive 50mm overlay (section denoted by subscript).
- 3 New section AMGB = pavement type (Asphalt Mix on Granular Base).
- 4 (2-lane) = speed-flow type given in parentheses.

The Road Works Standards and associated works assigned to each alternative are summarised in Table A4.5. Each alternative includes a Maintenance Standard (Routine + 50 mm overlay) providing condition responsive routine works and condition responsive overlay. This is effective from the year 2000 for the base case. The 'effective from' date for Section E is the year of opening of the bypass (applies Alternatives 2, 3 and 4).

Note that, for Section E, the 'effective from' date could also have been specified as the year 2000, giving the same resulting works. This is because the maintenance standard would not be applied during the construction period. (Based on the works ranking hierarchy, new section construction supersedes works associated with the maintenance standard).

The Maintenance Standard includes the following works:

- Crack Sealing if Area of Wide Structural Cracking $\geq$ 5% Carriageway Area.
- Patching if Severely Damaged Area $\geq$ 5% Carriageway Area.
- 50 mm overlay if Roughness $>$ 6 IRI.
- 50 mm overlay if 'All Structural Cracking' $\geq$ 15% Carriageway Area.

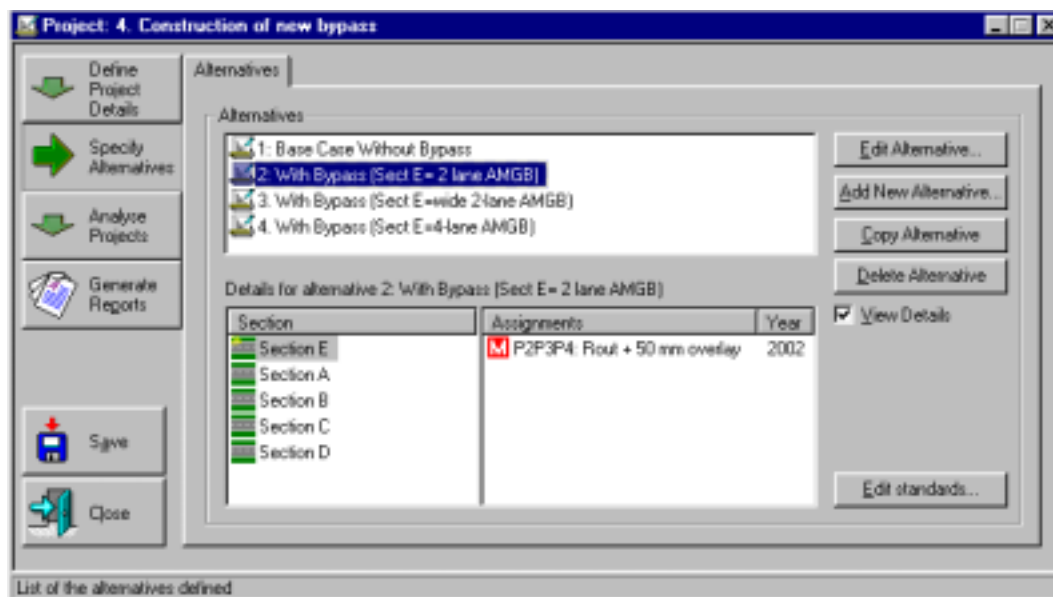
Table A4.5
Details of road works standards for each project alternative

Project alternative		Road Works Standard	"Effective from" year	Maintenance Works/Improvement Type
Alternative 1 Base Case - without bypass	M	Routine + 50mm Overlay (R&OV50) Sections A, B, C and D	2000	Crack Sealing (CRKSL)
				Patching (PATCH)
				Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 2 Bypass completed in 2001 Opening year 2002	M	Routine + 50mm Overlay (R&OV50) Sections A,B,C and D Section E (construction 2000-2001) . . .	2000	Crack Sealing (CRKSL)
				Patching (PATCH)
			2002	Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 3 Bypass completed in 2002 Opening year 2003	M	Routine + 50mm Overlay (R&OV50) Sections A,B,C and D Section E (construction 2000-2002) . . .	2000	Crack Sealing (CRKSL)
				Patching (PATCH)
			2003	Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)
Alternative 4 Bypass completed in 2003 Opening year 2004	M	Routine + 50mm Overlay (R&OV50) Sections A,B,C and D Section E (construction 2000-2003) . . .	2000	Crack Sealing (CRKSL)
				Patching (PATCH)
			2004	Overlay 50 mm at 6 IRI (OVL50)
				Overlay 50 mm at 15% cracking (OV50)

Specification of bypass details

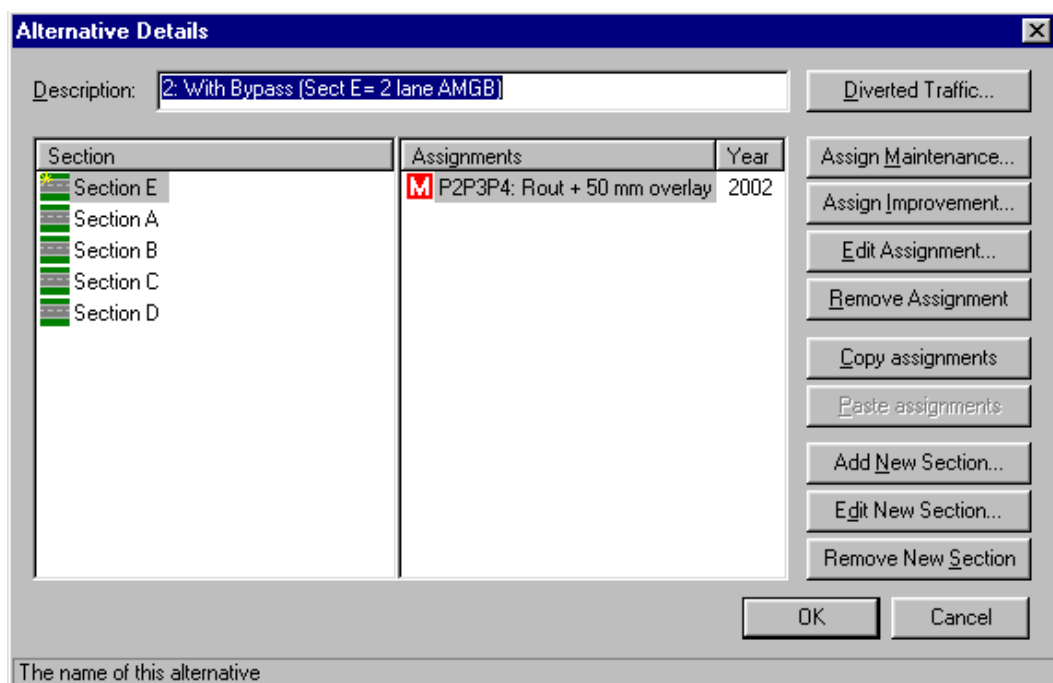
The bypass is modelled as a section alternative within a project alternative. Before the bypass is created, the **Alternatives** screen shows only Sections A-D. The bypass (Section E) was created by selecting the appropriate Alternative, then **Edit Alternative/Add New Section**.

■ Alternatives



Clicking on the **Edit Alternative** button gives:

■ Alternative Details



This shows the maintenance standard assigned to Section E under Alternative 2. The assignments for a particular section can be reviewed by selecting that section. The pre-defined maintenance standards were assigned via the **Assign Maintenance** button.

■ New Construction Section Option

The following screen is displayed by clicking the **Edit New Section** button on the Alternative Details screen (see previous page).

New Construction Section Option

General

Name:

Start Year:

Costs

Economic Cost: **Financial** Cost: per km

Duration: years

Residual value:

Annual cost stream: %

Normal Traffic...
Section Details...

OK
Cancel

The name of the new construction section option

This gives Section E details that have been specified for Alternative 2:

□ Details of Bypass

General Name of new section (Section E).

Start year of construction of Section E is 2000.

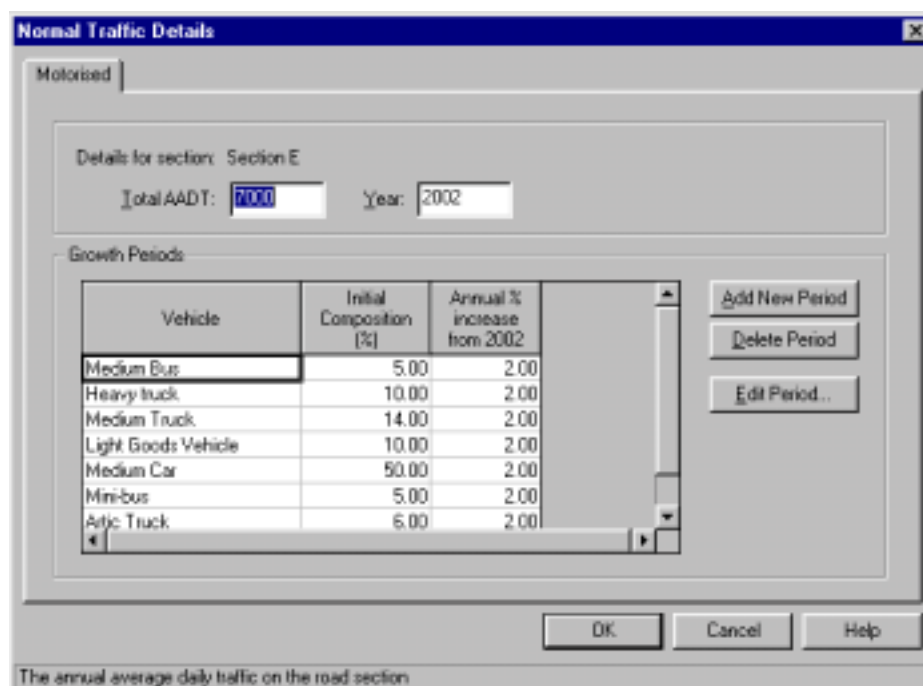
Costs Economic and financial costs specified as shown (US Dollars per km).

Duration of construction is 2 years.

Allocation of construction cost within construction period is 50% each year.

■ Normal Traffic

The following screen is displayed by clicking the **Normal Traffic** button on the **New Construction Section Option** screen (see previous page).



Normal Traffic Details

Motorised

Details for section: Section E

Total AADT: 7000 Year: 2002

Growth Periods

Vehicle	Initial Composition (%)	Annual % increase from 2002
Medium Bus	5.00	2.00
Heavy truck	10.00	2.00
Medium Truck	14.00	2.00
Light Goods Vehicle	10.00	2.00
Medium Car	50.00	2.00
Mini-bus	5.00	2.00
Artic Truck	6.00	2.00

Buttons: Add New Period, Delete Period, Edit Period...

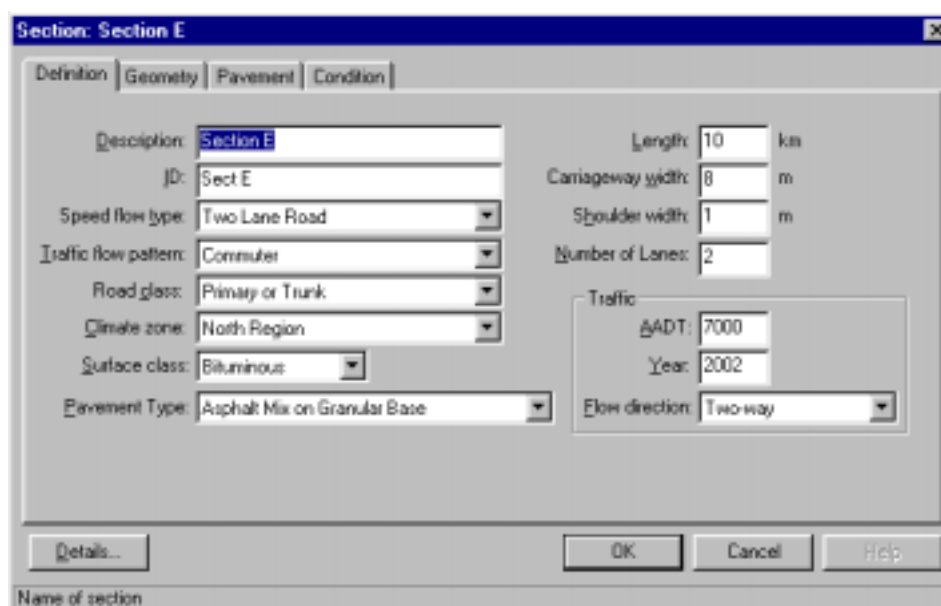
Buttons: OK, Cancel, Help

The annual average daily traffic on the road section

This screen defines the AADT and vehicle composition using the bypass in its opening year (2002 for Alternative 2). The traffic data for this case study are summarised in Table A4.1 and Table A4.2.

■ Section Details

The following screen is displayed by clicking the **Section Details** button on the **New Construction Section Option** screen.



Section: Section E

Definition | Geometry | Pavement | Condition

Description: Section E

ID: Sect E

Speed flow type: Two Lane Road

Traffic flow pattern: Commuter

Road class: Primary or Trunk

Climate zone: North Region

Surface class: Bituminous

Pavement Type: Asphalt Mix on Granular Base

Length: 10 km

Carriageway width: 8 m

Shoulder width: 1 m

Number of Lanes: 2

Traffic

AADT: 7000

Year: 2002

Flow direction: Two-way

Buttons: Details..., OK, Cancel, Help

Name of section

The Tab pages **Definition/Geometry/Pavement/Condition** are used to specify the bypass characteristics (by Alternative). As with all section data specification, care is needed to ensure consistency of data on the different Tab pages. For this case study:

- ❑ Surfacing material type (**Pavement** Tab) should be consistent with Pavement Type (**Definition** Tab).
 - ❑ AADT/year (**Definition** Tab) refers to bypass opening year.
 - ❑ Dates of previous works (**Pavement** Tab) refer to last year of bypass construction period.
 - ❑ Date of condition (**Condition** Tab) refers to last year of bypass construction period.
- **Diverted Traffic**

Once Section E has been defined for a specific project alternative via the **New Construction Section Option** screen, it is listed in the **Alternative Details** screen.

Section	Assignments	Year
Section E	M P2P3P4: Rout + 50 mm overlay	2002
Section A		
Section B		
Section C		
Section D		

The top button on this screen allows the user to specify Diverted Traffic. In the context of this case study, "Diverted traffic" refers to traffic using sections A,B C and D after the opening of the bypass.

Section	Initial AADT	Year	New AADT in 2002
Section A	5600.00	1998	2062.00
Section B	3200.00	1998	7464.00
Section C	3200.00	1998	464.00
Section D	8800.00	1998	2526.00

The Diverted Traffic screen indicates the initial AADT (1998) on sections A-D, and the new "diverted" AADT values in 2002 (the bypass opening year under Alternative 2). The vehicle composition and growth rates for a particular section (in 2002) may be reviewed by double-clicking the relevant cell under the "New AADT in 2002" column or by selecting the appropriate cell and clicking the **Edit Diversion Details** button.

■ Diverted Traffic Details

The screen below shows the vehicle composition and growth rates (in 2002) for Section A under Alternative 2.

Vehicle	New % Composition	Growth
Medium Bus	5.00	2.00
Heavy truck	3.00	2.00
Medium Truck	7.00	2.00
Light Goods Vehicle	10.00	2.00
Medium Car	70.00	2.00

1.3 Run HDM-4 and examine the results

1.3.1 Setup Run

The **Setup Run** Tab page is accessed via the **Analyse Projects** button. For this case study the base alternative for economic analysis is Alternative 1 (without the bypass) and the discount rate is 6%.

1.3.2 Run Analysis

The HDM-4 analysis commences when '**start**' is selected, and produces the output necessary for report generation.

1.3.3 Deterioration/Works Effects

The Timing of Works (by section) report can be used to check the works that would be implemented under each project alternative. Note that for alternatives 2-4, works are listed for the bypass (new section: Section E). The progression of defects on the existing roads (Sections A-D) and the new bypass (Section E) is given on the Deterioration Summary Report. This can be used to check when condition responsive works should be triggered.

1.3.4 Road User Effects

The reports produced under Road User Effects can be used to check the impact of the new bypass on vehicle speeds and volume capacity ratio (VCR) on the sub-network under study. The traffic flows on each section (with and without the bypass), derived externally, have been defined in Table A4.1. The construction of the bypass is expected to reduce the daily volume of traffic on sections A, C and D.

1.3.5 Cost Streams

The economic analysis (see Benefit Cost Ratio report) produces the economic indicators for each project alternative. The results for this case study indicate that Alternative 2 (bypass with 8m carriageway) is the most cost-effective, having the highest values of NPV, Benefit/Cost ratio, and IRR.